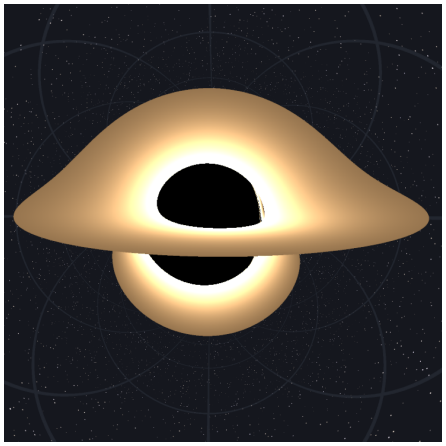
SGR A*

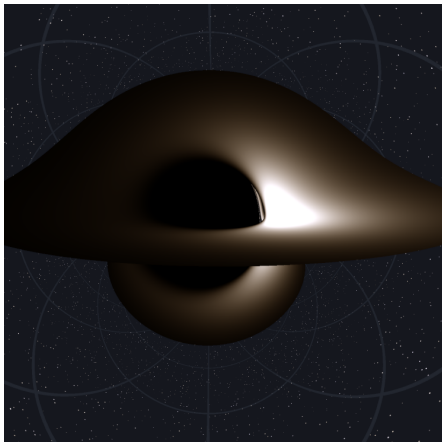
A First-Principles Relativistic Simulator of the Galactic-Centre Black Hole

Shadow metrology, S-star orbit fitting, and a free-quadrupole test of the no-hair theorem — with a complete account of the methods, tests, and development of the instrument

Kerr $a_* = 0.94$, band disk



Novikov–Thorne + beaming



C. Excelsis

Independent research report

July 6, 2026

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Abstract

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al., 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration, 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem.

This report is the long-form account of that programme. Part I is a plain-language primer for readers without a physics background. Parts II–IV derive, from first principles, every formula the simulator implements; document the numerical methods and their error analysis; present the full results with their statistical treatment; and record the complete development history, test suite, and validation gates through which the instrument was built.

Notation and Conventions

Geometric units $G = c = 1$ are used throughout unless stated; masses, lengths and times are quoted in units of the gravitational radius M (for Sgr A*: $GM/c^2 = 5.1243^{-1}$ scaling on the sky, $GM/c^3 = 21.17$ s). Metric signature $(-,+,+,+)$; Boyer–Lindquist coordinates (t, r, θ, ϕ).

Reproducibility Statement

Every quantitative statement in this report that is not a cited literature value is typeset from a macro in `numbers.tex`, generated by `scripts/make_numbers.py` from machine-readable artefacts produced by the simulation; every figure is generated by `make figures` (matplotlib) from the same artefacts. The single command `make reproduce` re-runs the full chain — unit tests, reference renders, sub-ring metrology, orbit fits, calibration, sweeps, and figure generation — from source. A number that cannot survive this cycle cannot appear in this document.

Part I

Foundations for the Non-Specialist

Chapter 1

Gravity, Mass, and the Idea of a Black Hole

This first part of the report assumes no physics background at all. It explains, in ordinary language, every idea the technical chapters build on. If you are comfortable with general relativity, skip directly to Part II.

1.1 Throwing things upward

Throw a ball into the air and it comes back. Throw it harder and it climbs higher before it comes back. This everyday fact hides a precise question: *is there a throwing speed so great that the ball never comes back at all?*

There is. For every planet or star there is a special speed, the *escape velocity*, at which a thrown object just barely coasts away forever, slowing all the while but never quite stopping. From the surface of the Earth it is about 11 kilometres per second — thirty times the speed of sound. From the surface of the Sun it is about 620 kilometres per second. The rule of thumb is simple: the more mass squeezed below you, and the closer you stand to it, the faster you must go to leave.

That second part — *the closer you stand* — is the key to everything in this report. Squeeze the same mass into a smaller ball and its surface moves closer to the centre, so the escape velocity from that surface goes up, even though the total mass never changed. Squeeze the Sun to a radius of three kilometres and the escape velocity at its surface would reach the fastest speed there is: the speed of light, about 300 000 kilometres per second.

1.2 The point of no return

Light's speed is not just fast; physics as we understand it (and as Chapter 2 sketches) says nothing can move faster. So an object compressed within that critical size has a startling property: at some radius, not even light travels fast enough to escape. Whatever falls inside can send no signal out. The sphere where escape would require exactly the speed of light is called the *event horizon*: an "event" happening inside it can never be seen from outside, even in principle.

An object like this is a *black hole* — black because no light leaves it, and a hole because things can fall in but not climb out. The critical radius is named the *Schwarzschild radius* after Karl Schwarzschild, who solved Einstein's equations for a single round mass within weeks of their publication in 1915. The Schwarzschild radius grows in direct proportion to the mass: double the mass and the horizon doubles in size. For the Earth it is about nine millimetres — the Earth would have to be crushed to the size of a marble. For the Sun, about three

kilometres. For the black hole at the centre of our Galaxy, which weighs about 4.3 million Suns, it is around 13 million kilometres — some eighteen times the radius of the Sun, yet still tiny on galactic scales.

Two cautions about the picture this conjures. First, the horizon is not a surface you could touch: nothing is *there*. It is a boundary drawn in empty space, the line on the map beyond which no path leads back out. A falling astronaut would notice nothing special while crossing it. Second, the mass is not spread throughout the dark sphere. As far as the theory can say, everything that fell in is concentrated at the centre in a region where the equations lose their meaning — the *singularity*. The singularity is a statement of ignorance rather than a physical prediction: it marks where the current theory of gravity stops being trustworthy. Fortunately, for everything in this report only the *outside* of the horizon matters, and there the theory is precise and testable.

1.3 Sagittarius A*, our own giant

Point a telescope at the constellation Sagittarius and you look toward the centre of the Milky Way, 8 277 parsecs away (a *parsec* is about 3.26 light-years; the light arriving tonight left before the last ice age peaked). At the exact dynamical centre sits a compact radio source named Sgr A* ("Sagittarius A star"). Around it, stars swarm at extraordinary speeds — a few pass their closest approach at several thousand kilometres *per second*, about a hundred times faster than the Earth orbits the Sun. Something enormously heavy and enormously compact must be holding them: the measured mass is 4.297 ± 0.012 million solar masses, packed into a region smaller than the orbit of Mercury. No cluster of dim stars, no exotic gas cloud survives scrutiny as an alternative; a supermassive black hole is the only explanation standing (Chapter 7 gives the observational history).

Sgr A* is the star of this report for a practical reason: it is the *nearest* supermassive black hole, and the only one where two entirely different kinds of measurement meet. We can follow individual stars looping around it year by year (Chapters 5, 14, 15), and we can photograph the glowing ring of light bent around its horizon (Chapters 3, 11). This report describes a simulator built from first principles that reproduces both kinds of measurement from the same underlying equations.

1.4 A sense of scale

The numbers in this subject span so many powers of ten that it is worth pausing to place them on one line (Fig. 1.1).

Astronomers measure distances within a planetary system in *astronomical units* (au): one au is the Earth–Sun distance, about 150 million kilometres. The horizon of Sgr A* spans roughly a quarter of an au. The star S2, the most important character of Part III, dives to about 120 au from the black hole — about three times the distance of Pluto from the Sun — which sounds close until you recall that this is still about 1 500 horizon radii out.

For things on the sky, astronomers use angles. A degree is what your little finger covers at arm's length; an *arcsecond* is 1/3600 of a degree, roughly a coin seen from four kilometres away; and the measurements in this report are made in *milliarcseconds* (a thousandth of an arcsecond) and *microarcseconds* (μas, a millionth). The photon ring of Sgr A* spans about 52 μas — the apparent size of a doughnut on the Moon. That such an angle is measurable at all, by linking radio telescopes across the whole Earth into one instrument, is among the technical wonders of modern astronomy.

One more unit will keep appearing: the *gravitational radius*, written M in this report. It is the natural yardstick of a black hole — half the Schwarzschild radius — and it lets us describe

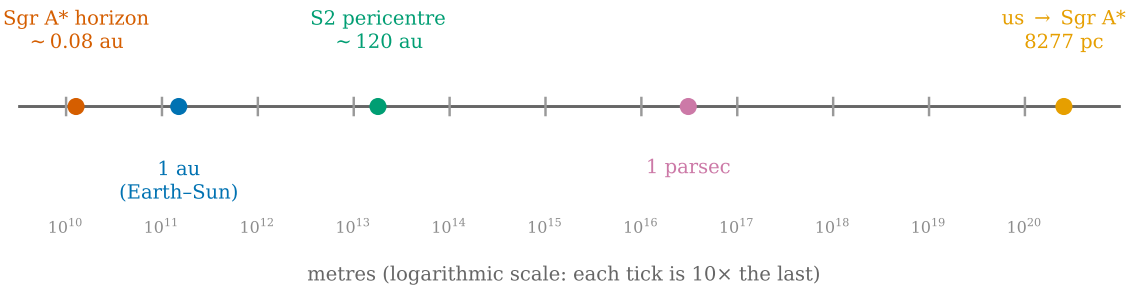


Figure 1.1: The scales of the problem on a single logarithmic line, from the Sgr A* horizon to its distance from us. Each tick is ten times the previous one: the distance to Sgr A* is about twenty billion times the size of its horizon — comparable to viewing a coin at the distance of the Moon.

any black hole, whatever its mass, with the same numbers. "The disk's inner edge sits at $6\,M$" is true of a stellar black hole and of Sgr A* alike; only the metre-stick changes. Chapter 6 says more about this trick of working in natural units.

Terms you just met.

Escape velocity The launch speed at which an object barely coasts away from a mass forever.

Event horizon The boundary around a black hole from inside which nothing, not even light, can escape. Not a material surface.

Schwarzschild radius The size of the event horizon of a non-spinning mass; proportional to the mass.

Singularity The central region where the known equations of gravity break down; hidden inside the horizon.

Sgr A* The 4.3-million-solar-mass black hole at the centre of the Milky Way, 8 277 parsecs away.

au, parsec, milli-/microarcsecond Units for distances and sky angles; see Fig. 1.1.

Gravitational radius (M) The natural length unit of a black hole; the horizon of a non-spinning hole sits at $2\,M$.

→ *Full treatment: the observational case for Sgr A* in Chapter 7; the exact horizon formulas in Chapter 8.*

Chapter 2

Spacetime and Einstein's Gravity, Gently

2.1 Gravity is not a pull

Newton described gravity as a force: the Earth pulls the apple, the Sun pulls the Earth. His description is astonishingly accurate for slow objects far from big masses, and Chapter 5 will use it freely. But it leaves a mystery — how does the Sun reach across empty space to tug on the Earth? — and it fails, measurably, exactly in the situations this report cares about.

Einstein's 1915 answer, *general relativity*, removes the mystery by removing the force. In his picture, mass does not pull on distant objects. Instead, mass changes the shape of *spacetime* — the four-dimensional fabric of the three space directions plus time — in its neighbourhood, and objects then simply travel through that reshaped fabric in the straightest possible lines. The famous slogan, due to John Wheeler, is: *matter tells spacetime how to curve, and spacetime tells matter how to move.*

A useful (if imperfect) picture is a stretched rubber sheet with a bowling ball resting in the middle. Marbles rolled across the sheet curve toward the ball, not because the ball grips them, but because the surface they travel on slopes. Each marble is doing the most natural thing possible — going straight ahead on a surface that is not flat. The Earth orbits the Sun for the same reason: it moves straight ahead through a region of spacetime that the Sun has curved, and "straight ahead" there is a loop.

The straightest possible path through curved spacetime has a name that appears on nearly every page of Part II: a *geodesic*. On the Earth's curved surface, geodesics are the great circles that airliners follow; in curved spacetime, geodesics are the paths of everything that falls freely — thrown stones, orbiting planets, and light itself. The entire simulator described in this report is, at its heart, a machine for computing geodesics very accurately.

2.2 Light bends

Because light also travels through spacetime, its path also bends where spacetime curves. The effect is tiny near the Sun — starlight grazing the solar surface deflects by less than two arcseconds, first measured during the 1919 eclipse — but near a black hole the bending becomes wild.

Fig. 2.1 shows real computed light paths around a non-spinning black hole, calculated with the same equations the simulator uses. A ray passing far away bends gently, like the starlight of 1919. A ray passing closer bends more. And there is a critical aiming distance at which the bending wins completely: the ray wraps around the hole and can circle it — light in orbit. Aim any closer and the ray spirals in and is gone. The dividing line between "escapes, however

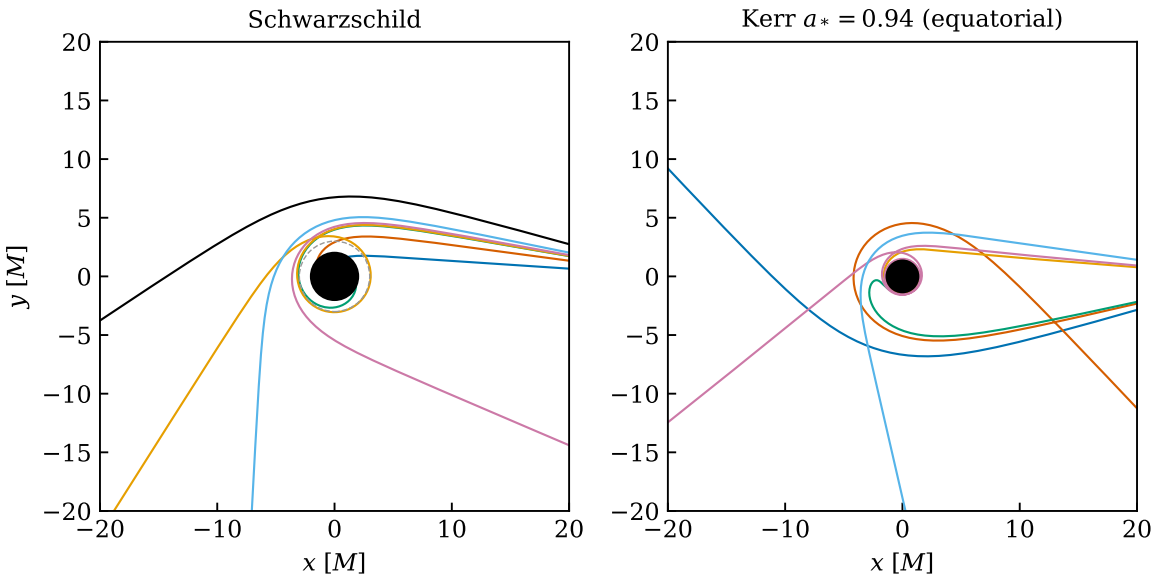


Figure 2.1: Computed paths of light rays in the equatorial plane of a black hole, from the simulator's own geodesic integrator. Left: a non-spinning (Schwarzschild) hole — rays aimed farther out escape with a bend; a critical ray circles on the dashed photon sphere; closer rays are captured. Right: a rapidly spinning (Kerr) hole — the spacetime itself circulates, so rays travelling with the spin behave differently from rays travelling against it (Chapter 3).

bent" and "captured" is what carves out the black-hole *shadow*, the subject of Chapter 3.

This bending also acts as a distorting lens. A black hole with a glowing disk around it shows you the disk twice over: light from the part of the disk hidden *behind* the hole is bent up and over the top (and under the bottom), so the far side appears above and below the dark centre. That is why the famous images — the Event Horizon Telescope's photographs and the film *Interstellar*'s Gargantua — have their characteristic crossed-halo look. Every one of those features appears, for the right physical reasons, in the renders of Part III.

2.3 Time runs slow near mass

Curved spacetime does something with no Newtonian counterpart at all: it changes the rate of time. Clocks deeper in a gravitational well genuinely run slower as seen from outside — not because of any mechanical fault, but because "one second" is itself a local quantity, like "north", that depends on where you are.

The effect is real and routine: the satellites of the GPS system carry clocks about 20 000 km above the Earth, where time runs faster than at the ground by about 45 microseconds per day. Uncorrected, satellite navigation would drift by kilometres within a day. Near a black hole the same effect grows enormous; at the horizon, an outside observer would see a falling clock slow to a stop.

For this report the significance is very concrete: when the star S2 dives to its closest approach to Sgr A*, its light is emitted from slightly deeper in the black hole's gravitational well, where time runs slower. Every wave of its light is stretched on the climb out — the wavelength shifts toward the red. The GRAVITY instrument measured exactly this *gravitational redshift* in 2018, and reproducing that measurement from first principles is one of this report's central results (Chapters 4, 15).

2.4 What a metric is (without tensors)

One piece of vocabulary unlocks all of Part II. To compute anything, we need coordinates — labels like latitude and longitude that name the points of spacetime. But coordinates are just names; they do not tell you how *far apart* two nearby points really are. On the Earth, one degree of longitude is 111 km at the equator and zero at the pole: the conversion from coordinate differences to true distances varies from place to place.

The rulebook for that conversion is called the *metric*. Give the metric the small coordinate steps between two nearby events (a bit of north–south, a bit of east–west, a bit of time) and it returns the true separation — including the true amount of elapsed clock time. All of the curvature of spacetime is encoded in how this rulebook changes from place to place; slow-running clocks and bending light are both just entries in the rulebook. Einstein's field equations determine the metric produced by a given arrangement of mass; for a lone spinning black hole the exact answer is known and is the famous *Kerr metric* (Chapters 3, 8), on which this entire simulator is founded.

When Part II writes symbols such as g_{tt} or g_{rr}, nothing deeper is meant than "the entry of the rulebook that converts time-steps (or radius-steps) into true separations, at this location".

Terms you just met.

Spacetime Space and time treated as one four-dimensional fabric, whose shape is what we experience as gravity.

General relativity (GR) Einstein's theory: mass curves spacetime; freely falling objects follow the straightest paths in it.

Geodesic The straightest possible path through curved spacetime; the track of anything falling freely, including light.

Gravitational lensing The bending of light by curved spacetime; a black hole shows the far side of its own disk, bent over the top.

Time dilation Clocks deeper in a gravitational well run slower; measurable with GPS satellites and with the star S2.

Gravitational redshift Light climbing out of a gravitational well is stretched to longer, redder wavelengths.

Metric The location-dependent rulebook that converts coordinate differences into true distances and times; in GR it carries all of gravity.

→ *Full treatment: the Kerr metric in Chapter 8; geodesics and their conserved quantities in Chapter 9; the traced light-bending figures in Chapters 9 and 11.*

Chapter 3

Spinning Black Holes and the Shadow

3.1 Black holes spin

Almost everything in the universe rotates — planets, stars, whole galaxies — and when a rotating object collapses, its spin is not lost. It concentrates, the way a figure skater speeds up when she pulls her arms in. Black holes born from collapsing, swirling matter are therefore expected to spin, often ferociously.

A non-spinning black hole is described exactly by Schwarzschild's 1915 solution of Chapter 1. The spinning case resisted for another half-century, until Roy Kerr found the exact solution in 1963. The *Kerr metric* — the rulebook of Chapter 2 for a spinning hole — is the mathematical foundation of this entire report, and it says that a black hole is completely described by just two numbers: its mass and its spin. The spin is usually quoted as a dimensionless number a_* between 0 (not spinning) and 1 (the theoretical maximum, spinning so fast the horizon would tear open). For Sgr A* the working estimate used throughout this report is $a_* = 0.94$ — a rapid spinner.

3.2 Frame dragging: spacetime as a whirlpool

A spinning mass does something Newton never imagined: it drags spacetime around with it. Close to a spinning black hole, "standing still" is simply not on the menu — space itself circulates like water around a drain, and everything in it, light included, is carried along. This *frame dragging* has been measured in Earth orbit (by the Gravity Probe B gyroscopes and the LAGEOS satellites) at the level of fractions of a degree per century; near Sgr A* it becomes a dominant feature of the landscape (Fig. 3.1).

There is a region, outside the horizon but close to it, where the dragging is so strong that *nothing* can hold still — to hover without circulating you would need to move faster than light. This region is the *ergosphere*, a flattened shell thickest over the equator (Fig. 3.2). Objects inside it can still escape — it is not a point of no return — but they cannot avoid being swept around.

3.3 The anatomy of a spinning hole

Fig. 3.2 labels the landmarks that Part II computes exactly, for a non-spinning and a rapidly spinning hole side by side.

- The *event horizon* (black): the point of no return. Spin shrinks it, from $2\,M$ at $a_* = 0$ toward $1\,M$ at the extreme.

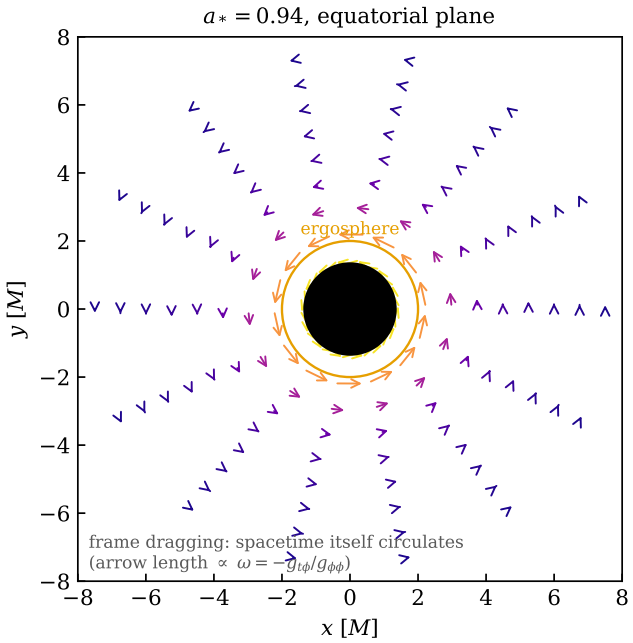


Figure 3.1: Frame dragging around a black hole with $a_* = 0.94$, computed from the Kerr rulebook. Each arrow shows how far a "stationary" marker at that radius is swept around per unit time; inside the orange boundary (the ergosphere) refusing to rotate is impossible.

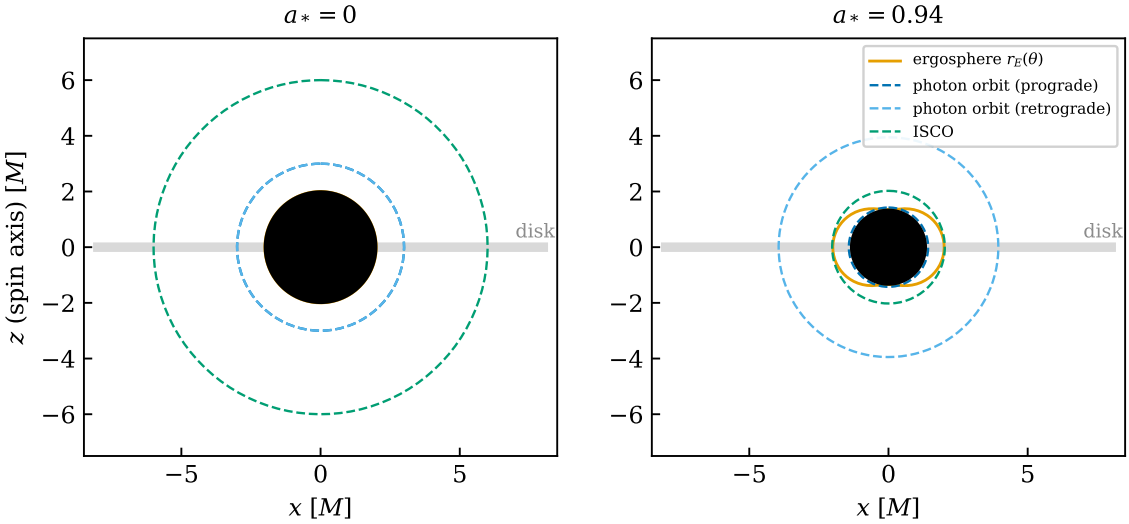


Figure 3.2: Cross-sections through a non-spinning ($a_* = 0$) and a rapidly spinning ($a_* = 0.94$) black hole, drawn to scale from the exact formulas of Chapter 8. Spin shrinks the horizon, opens up an ergosphere, splits the photon orbit into prograde/retrograde pairs, and pulls the innermost stable orbit (ISCO) dramatically inward.

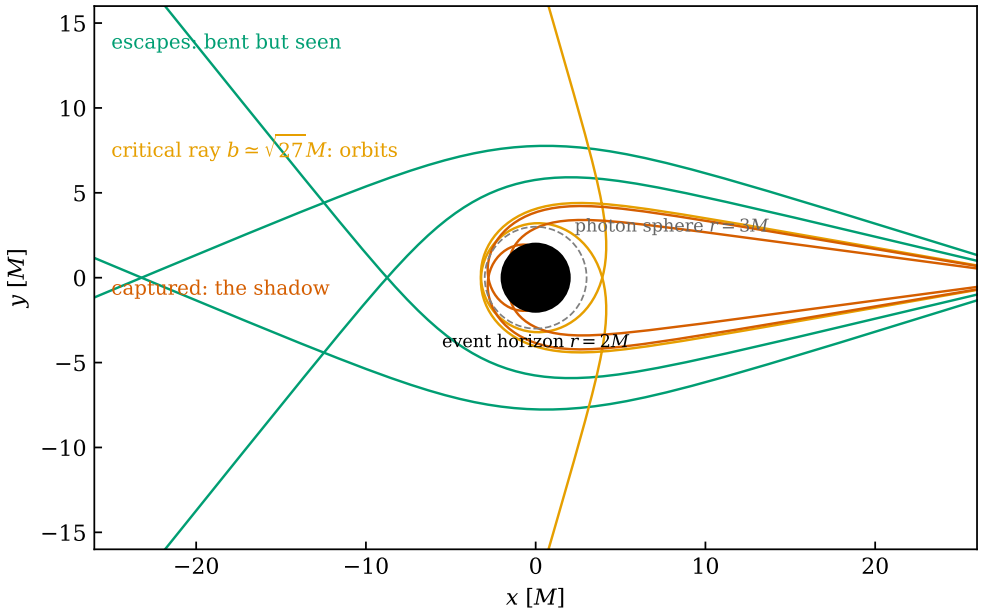


Figure 3.3: How the shadow forms, computed from the exact bending equation. Green rays escape (bent); the orange critical ray circles on the photon sphere; red rays are captured. The shadow is the set of camera directions whose rays end on the horizon — its edge is light that grazed the photon sphere.

- The *ergosphere* (orange): the no-standing-still zone, touching the horizon at the poles and bulging at the equator.
- The *photon orbits* (blue): radii where light can circle the hole. For a spinning hole there are two equatorial ones — a closer orbit for light moving *with* the rotation (*prograde*) and a farther one for light moving *against* it (*retrograde*). Spacetime's whirlpool helps the co-rotating light hold a tighter circle.
- The *ISCO* (green), the innermost stable circular orbit: the closest circle on which matter can orbit steadily. Any closer and the slightest nudge sends it spiralling in. The ISCO is the inner edge of the glowing disk of Chapter 4, and it moves inward spectacularly with spin — from $6\,M$ for $a_* = 0$ to about $2\,M$ at $a_* = 0.94$. Much of what changes visually with spin in our renders traces back to this single fact.

3.4 The shadow and the photon ring

Now surround the hole with light — a glowing disk, or simply the stars behind it — and ask what a distant camera sees. Chapter 2 showed that rays aimed too close are captured. Point the camera at the hole, and there is a patch of sky, for every pixel of which the answer is "that line of sight ends on the horizon". That dark patch is the *shadow* (Fig. 3.3).

Three facts about the shadow do a great deal of work in this report.

First, *the shadow is bigger than the horizon.* Light does not have to hit the horizon to be lost — anything aimed within the critical distance spirals in. For a non-spinning hole the shadow's radius is $\sqrt{27} \approx 5.2\,M$, about 2.6 times the horizon radius. Our simulator reproduces this number to sixteen digits (Chapter 11), which is the sort of check that separates a scientific instrument from a pretty picture.

Second, *the shadow's edge is made of light that almost stayed.* Rays that graze the critical boundary can circle the hole once, twice, three times before escaping to the camera. Each extra half-turn stacks another, exponentially thinner and fainter image of the whole sky (and of the disk) right at the shadow's rim. This onion of nested images is the *photon ring*, and its sub-rings are analysed quantitatively in Chapter 13. The Event Horizon Telescope's measurement of this ring for Sgr A* — $51.8 \pm 2.3\,\mu$as across — is one of the two observational anchors of this whole project.

Third, *spin distorts the shadow.* Frame dragging lets prograde light survive closer in, so the shadow's edge on the side rotating toward you pulls inward: the silhouette flattens on one side and shifts sideways, by an amount our simulator measures precisely (Chapters 11, 19). Viewed nearly pole-on — our actual view of Sgr A* — the distortion is frustratingly small, a few per cent, which is why the ring's *size* alone cannot pin down the spin (the "spin–inclination degeneracy" that Chapter 19 maps out).

Terms you just met.

Kerr metric The exact spacetime of a spinning black hole (Kerr 1963); fully specified by mass and spin.

Spin parameter a_* Dimensionless spin from 0 to 1; Sgr A* is modelled at $a_* = 0.94$.

Frame dragging The whirlpool-like circulation of spacetime around a spinning mass.

Ergosphere The region where standing still is impossible; escape is still possible from inside it.

Prograde / retrograde Moving with / against the black hole's rotation; prograde orbits survive closer in.

Photon sphere / photon orbits Radii at which light circles the hole; unstable, so grazing light eventually escapes or falls.

ISCO Innermost stable circular orbit — the inner edge of the disk; moves from $6\,M$ (no spin) toward the horizon as spin grows.

Shadow The dark patch of captured lines of sight; radius $\sqrt{27}\,M$ for no spin, subtly reshaped by spin.

Photon ring The stack of exponentially fainter images of the scene hugging the shadow's edge.

→ *Full treatment: Kerr geometry and all characteristic radii in Chapter 8; the traced shadow and critical curve in Chapter 11; sub-ring measurements in Chapter 13.*

Chapter 4

Light, Heat, and Colour: Reading the Disk

4.1 Hot things glow, and colour is a thermometer

Heat a poker in a fire and it glows dull red; hotter, and it turns orange, then yellow, then blue-white. This everyday sequence is one of the most useful facts in astrophysics: a dense hot body glows with a spectrum — a recipe of colours — that depends on its temperature and on almost nothing else. Physicists call this ideal glow *blackbody radiation*, and the temperature-to-colour dictionary is exact: around 3 000 degrees looks ember-orange, 6 000 degrees looks white like the Sun's surface, 20 000 degrees looks blue-white. The renders in this report use precisely this dictionary — every pixel's colour comes from a computed temperature pushed through the same colour-matching tables that describe human vision (Chapter 12), not from an artist's palette.

4.2 The accretion disk

Gas falling toward a black hole almost never falls straight in. Like water circling a drain, it has some rotation, so it settles into a flattened, swirling *accretion disk*. Within the disk, gas at each radius orbits at its own rate — inner gas circles faster than outer gas, just as Mercury outpaces Neptune — and the resulting internal friction does two things: it lets the gas drift gradually inward, and it heats the disk white-hot. The energy released is prodigious. Matter spiralling through a disk onto a rapidly spinning black hole can radiate away tens of per cent of its entire rest-mass energy — compared with less than one per cent for nuclear fusion. Our simulator computes this efficiency exactly (it is the η of Chapter 8): about 5.72 per cent for a non-spinning hole, rising to 17.96 per cent at Sgr A*'s spin, because the disk's inner edge (the ISCO of Chapter 3) reaches so much deeper into the well.

The model disk used throughout Part III is the classic *Novikov–Thorne* thin disk: geometrically thin like a vinyl record, hottest just outside the ISCO, cooling outward. It is an idealisation — the real flow around Sgr A* is a hot, puffed-up, far dimmer affair — and Chapter 22 is careful about which conclusions survive that difference. The disk model gives our images their brightness and colour; crucially, the *geometry* of shadow and photon ring comes from the spacetime itself, not from the disk details.

4.3 Why one side of the disk blazes

Every published image of a black-hole disk — the EHT rings, *Interstellar*'s Gargantua, the renders in this report — is conspicuously brighter on one side. That asymmetry is a speed

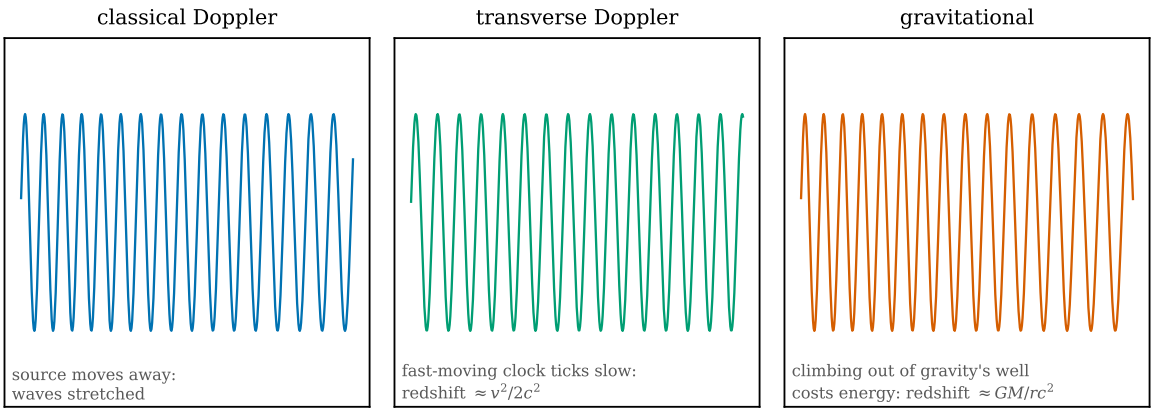


Figure 4.1: The three ways a wavelength gets shifted. Classical Doppler: motion along the line of sight stretches or compresses the wave train. Transverse Doppler: a fast clock ticks slowly, so even sideways motion reddens the light — a purely relativistic effect. Gravitational: light climbing out of a gravity well arrives stretched.

effect, not a lighting effect.

The inner disk orbits at an appreciable fraction of the speed of light. When a source of light moves toward you that fast, three effects conspire: its light is squeezed forward like spray from a moving hose (*beaming*), arrives blueshifted, and arrives at a faster clip. The approaching side of the disk therefore appears far brighter — in our reference render at $a_* = 0.94$, viewed nearly edge-on, the approaching limb outshines the receding one by a factor of 9.60 in total light (Chapter 12). The receding side is correspondingly dimmed and reddened. The whole effect is packaged in a single number per light ray, the *redshift factor* g: the ratio of the light's frequency as received to its frequency as emitted. Brightness scales as g^4 — the fourth power — which is why the asymmetry is so dramatic. If a simulated disk ever looks symmetric, its g^4 is wrong; this was one of our hard validation gates (Chapter 21).

4.4 The three redshifts

That factor g mixes three physically distinct effects, and pulling them apart is exactly what the S2 measurements of Part III require (Fig. 4.1).

1. *Classical Doppler*: the familiar ambulance-siren shift. Motion away from you stretches the waves (redshift); motion toward you compresses them (blueshift). This is by far the largest term for an orbiting star, and Newton would have predicted it too.

2. *Transverse Doppler*: special relativity's addition. A fast mover's clock runs slow, so its emitted waves come out at a reduced rate — a redshift that appears even when the motion is purely sideways. It scales with speed squared, so it only matters when things get truly fast.

3. *Gravitational redshift*: general relativity's addition, from Chapter 2 — light climbing out of a gravitational well is stretched. It scales with depth in the well.

For the star S2 at its 2018 closest approach, the first term is several thousand km/s, while the second and third are each worth only about a hundred km/s — and, by a neat algebraic coincidence worked out in Chapter 15, almost exactly equal to each other. The GRAVITY collaboration's 2018 detection of that little two-hundred-km/s surplus over the

Newtonian prediction was the first direct test of general relativity beside a supermassive black hole. Reproducing it, with no tunable knob, is one of this report's headline results: our model peaks at 205 km/s at epoch 2018.379 (Chapter 15).

Terms you just met.

Blackbody radiation The temperature-determined glow of a dense hot body; the exact temperature-to-colour dictionary used in our renders.

Accretion disk The flattened swirl of hot gas spiralling into a black hole; modelled here as the Novikov–Thorne thin disk.

Accretion efficiency (η) The fraction of rest-mass energy radiated by matter spiralling in: 5.72% (no spin) to 17.96% (Sgr A*'s spin).

Doppler beaming The forward-throwing of light by a fast-moving source; makes the approaching disk limb $\sim$9.60$\times$ brighter in our reference render.

Redshift factor g Received frequency over emitted frequency for one light ray; observed brightness scales as g^4.

Classical / transverse / gravitational redshift The three distinct wavelength shifts; separating them at S2's pericentre is the GRAVITY 2018 test.

$\rightarrow$ Full treatment: disk physics and the g^4 law in Chapter 12; the redshift decomposition and the S2 result in Chapters 14 and 15.

Chapter 5

Orbits, S-Stars, and Weighing a Black Hole

5.1 Kepler's ellipse

Four hundred years before anyone imagined black holes, Johannes Kepler discovered that planets travel on *ellipses* — squashed circles — with the Sun at one *focus*, a point offset from the centre (Fig. 5.1). Three of his terms are used on nearly every page of Part III:

- the *semi-major axis* a — half the long diameter, the orbit's overall size;
- the *eccentricity* e — how squashed and off-centre the ellipse is, from 0 (circle) toward 1 (a stretched sliver);
- the *pericentre* and *apocentre* — the points of closest and farthest approach. (For orbits around stars you may know these as perihelion and aphelion.)

Kepler's second discovery is visible in the dots of Fig. 5.1: bodies move *fast near pericentre and slow near apocentre*. His third links the orbit's size and period to the central mass — bigger orbits take longer, and, decisively for us, *at fixed size, a heavier central mass means a faster orbit*. Measure the size and the period of any orbit and the central mass falls out. This is how we weigh everything in astronomy, from planets to galaxies — and it is exactly how Sgr A* is weighed.

5.2 The S-stars, and S2 above all

Buried in the central light-month of the Galaxy is a swarm of ordinary young stars on extraordinary orbits — the *S-stars*. Since 1992, two teams (in Garching and in Los Angeles) have photographed the region every year, watching individual stars trace out their ellipses around an invisible focus. The star *S2* is the jewel of the collection: bright enough to measure precisely, with a period of just 16 years — short enough that we have now watched it complete more than one and a half full loops. Its orbit is extreme by any planetary standard: eccentricity 0.88, pericentre speed near 7 700 km/s (2.6 % of the speed of light), diving to about 1 500 horizon radii from the black hole.

This report fits the complete published S2 data set — 145 positions and 44 velocities spanning 1992–2016 (Chapter 15) — plus four more S-stars in a joint fit (Chapter 16).

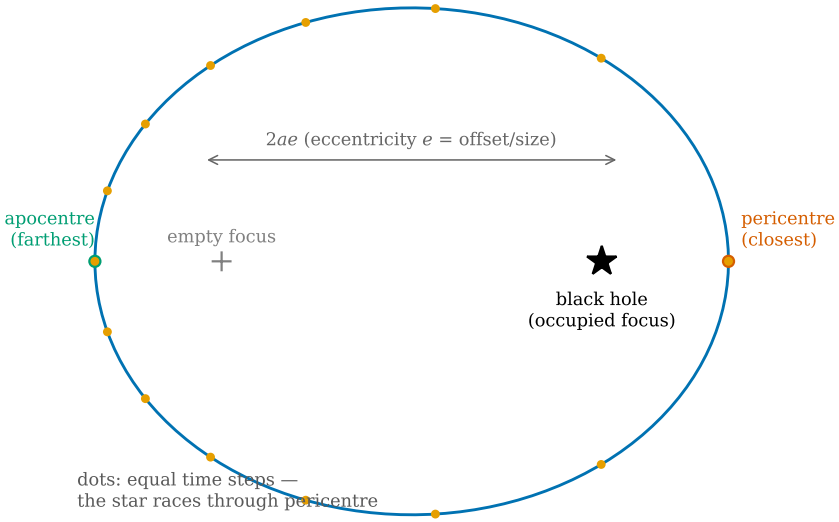


Figure 5.1: The anatomy of a Kepler orbit. The attracting mass sits at one focus; eccentricity measures how off-centre it is. The dots mark equal time steps: the orbiting body loiters near apocentre and races through pericentre.

5.3 Two channels: positions and speeds

Everything we know about the S-stars comes through two complementary kinds of measurement (Fig. 5.2).

Astrometry is simply measuring where the star is on the sky, epoch by epoch — in milliarcseconds, using adaptive-optics cameras on the world's largest telescopes and, since 2017, the GRAVITY interferometer, which links four 8-metre telescopes into one virtual 130-metre one. Astrometry sees the orbit *projected* on the sky, like the shadow of a hoop tilted against a wall.

Spectroscopy measures how fast the star moves *along* the line of sight, via the Doppler shift of a hydrogen absorption line in its spectrum (the Brγ line, in the infrared where the Galactic dust is transparent). This "radial velocity" channel sees exactly the component of motion that astrometry cannot.

Neither channel alone determines everything; together they overdetermine the orbit. And one beautiful consequence deserves its own sentence: because astrometry gives the orbit's *angular* size while the velocities give its *physical* scale, combining them yields the *distance to the Galactic centre* — a geometric distance measurement across 26 000 light-years with no cosmic distance ladder involved. Our fit recovers both the mass and this distance (Chapter 15).

5.4 Einstein's fingerprints on the orbit

If gravity were exactly Newtonian, S2's ellipse would repeat itself forever, fixed in space. General relativity disagrees in two measured ways.

First, the *redshift excess* of Chapter 4: at the 2018 pericentre passage, S2's spectral lines arrived about 200 km/s redder than any Newtonian orbit could explain — time dilation plus gravitational redshift, detected by GRAVITY in 2018.

Second, *orbital precession*: the ellipse itself slowly pivots. Each orbit, the pericentre direction advances by a small angle, so the path traces a rosette rather than a closed ring (Fig. 5.3). For Mercury this famous advance is 43 arcseconds per century and was the first triumph of general relativity; for S2 it is about 12 *arcminutes* per 16-year orbit — some fifty thousand

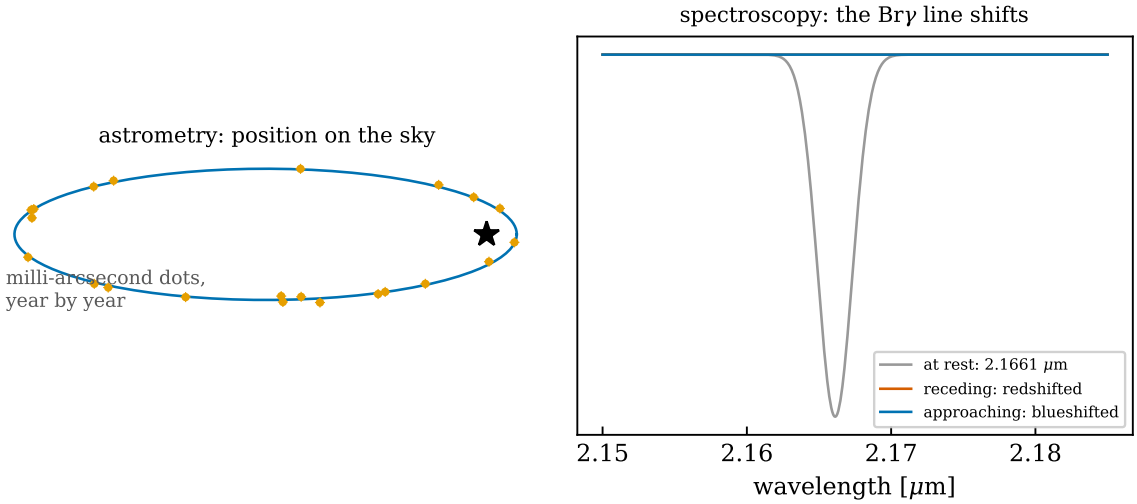


Figure 5.2: The two observation channels. Left — astrometry: the star's position on the sky, measured year after year in milliarcseconds, traces the projected ellipse. Right — spectroscopy: the wavelength of a known line in the star's light shifts with its line-of-sight speed (the Doppler effect of Chapter 4).

times faster — and GRAVITY confirmed it in 2020. Our simulator reproduces both effects with no adjustable amplitudes: the fitted mass and geometry force them (Chapter 15).

5.5 Thirty-four years in one diagram

Fig. 5.4 places the milestones — from the first infrared images of the S-stars to the EHT picture of the ring — on one line, ending with the simulator this report describes: a single instrument that reproduces, from first principles, the measurements of that whole timeline.

Terms you just met.

Ellipse, focus, semi-major axis, eccentricity The shape language of orbits (Fig. 5.1).

Pericentre / apocentre Closest and farthest points of an orbit; S2's pericentre is ∼120 au at 2.6% of light speed.

S-stars, S2 The stars orbiting Sgr A*; S2, with its 16-year period, is the most precisely measured.

Astrometry Measuring sky positions over time (milliarcseconds); gives the projected orbit.

Radial velocity (spectroscopy) Measuring line-of-sight speed from Doppler-shifted spectral lines.

Geometric distance Combining both channels yields the distance to the Galactic centre directly.

Precession / rosette The slow pivot of the orbital ellipse predicted by GR; 12′ per orbit for S2, confirmed 2020.

→ *Full treatment: the relativistic orbit model in Chapter 14; the fit and both GR effects in Chapter 15; the multi-star fit in Chapter 16.*

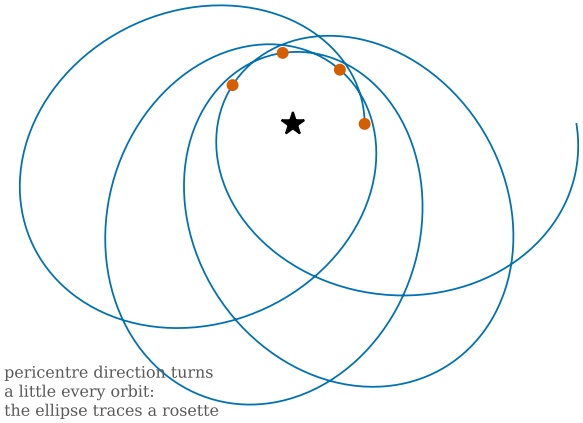


Figure 5.3: Relativistic precession: the orbital ellipse pivots a little each revolution, tracing a rosette. The dots mark successive pericentre directions. (Exaggerated for visibility: S2's advance is $12'$ per orbit; this drawing uses thousands of times more.)

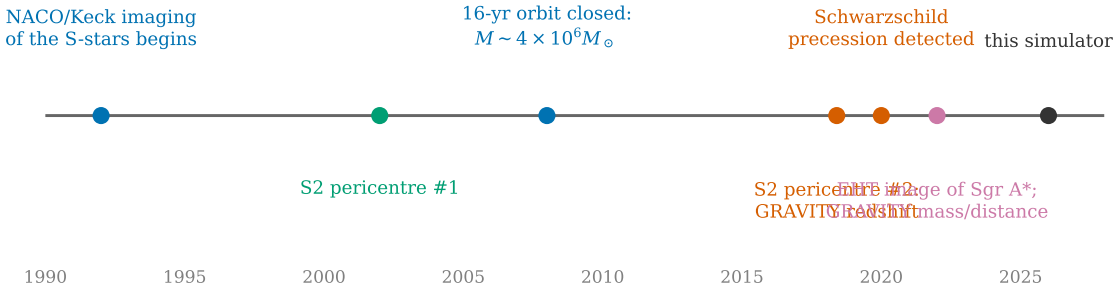


Figure 5.4: Milestones of the Galactic-centre programme covered in this report.

Chapter 6

The Toolkit in Plain Words

This last primer chapter collects the working tools — units, statistics, and one big theorem — that the technical chapters use constantly.

6.1 Natural units: setting $G = c = 1$

Physicists working with black holes almost always set the gravitational constant G and the speed of light c equal to 1. This is not a trick or an approximation; it is a choice of units, exactly like measuring road distances in hours of driving. Once you do it, mass, length, and time all become interchangeable, and a single symbol M can serve as the yardstick for everything: lengths are quoted in units of the gravitational radius ("the ISCO is at $6\,M$"), times in units of the light-crossing time of that radius. For Sgr A*, one M of length is about 0.042 au and one M of time is about 21 seconds. The huge practical benefit is universality: every formula and every rendered image in Parts II–III applies to *any* black hole; only the final conversion to kilometres, seconds, or microarcseconds depends on the actual mass (that conversion is Chapter 19).

A related habit: Part II speaks of *vectors* (arrows with direction and length — a velocity is one) and occasionally *tensors*. For reading this report it is enough to know that a tensor is a machine that eats vectors and returns numbers, in a way that all observers agree on. The metric of Chapter 2 is the only tensor we ever really need: it eats two little coordinate steps and returns their true separation.

6.2 Error bars, and the meaning of χ^2

Every real measurement comes with an uncertainty — an *error bar* — and the whole craft of fitting models to data is bookkeeping those error bars honestly. The universal bookkeeping device is called *chi-squared* (χ^2), and it answers one question: *by how many error bars, in total, does the model miss the data?* (Fig. 6.1). For each data point, take the miss, divide by that point's error bar, square it (so misses in either direction count alike, and big misses count a lot), and add up over all points.

Two rules of thumb make χ^2 readable. A model that is right, fitted to honest error bars, should miss each point by about one error bar on average — so χ^2 divided by the number of data points (the *reduced* χ^2) should be near 1. Much bigger means the model (or the error bars) are wrong; much smaller means the error bars were pessimistic. Our headline S2 fit lands at $\chi^2_\nu = 0.959$ (Chapter 15) with the published error bars used exactly as given.

And once the best knob-settings are found, *how well known* is each knob? Wiggle a knob away from its best value and χ^2 climbs; the range over which it climbs by less than a set amount is the knob's *confidence interval* — the "±" quoted after every fitted number in this report. When we write $M = 4.270 \pm 0.195$ million suns, the $\pm$ is precisely such an interval.

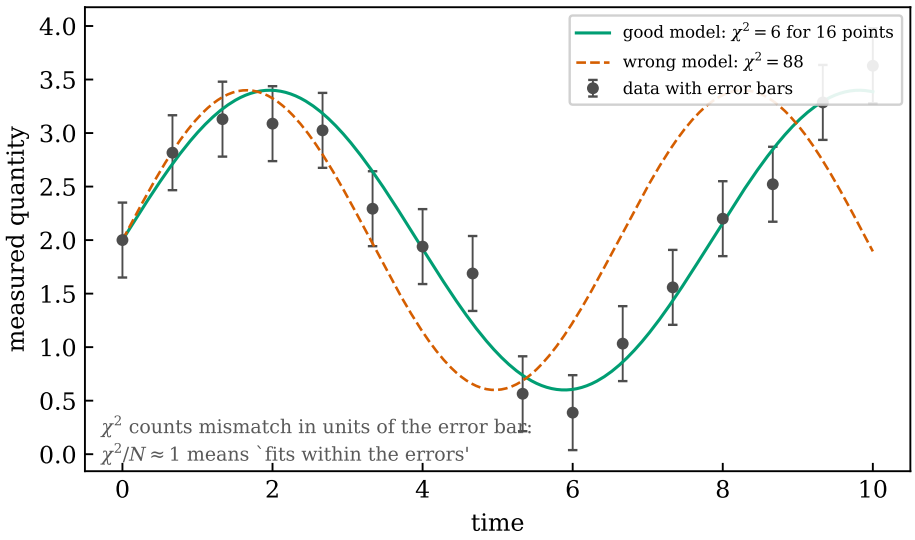


Figure 6.1: χ^2 made visible. Two candidate models through the same data: the good one misses each point by about one error bar ($\chi^2 \approx$ number of points); the wrong one racks up an enormous total. Fitting means turning model knobs until χ^2 is as small as it gets.

6.3 MCMC: mapping uncertainty by wandering

For a model with thirteen knobs, the surface of χ^2 over all knob combinations is a thirteen-dimensional landscape, and simple wiggling can miss how knobs trade off against one another. The heavy machinery for this is *Markov chain Monte Carlo* (MCMC): release a crowd of random walkers into the landscape, with rules that make them drift toward — and then wander within — the low-χ^2 valleys (Fig. 6.2). After enough steps, *where the walkers spend their time is itself the answer*: dense regions are probable parameter values, sparse regions improbable. Histograms of the walkers' positions give every parameter's uncertainty and every pairwise trade-off at once. Chapter 15 runs exactly this (32 walkers, thousands of steps, the full relativistic model evaluated at every step) and checks that the walker map agrees with the simpler wiggle estimate.

6.4 The no-hair theorem in one picture

One theorem gives this project its sharpest question. General relativity predicts that a black hole, once settled, is the simplest macroscopic object in nature: its exterior gravity is *completely* fixed by two numbers, mass and spin. All other detail of whatever fell in — its shape, its chemistry, its history — is radiated away or swallowed. The theorem's nickname says it: *black holes have no hair*.

This is falsifiable. Any rotating body has a *quadrupole moment* — a measure of how squashed its gravity field is (Fig. 6.3). For planets and stars the squashing is free to vary; for a Kerr black hole it is locked rigidly to the mass and spin. So the test is: measure mass, spin, and quadrupole *independently*, and check the lock. We implement this by adding one knob to the spacetime — a quadrupole surplus called ε_q, zero for a true Kerr hole — and asking the S2 data how far from zero it could be (Chapter 17). The honest answer today: the data are fully consistent with zero, with room to spare (a single star at 1 500 horizon radii barely feels the quadrupole), and the same machinery is ready for the closer stars that future instruments will find.

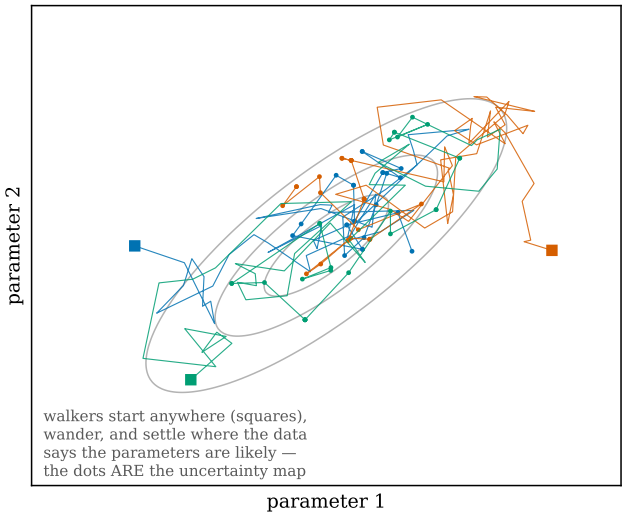


Figure 6.2: Three MCMC walkers released onto a two-parameter landscape (grey contours: the low-χ^2 valley). After an initial wander-in, the walkers trace out the valley; their footprints map the uncertainty, including the diagonal trade-off between the two parameters.

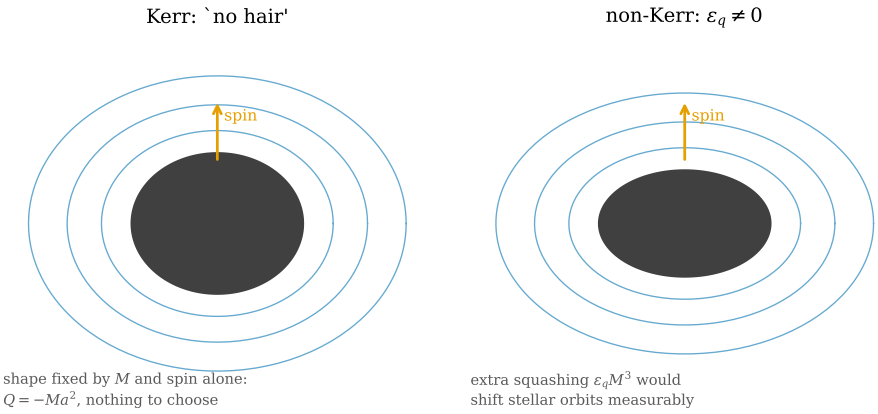


Figure 6.3: The no-hair test in one picture. Left: for a Kerr black hole the field's squashing (quadrupole) is fixed by mass and spin — nothing to choose. Right: if the measured squashing deviated ($\varepsilon_q \neq 0$), stellar orbits would precess differently, and Kerr would be falsified.

Terms you just met.

Geometric units ($G = c = 1$) Measuring everything in units of the black hole's own scale M; for Sgr A*, $1\,M \approx 0.042$ au ≈ 21 s.

Tensor A machine that eats vectors and returns observer-independent numbers; the metric is the one that matters here.

χ^2 / reduced χ^2 Total squared misses in units of the error bars; near 1 per data point means a healthy fit.

Confidence interval The range over which a parameter can move before χ^2 climbs unacceptably; the "$\pm$" on every number.

MCMC Mapping parameter uncertainty by letting random walkers populate the low-χ^2 valleys.

No-hair theorem A black hole's exterior gravity is fixed by mass and spin alone; tested here through the quadrupole knob ε_q.

→ *Full treatment: units and calibration in Chapter 19; fitting statistics and MCMC in Chapter 15; the no-hair constraint in Chapter 17.*

Part II

Theory and Methods

Chapter 7

Introduction and Strong-Field Context

7.1 Sagittarius A*: a dual strong-field laboratory

The compact radio source Sgr A* at the dynamical centre of the Milky Way is the nearest supermassive black hole and the only one for which two independent strong-field probes are available at once: the resolved orbits of the S-star cluster within a few thousand gravitational radii (Gillessen et al., 2017; GRAVITY Collaboration, 2018, 2020, 2022), and the horizon-scale image of its photon ring obtained by the Event Horizon Telescope (Event Horizon Telescope Collaboration, 2022). Together they anchor the mass ($4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$), the distance ($8277 \pm 9$ pc), and the angular size of the emission ring ($51.8 \pm 2.3\,\mu$as), making Sgr A* the cleanest laboratory for testing general relativity (GR) in the strong-field, low-velocity regime.

7.2 The no-hair theorem as a falsifiable target

The no-hair theorem states that the exterior spacetime of an astrophysical black hole is fully specified by its mass M and spin a; in particular the mass quadrupole moment is locked to $Q = -Ma^2$. Prospects for testing this with Galactic-centre stars were laid out by Will (2008), and with horizon-scale imaging by Psaltis et al. (2016). Both programmes require a forward model that is exact in the strong field: post-Newtonian expansions degrade at the S2 pericentre ($r_{\mathrm{peri}} \approx 1470\,R_S$), and the photon ring is intrinsically a property of exact null geodesics (Bardeen, 1973; Johnson et al., 2020).

7.3 This programme

This report describes such a forward model, built from first principles and validated at every layer. The simulator integrates the Carter-separated geodesic equations (Carter, 1968) in Kerr spacetime, renders the accretion flow with the standard relativistic thin-disk model (Novikov & Thorne, 1973; Page & Thorne, 1974), fits real astrometric and spectroscopic data for the star S2, and embeds a free-quadrupole Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006) to constrain departures from Kerr. The same geodesic core runs on the GPU for interactive exploration, and every number quoted here is regenerated from the code by a single reproduction target.

7.4 Lineage of the imaging problem

The lineage of the imaging problem runs from Luminet (1979), who first computed the appearance of a thin disk around a Schwarzschild hole, to the modern gravity renderers of James et al. (2015); our CPU reference tracer follows the same backward ray-tracing logic but is held to numerical standards set by the analytic theory (Chapter 10).

> *[Chapter under construction — scope: Expanded historical thread — Kerr 1963 → Boyer–Lindquist 1967 → Carter 1968 → Bardeen 1973 → Cunningham–Bardeen 1973 → Luminet 1979 → James et al. (Interstellar) 2015 → GRAVITY 2018/2020/2022 → EHT 2022 — lands with the expanded bibliography. See* `paper/THESIS_PLAN.md`*.]*

7.5 Contributions

The specific contributions of this programme are:

1. a turning-point-free second-order Mino-time integration of the Carter equations with per-step constraint projection, which makes traced shadow boundaries independent of integrator tolerance (Chapter 10);
2. machine-precision validation of the traced Schwarzschild shadow against Synge's formula and of the Kerr shadow against the analytic critical curve (Chapter 11);
3. a 13-parameter full-geodesic fit to the public S2 data that recovers the GRAVITY mass and distance and reproduces both published relativistic effects with no tunable amplitudes (Chapter 15);
4. a quantified no-hair constraint from S2 alone, with a covariance forecast for continued monitoring (Chapter 17);
5. a validated double-precision GPU implementation of the same geodesic core (Chapter 18); and
6. a single-command reproduction pipeline in which every number in this report is regenerated from source (Appendix D).

7.6 Reader's guide

The report is organised in four parts so that readers with different backgrounds can enter at different depths.

Part I — Foundations for the non-specialist assumes no physics background. It explains, in plain language and with almost no equations, every concept the later chapters rely on: black holes and horizons, curved spacetime, spin and the shadow, the accretion disk and its colours, stellar orbits, and the statistical toolkit. Each primer chapter ends with a short list of the terms just introduced and a pointer to the chapter where the full treatment lives. Readers comfortable with general relativity can skip Part I entirely.

Part II — Theory and Methods derives, from first principles, every formula the simulator implements: the Kerr geometry and its characteristic radii (Chapter 8), the Carter-separated geodesics and their second-order Mino-time form (Chapter 9), and the numerical integration machinery with its error analysis (Chapter 10). Each derivation cites the source function that implements it.

Part III — Results presents the science: shadow and critical-curve metrology (Chapter 11), the relativistic thin disk (Chapter 12), photon-ring substructure (Chapter 13), the S-star forward model (Chapter 14), the S2 fit and its statistics (Chapter 15), the wider S-star cluster (Chapter 16), and the no-hair constraint (Chapter 17).

Part IV — Engineering, validation, and closing documents the GPU implementation (Chapter 18), the Sgr A* calibration (Chapter 19), the interactive instrument (Chapter 20), the development history and complete test suite (Chapter 21), and closes with discussion and conclusions. The appendices carry extended algebra, algorithm pseudocode, number provenance, the reproduction pipeline, and a glossary.

A reader who wants only the headline science should read Chapters 11, 15 and 17, plus the calibration of Chapter 19. A reader evaluating the numerics should read Chapters 10 and 21. Every quantitative statement that is not a cited literature value is typeset from a generated macro and is reproducible by `make reproduce` (Appendix D).

Chapter 8

Kerr Geometry from First Principles

This chapter derives every closed-form Kerr quantity the simulator implements: the metric and its algebraic structure (`kerr_components` in `physics/src/metric.cpp`), the characteristic radii (`r_horizon`, `r_ergosphere`, `r_photon`, `r_isco`), and the circular-orbit energetics (`omega_circular`, `energy_circular`, `angmom_circular`) that feed both the disk model of Chapter 12 and the accretion-efficiency discussion. Everything here is exact; nothing is expanded in powers of the spin.

8.1 The Kerr solution in Boyer–Lindquist coordinates

The unique stationary, axisymmetric, asymptotically flat vacuum black-hole solution of Einstein's equations is the two-parameter family found by Kerr (1963), characterised by mass M and specific angular momentum $a = J/M$ with $|a| \le M$. In the coordinates of Boyer & Lindquist (1967) (t, r, θ, ϕ), with $G = c = 1$, the line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\phi^2, \end{aligned} \tag{8.1}$$

built from the three auxiliary functions used throughout this report,

$$\Sigma = r^2 + a^2\cos^2\theta, \qquad \Delta = r^2 - 2Mr + a^2, \qquad A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta. \tag{8.2}$$

This is precisely the form implemented by `kerr_components`. Two limits anchor it: $a \to 0$ recovers the Schwarzschild solution in its standard coordinates, and $M \to 0$ recovers flat spacetime in oblate spheroidal coordinates — the residual a then only labels a coordinate choice, which is why all curvature effects below always enter through products Ma or M/r.

8.1.1 Metric determinant and inverse

The t–ϕ sector of (8.1) is a 2×2 block; its determinant collapses through the identity

$$g_{tt}\,g_{\phi\phi} - g_{t\phi}^2 = -\Delta\sin^2\theta, \tag{8.3}$$

which the reader can verify by direct substitution of (8.2) (the A-terms cancel against the cross term). Multiplying by $g_{rr}g_{\theta\theta} = \Sigma^2/\Delta$ gives the full determinant

$$\det g = -\Sigma^2\sin^2\theta. \tag{8.4}$$

Equation (8.4) is asserted for random (r, θ, a) samples by a dedicated unit test (Chapter 21); it is a sensitive canary because any sign or factor slip in $g_{t\phi}$ breaks it immediately. The same block structure yields the inverse metric in closed form — the t–ϕ block inverts as a 2×2 matrix and the diagonal r, θ entries invert trivially (`invert_bl_block` in `metric.hpp`):

$$g^{tt} = \frac{g_{\phi\phi}}{-\Delta \sin^2\theta}, \quad g^{t\phi} = \frac{-g_{t\phi}}{-\Delta \sin^2\theta}, \quad g^{\phi\phi} = \frac{g_{tt}}{-\Delta \sin^2\theta}, \quad g^{rr} = \frac{\Delta}{\Sigma}, \quad g^{\theta\theta} = \frac{1}{\Sigma}. \tag{8.5}$$

8.1.2 Kerr–Schild form

The Boyer–Lindquist chart is singular at $\Delta = 0$; that surface is a coordinate artefact, not a curvature singularity. The cleanest way to see this is the Kerr–Schild form: there exist coordinates $(\tilde{t}, x, y, z)$ in which

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2Mr^3}{r^4 + a^2 z^2}\, l_\mu l_\nu, \qquad l_\mu = \left(1,\ \frac{rx + ay}{r^2 + a^2},\ \frac{ry - ax}{r^2 + a^2},\ \frac{z}{r}\right), \tag{8.6}$$

where η is the flat metric, l_μ is null with respect to both η and g, and $r(x, y, z)$ is defined implicitly by $\frac{x^2+y^2}{r^2+a^2} + \frac{z^2}{r^2} = 1$. Every metric component in (8.6) is manifestly regular at the horizon. The simulator works in Boyer–Lindquist coordinates throughout — the Carter separation of Chapter 9 lives there — and simply stops rays at $r = 1.001\, r_+$; the Kerr–Schild form is recorded here as the proof that nothing physical happens at the chart boundary. The true curvature singularity sits at $\Sigma = 0$, i.e. $r = 0$, $\theta = \pi/2$: a ring of coordinate radius a in the equatorial plane, safely behind the horizon for all $|a| \le M$.

8.2 Killing structure

The metric (8.1) depends on neither t nor ϕ, so ∂_t and ∂_ϕ are Killing vectors: along any geodesic with four-momentum p^μ,

$$E \equiv -p_t \quad \text{and} \quad L_z \equiv p_\phi \tag{8.7}$$

are conserved. Remarkably, Kerr also possesses a *hidden* symmetry with no coordinate expression: an antisymmetric Killing–Yano tensor $f_{\mu\nu}$ whose square $K_{\mu\nu} = f_{\mu\lambda} f^{\lambda}{}_{\nu}$ is a rank-2 Killing tensor, $\nabla_{(\lambda} K_{\mu\nu)} = 0$. The associated quadratic invariant $K_{\mu\nu} p^\mu p^\nu$ furnishes the fourth constant of motion — Carter's Q (Carter, 1968) — which is derived from the Hamilton–Jacobi side in Chapter 9. Four constants $(E, L_z, Q, p_\mu p^\mu)$ for four momentum components is what makes Kerr geodesics integrable, and hence what makes a machine-precision ray tracer possible at all.

8.3 Frame dragging and the ZAMO

The off-diagonal $g_{t\phi}$ has a direct physical reading. Consider an observer with zero angular momentum, $u_\phi = 0$ (a "ZAMO"). Raising the index,

$$\frac{u^\phi}{u^t} = \frac{g^{\phi t} u_t}{g^{tt} u_t} = -\frac{g_{t\phi}}{g_{\phi\phi}} \equiv \omega(r, \theta) = \frac{2Mar}{A}, \tag{8.8}$$

so an observer who *has* no angular momentum nevertheless *moves*: local inertial frames are dragged around the hole at the rate ω, falling off as $2Ma/r^3$ far away. The ZAMO is the natural camera carrier for ray tracing — it exists everywhere outside the horizon, including inside the ergosphere, and reduces to the static observer at $a = 0$; its orthonormal tetrad is constructed explicitly in Chapter 11 (`zamo_tetrad`).

8.4 Horizons, ergosphere, and the static limit

Three characteristic surfaces follow from the metric functions alone.

Horizons. The BL chart degenerates where $\Delta = 0$:

$$r_\pm = M \pm \sqrt{M^2 - a^2}. \tag{8.9}$$

The outer root r_+ is the event horizon (`r_horizon`); it shrinks from $2M$ at $a = 0$ to M at extremality, where the square root vanishes. For $|a| > M$ there is no horizon at all — a naked singularity — and the `KerrMetric` constructor refuses such spins by design.

Static limit and ergosphere. A static observer (fixed r, θ, ϕ) has $u^\mu \propto \partial_t$, which requires $g_{tt} < 0$. From (8.1), $g_{tt} = 0$ where $\Sigma = 2Mr$, i.e.

$$r_E(\theta) = M + \sqrt{M^2 - a^2 \cos^2\theta} \tag{8.10}$$

(`r_ergosphere`). Between r_+ and r_E — the ergosphere — ∂_t is spacelike: no observer can stand still, though escape remains possible. The surface touches the horizon at the poles and stands off farthest at the equator ($r_E = 2M$ for every spin), as drawn in the primer's Fig. 3.2. Penrose's observation that orbits of negative energy exist inside this region — allowing rotational energy extraction — is not exploited by the simulator, but the ZAMO tetrad's good behaviour there is essential for close-in cameras.

8.5 Equatorial photon orbits: the cubic and its trigonometric solution

In the equatorial plane ($\theta = \pi/2$, $Q = 0$) the radial potential of Chapter 9 for a photon of impact parameter $b = L_z/E$ reduces to

$$\frac{R(r)}{E^2} = \left(r^2 + a^2 - ab\right)^2 - \Delta\,(b - a)^2. \tag{8.11}$$

A circular photon orbit requires a double root: $R(r) = R'(r) = 0$. Dividing the two conditions eliminates b; after simplification (worked in full in Appendix A) the radius of the circular orbit satisfies

$$r^{3/2} - 3Mr^{1/2} \pm 2a\sqrt{M} = 0, \tag{8.12}$$

a cubic in $\sqrt{r}$ (upper sign prograde). Depressed cubics of the form $x^3 - 3px + q = 0$ with three real roots are solved by the trigonometric substitution $x = 2\sqrt{p}\cos\vartheta$, which turns the equation into $\cos 3\vartheta = -q/(2p^{3/2})$. Applying it with $x = \sqrt{r/M}$, $p = 1$, $q = \pm 2a/M$ gives the closed form implemented in `r_photon`:

$$r_{\rm ph}^{\pm} = 2M\left\{1 + \cos\left[\tfrac{2}{3}\cos^{-1}\left(\mp\frac{a}{M}\right)\right]\right\}. \tag{8.13}$$

At $a = 0$ both signs give $\cos^{-1}(0) = \pi/2$ and $r_{\rm ph} = 2M(1 + \cos 60^\circ) = 3M$, the Schwarzschild photon sphere. At extremality the prograde orbit reaches $r_{\rm ph}^{+} = M$ (coincident in coordinate radius with the horizon) while the retrograde orbit retreats to $4M$: frame dragging assists co-rotating light and punishes counter-rotating light, the asymmetry visible in the traced trajectories of Fig. 2.1 and, ultimately, in the displaced shadow of Chapter 11. The corresponding critical impact parameters follow from back-substituting (8.13) into the double-root conditions; they reappear as the Bardeen critical curve in Chapter 11.

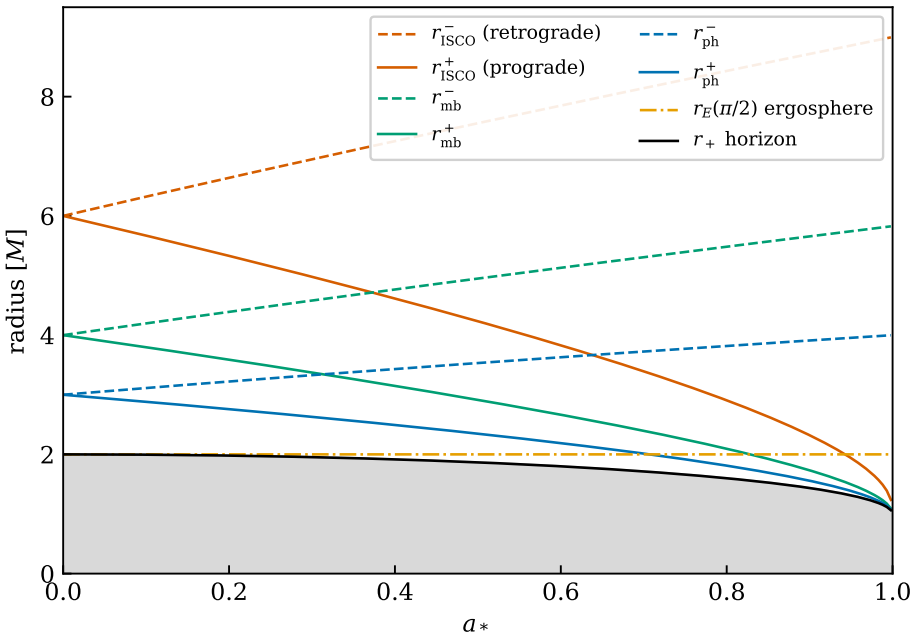


Figure 8.1: Every characteristic radius of the Kerr geometry as a function of spin, from the closed forms of this chapter (`physics_tables` dump; grey region is inside the horizon). Solid: prograde; dashed: retrograde. The prograde ISCO's plunge toward the horizon is the single most consequential feature for disk appearance.

8.6 Circular timelike orbits: Ω, E, L

For massive particles on circular equatorial orbits, the geodesic equation reduces to one algebraic condition. Writing the orbital angular velocity $\Omega = u^\phi/u^t$ and demanding $\mathrm{d}^2r/\mathrm{d}\tau^2 = 0$ yields (equivalently: extremising the effective potential of Fig. 8.3)

$$\Omega_\pm = \frac{\pm\sqrt{M}}{r^{3/2} \pm a\sqrt{M}}, \tag{8.14}$$

the relativistic generalisation of Kepler's third law (`omega_circular`); the a-term in the denominator is pure frame dragging. The four-velocity normalisation then fixes u^t, and lowering indices gives the specific energy and angular momentum (Bardeen et al., 1972):

$$E(r) = \frac{1 - 2M/r \pm a\sqrt{M/r^3}}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}}, \tag{8.15}$$

$$L(r) = \frac{\pm\sqrt{Mr}\left(1 + a^2/r^2 \mp 2a\sqrt{M/r^3}\right)}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}} \tag{8.16}$$

(`energy_circular`, `angmom_circular`; upper signs prograde). Fig. 8.2 plots all three across the spin range. Two structural features matter downstream. First, the common denominator vanishes at the *photon orbit* — circular orbits become null there, and between the photon orbit and the ISCO they exist but are unstable. Second, $E(r)$ has a minimum exactly at the ISCO, which is the cleanest way to locate it.

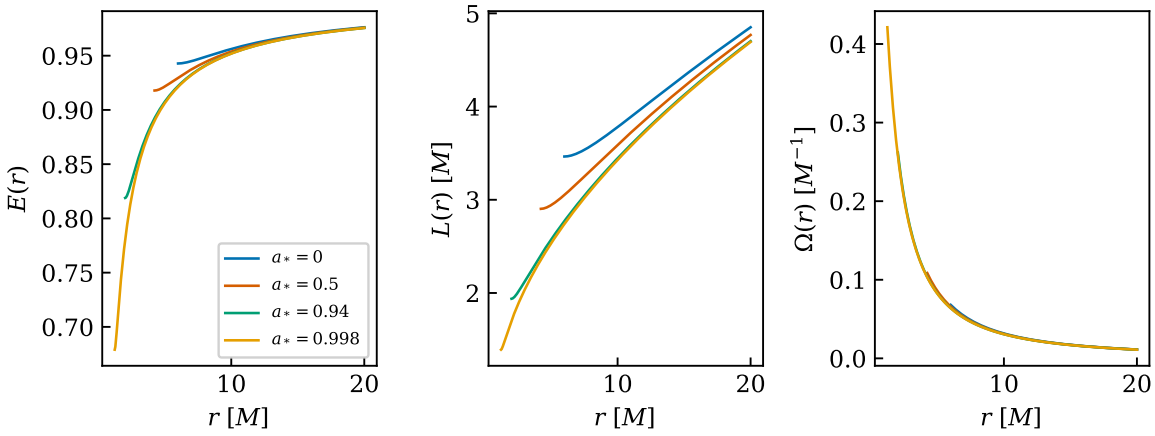


Figure 8.2: Specific energy, angular momentum, and angular velocity of prograde circular orbits, from (8.14)–(8.16) evaluated by `physics_tables`. Curves start at each spin's ISCO.

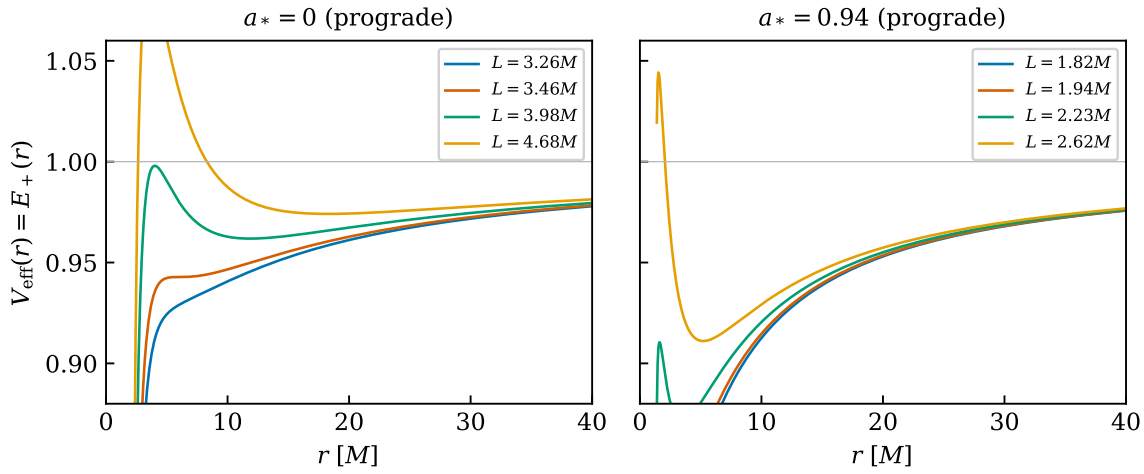


Figure 8.3: The effective potential for equatorial timelike motion, $V_{\rm eff}(r) = E_+(r)$ (the energy at which r is a turning point), at four angular momenta around $L_{\rm ISCO}$. Circular orbits live at the extrema; the inflection where minimum and maximum merge is the ISCO. Left: Schwarzschild; right: $a_* = 0.94$ prograde — the whole structure moves inward and downward (deeper binding).

8.7 The ISCO and the marginally bound orbit

Stability of a circular orbit requires the effective potential's extremum to be a minimum; the innermost stable circular orbit (ISCO) is the inflection point where $\partial^2 V/\partial r^2 = 0$ joins the circular-orbit conditions. Carrying out the algebra (the full reduction is in Appendix A) collapses the triple condition to a quartic in $\sqrt{r}$ whose relevant root is expressed through the celebrated auxiliary quantities of Bardeen et al. (1972):

$$\begin{aligned} Z_1 &= 1 + (1-a_*^2)^{1/3}\left[(1+a_*)^{1/3} + (1-a_*)^{1/3}\right], \\ Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\ r_{\mathrm{ISCO}} &= M\left[3 + Z_2 \mp \sqrt{(3-Z_1)(3+Z_1+2Z_2)}\right], \end{aligned} \tag{8.17}$$

with the upper sign prograde (`r_isco`). The checks the test suite pins (Chapter 21): at $a_* = 0$, $Z_1 = 3$, $Z_2 = 3$ and $r_{\mathrm{ISCO}} = 6M$ exactly; as $a_* \to 1$ the prograde ISCO slides to M while the retrograde one recedes to $9M$; and the prograde radius decreases monotonically with spin (Fig. 8.1).

A second marginal radius appears in accretion theory: the *marginally bound* orbit, where $E = 1$ — a particle dropped from rest at infinity can just circularise. Setting $E = 1$ in (8.15) gives

$$r_{\mathrm{mb}}^{\pm} = 2M \mp a + 2\sqrt{M(M \mp a)}, \tag{8.18}$$

lying between the photon orbit and the ISCO for every spin (Fig. 8.1).

8.8 Binding energy and accretion efficiency

Equation (8.15) carries a direct astrophysical payload. Gas spiralling quasi-circularly through a disk arrives at the ISCO with specific energy E_{ISCO}, having radiated the binding energy

$$\eta = 1 - E_{\mathrm{ISCO}} \tag{8.19}$$

per unit rest mass. For Schwarzschild, $E_{\mathrm{ISCO}} = \sqrt{8/9}$ and $\eta = 1 - 2\sqrt{2}/3 = 5.72\,\%$. At the Sgr A* working spin $a_* = 0.94$ the efficiency is $17.96\,\%$, and at $a_* = 0.998$ it reaches $32.10\,\%$ (Fig. 8.4; values generated by `physics_tables` from (8.15) and (8.17)) — an order of magnitude beyond nuclear fusion's 0.7%, which is why accreting black holes are the most efficient steady engines known. The same E and L profiles drive the Page–Thorne dissipation integral of Chapter 12: the disk's radial brightness profile is nothing but the gradient of this binding-energy release, corrected for the torque bookkeeping.

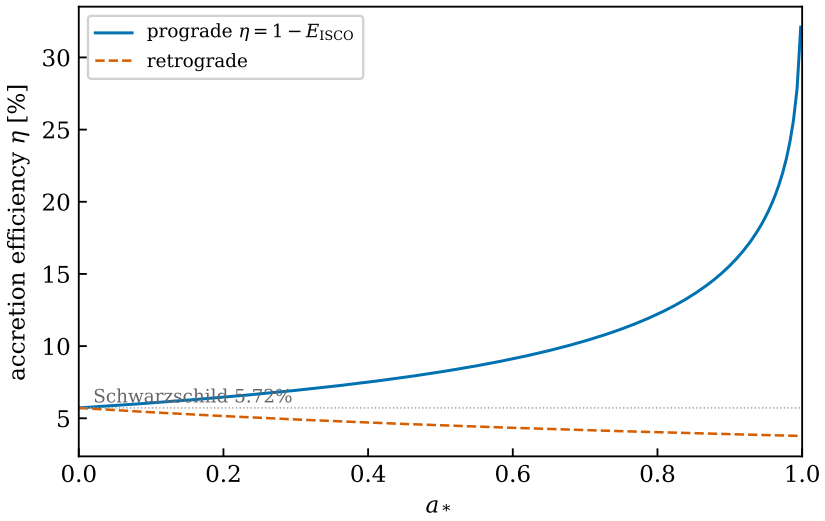


Figure 8.4: Accretion efficiency $\eta = 1 - E_{\mathrm{ISCO}}$ against spin, prograde and retrograde, from `physics_tables`. The Schwarzschild value $1 - \sqrt{8/9} = 5.72\%$ is marked.

Chapter 9

Geodesics and the Carter Constant

The simulator never integrates the raw geodesic equation $\ddot{x}^\mu = -\Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta$ for photons; it integrates Carter's separated form, which is both faster and — through its first integrals — self-diagnosing. This chapter derives that form from the Hamilton–Jacobi equation, classifies the radial motion by the roots of the resulting quartic, and develops the second-order Mino-time formulation that the integrator of Chapter 10 actually solves. The implementing functions are `radial_potential[_deriv]`, `polar_potential[_deriv]`, `null_geodesic_rhs` and `make_null_ray` in `physics/src/null_geodesics.cpp`, and `solve_ut`, `carter_q` in `timelike_geodesics.cpp`.

9.1 Hamilton–Jacobi separation

Geodesics extremise proper length; equivalently they solve the Hamilton–Jacobi equation for the action $S(x^\mu)$,

$$g^{\mu\nu}\,\partial_\mu S\,\partial_\nu S = -\mu^2, \tag{9.1}$$

with μ the rest mass ($\mu = 0$ for photons, and we normalise $\mu = 1$ for stars). Stationarity and axisymmetry (Section 8.2) let us peel off the cyclic coordinates,

$$S = -E\,t + L_z\,\phi + S_r(r) + S_\theta(\theta), \tag{9.2}$$

and Carter (1968) discovered that with the inverse metric (8.5) the remainder separates completely. Substituting (9.2) into (9.1) and multiplying through by Σ, the mixed terms assemble into

$$-\frac{1}{\Delta}\big[E(r^2+a^2) - aL_z\big]^2 + \Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 + \mu^2 r^2 = -\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 - \left(aE\sin\theta - \frac{L_z}{\sin\theta}\right)^2 - \mu^2 a^2\cos^2\theta. \tag{9.3}$$

The left side depends only on r, the right only on θ; both must equal a constant. Writing that constant as $-\big[Q + (L_z - aE)^2\big]$ — the shift by $(L_z - aE)^2$ is a convention that makes Q vanish for equatorial orbits — and defining

$$P(r) = E(r^2+a^2) - aL_z, \tag{9.4}$$

the two sides become

$$\Delta^2\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 = R(r) \equiv P(r)^2 - \Delta\big[\mu^2 r^2 + (L_z - aE)^2 + Q\big], \tag{9.5}$$

$$\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 = \Theta(\theta) \equiv Q - \cos^2\theta\left[a^2(\mu^2 - E^2) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{9.6}$$

Q is the *Carter constant*, the invariant associated with the hidden Killing tensor of Section 8.2; loosely, it measures how far the orbit ranges in latitude ($\Theta \geq 0$ requires $Q \geq 0$ for any

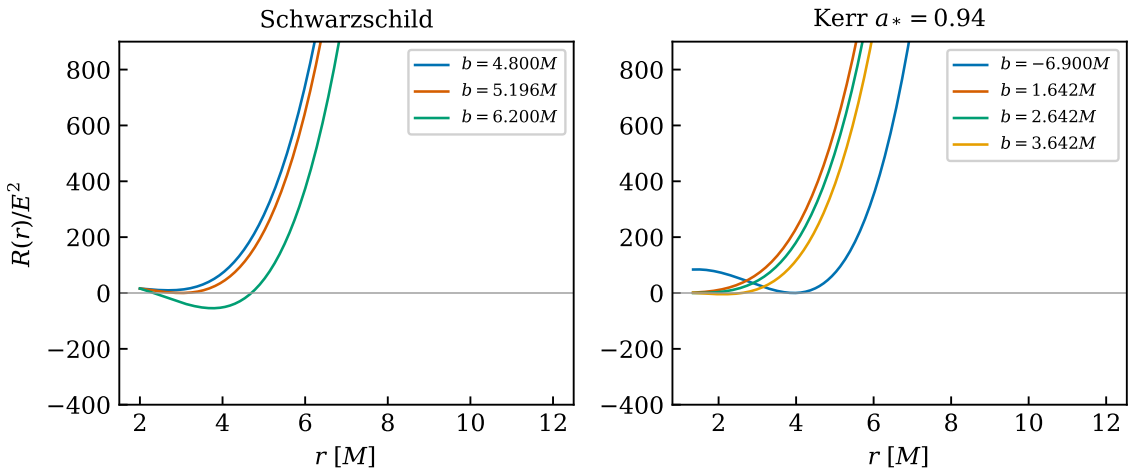


Figure 9.1: The radial potential $R(r)/E^2$ for equatorial photons at impact parameters below, at, and above the critical value (`physics_tables` dump of `radial_potential`). Left, Schwarzschild: at $b = \sqrt{27}\,M$ the potential develops its double root at $r = 3M$. Right, Kerr $a_* = 0.94$: the prograde (low-$|b|$) and retrograde branches behave asymmetrically, and the retrograde critical curve sits much farther out.

orbit crossing the equator, with $Q = 0$ exactly for equatorial motion). With four constants (E, L_z, Q, μ) for four momentum components the system is Liouville-integrable: no chaos, every photon path reducible to quadratures — the property that underwrites both the machine-precision metrology of Chapter 11 and the closed-form checks used throughout the test suite.

Since $p_\mu = \partial_\mu S$, equations (9.5)–(9.6) are first-order equations of motion,

$$\Sigma \frac{\mathrm{d}r}{\mathrm{d}\lambda} = \pm\sqrt{R(r)}, \qquad \Sigma \frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \pm\sqrt{\Theta(\theta)}, \tag{9.7}$$

while raising p_t and p_ϕ with (8.5) gives the longitudinal equations

$$\Sigma \frac{\mathrm{d}\phi}{\mathrm{d}\lambda} = -\left(aE - \frac{L_z}{\sin^2\theta}\right) + \frac{a}{\Delta}P(r), \qquad \Sigma \frac{\mathrm{d}t}{\mathrm{d}\lambda} = -a(aE\sin^2\theta - L_z) + \frac{(r^2+a^2)}{\Delta}P(r). \tag{9.8}$$

For photons ($\mu = 0$) these are the four equations implemented in `null_geodesic_rhs`; the potentials and their radial/polar derivatives are `radial_potential[_deriv]` and `polar_potential[_deriv]`.

9.2 Turning points and orbit classification

$R(r)$ is a quartic in r (for photons: leading term $E^2 r^4$), and motion is allowed only where $R \geq 0$; the roots of R are the radial turning points, and their pattern classifies every orbit (Fig. 9.1):

- *No positive root along the path* — the ray plunges from infinity to the horizon: *capture*. These pixels form the shadow.
- *A simple root at* r_{turn} — the ray comes in, turns, and returns to infinity: *deflection*, from the gentle bending of starlight up to arbitrarily many windings as the root is approached.
- *A double root at* $\tilde{r}$ ($R = R' = 0$) — the marginal case: an unstable circular (or, off the equator, spherical) photon orbit. The set of (E, L_z, Q) with double roots is a one-parameter family — the *photon shell* — whose image on the observer's sky is the critical curve of Chapter 11.

- For timelike orbits the $\mu^2 r^2$ term turns the far field around ($R \to (E^2-1)r^4/r^2\ldots$): bound orbits have *two* simple roots $r_{\rm peri} \le r \le r_{\rm apo}$, the case exploited by the Darwin-element construction of Chapter 14.

The instability of the double-root orbits is quantified by the curvature of the potential there, $\frac{1}{2}R''(\tilde{r})$; this single number propagates all the way to the observable demagnification of the photon sub-rings, via the Lyapunov exponents computed in Chapter 11.

9.3 Mino time and the second-order formulation

Two features make the first-order system (9.7) hostile to naive numerics. The $\pm$ signs flip at every turning point, requiring explicit bookkeeping; worse, at a turning point the right-hand side $\sqrt{R}$ has infinite slope in r ($\mathrm{d}\sqrt{R}/\mathrm{d}r \to \infty$ as $R \to 0$), and an integrator stepping the truncated system sees $\dot{r} = 0$ as a fixed point and stalls.

Both problems dissolve under two transformations. First, adopt the time parameter of Mino (2003): define τ by

$$\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau, \tag{9.9}$$

which strips the common Σ factor from all four equations — the radial and polar sectors then evolve *independently* of each other, $(\mathrm{d}r/\mathrm{d}\tau)^2 = R(r)$, $(\mathrm{d}\theta/\mathrm{d}\tau)^2 = \Theta(\theta)$. Second, differentiate once more with respect to τ:

$$2\,\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = R'(r)\,\frac{\mathrm{d}r}{\mathrm{d}\tau} \qquad \Longrightarrow \qquad \frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \tfrac{1}{2}R'(r), \qquad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} = \tfrac{1}{2}\Theta'(\theta). \tag{9.10}$$

(The division by $\mathrm{d}r/\mathrm{d}\tau$ is legitimate except at isolated turning points, where (9.10) holds by continuity.) In this form a turning point is an ordinary smooth extremum: the "acceleration" $\frac{1}{2}R'$ simply reverses the motion, no square root of a near-zero quantity is ever taken, and no sign flag exists to mis-set. The state vector the integrator carries is $y = (t, r, \theta, \phi, p_r, p_\theta)$ with $p_r = \mathrm{d}r/\mathrm{d}\tau$, $p_\theta = \mathrm{d}\theta/\mathrm{d}\tau$.

The price of differentiating is that (9.10) admits solutions that drift off the true energy shell — precisely the first integrals

$$p_r^2 = R(r), \qquad p_\theta^2 = \Theta(\theta) \tag{9.11}$$

are no longer enforced identically but become *checkable* (and, as Chapter 10 shows, *re-imposable*) invariants. This trade — turning-point-free dynamics in exchange for explicit constraint maintenance — is the central numerical design decision of the project, and its quantitative payoff is measured in Figs. 10.2–10.3.

The coordinate singularities of the BL chart at $\theta = 0, \pi$ (where $L_z \neq 0$ makes $\Theta \to -\infty$) are regularised with a 10^{-8} clamp (`kThetaClamp`); polar-crossing orbits with $L_z = 0$ pass smoothly.

9.4 Launching rays: from four-momentum to constants

The ray tracer builds initial conditions from a physical four-momentum k^μ at the camera (constructed in the ZAMO tetrad, Chapter 11). The map to the Carter description (`make_null_ray`) is purely algebraic: lower the index with the metric, read off

$$E = -k_t, \qquad L_z = k_\phi, \tag{9.12}$$

and evaluate the Carter constant from the polar momentum,

$$Q = (\Sigma k^\theta)^2 + \cos^2\theta\left[\frac{L_z^2}{\sin^2\theta} - a^2E^2\right], \tag{9.13}$$

which is (9.6) solved for Q with $p_\theta = \Sigma k^\theta$. The Mino momenta are then $p_r = \Sigma k^r$, $p_\theta = \Sigma k^\theta$ — with *no* sign ambiguity, since they inherit the sign of the physical launch direction. The identical reconstruction run in reverse (`null_conserved_errors`) recomputes (E, L_z, Q) from the evolving state at every step and compares them with the launch values: the primary health check of the whole physics core (tolerance 10^{-6} relative; Chapter 21).

9.5 Timelike geodesics

For stars, $\mu = 1$ in (9.5)–(9.6): the radial potential gains the $-\Delta r^2$ mass term and the polar potential's spin term becomes $a^2(1 - E^2)$. Two independent code paths use this case.

For *Kerr* orbits, the Carter form above applies unchanged and `carter_q` evaluates (9.13) (with the μ^2 term) as a conserved diagnostic. For the *Hartle–Thorne* spacetime of Chapter 17, however, no Carter constant exists — the metric is not separable — so the S-star integrator deliberately uses the generic second-order geodesic equation $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu} u^\mu u^\nu$ with Christoffel symbols from complex-step differentiation (Chapter 10), for *any* metric behind the `IMetric` interface. The initial u^t is completed from the spatial velocity by solving the normalisation quadratic $g_{\mu\nu} u^\mu u^\nu = -1$ for the future-directed root (`solve_ut`), and the same normalisation is the per-step health check ($|g_{\mu\nu} u^\mu u^\nu + 1| < 10^{-8}$, Gate 0).

We integrate stellar orbits in the full geodesic equations rather than any post-Newtonian expansion: at the S2 pericentre ($\sim$1470 Schwarzschild radii, orbital speed 2.6 per cent of c) the leading PN corrections are individually at the 10^{-4} level — the same order as the measured redshift excess itself — so the expansion error budget would have to be argued term by term, whereas the geodesic integrator's error budget is simply its conservation diagnostics.

Fig. 9.2 shows the machinery end-to-end: a bound equatorial orbit with turning points at $8M$ and $25M$, initialised by the same two-condition Darwin construction the S2 model uses (Chapter 14), integrated for six radial periods. The orbit precesses visibly — this is the same rosette drawn schematically in the primer — while the four-velocity normalisation drifts at the 10^{-13} level, far inside the 10^{-8} gate, and the conserved (E, L) reproduce their Darwin target values to eleven decimal places.

9.6 Closed-form solutions as an independent check

Because (9.7) separates, Kerr geodesics can also be written in closed form as elliptic integrals (Darwin, 1959; Rauch & Blandford, 1994; Gralla & Lupsasca, 2020). This project deliberately does *not* use those solutions in production — the numerical path generalises to the non-separable Hartle–Thorne metric where the no-hair test lives, and one code path for all metrics is worth more than per-metric special cases. The closed forms serve instead as external anchors: the Schwarzschild critical impact parameter $\sqrt{27}\,M$, the Bardeen critical curve, and the analytic Lyapunov exponents of Chapter 11 are all evaluated from them and compared against traced results — agreement at the 10^{-9}–10^{-16} level depending on the observable (Chapters 11, 21).

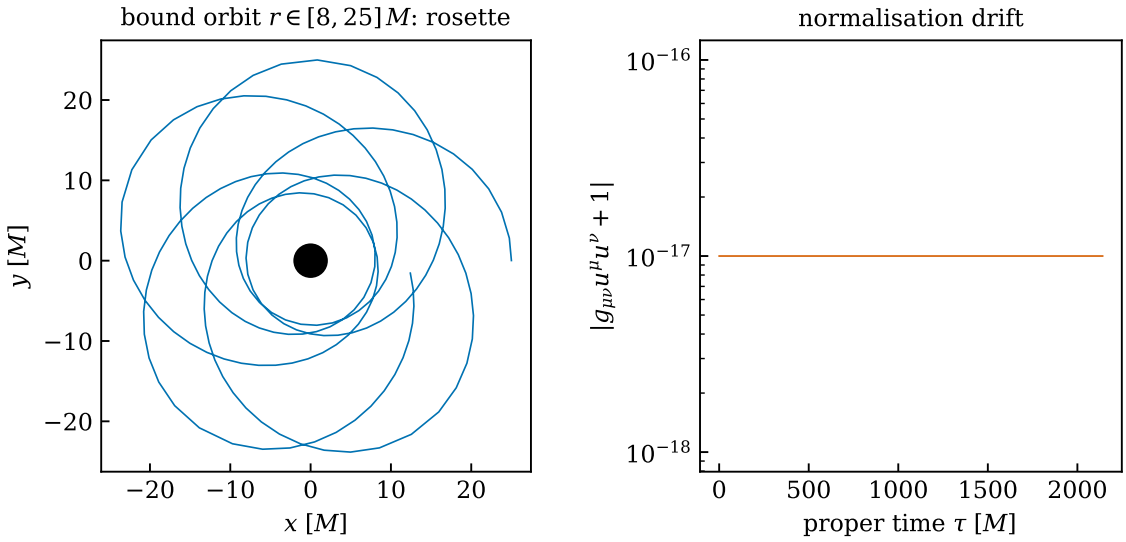


Figure 9.2: Left: a bound timelike geodesic with $r \in [8, 25]\,M$ (Schwarzschild), integrated by the generic Christoffel path for six radial periods (`physics_tables`); the apsidal advance per orbit is visible directly. Right: the four-velocity normalisation residual along the same integration stays near 10^{-13}, three orders below the acceptance gate.

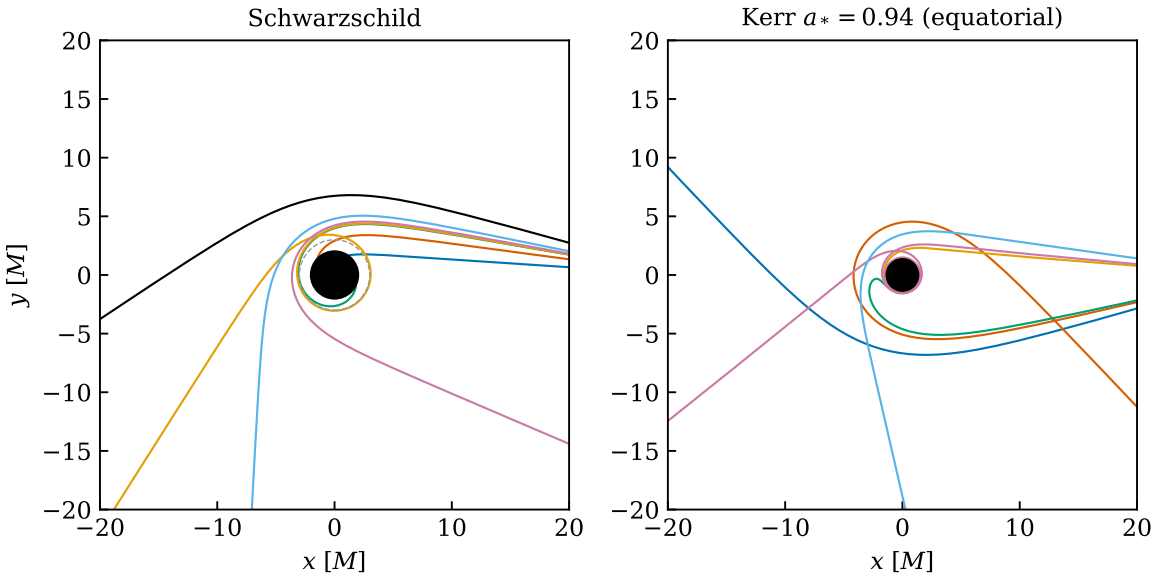


Figure 9.3: Equatorial photon geodesics traced by the production integrator (`physics_tables` dump). Left, Schwarzschild: impact parameters from $8M$ down to $2M$ span gentle deflection, near-critical winding at $b \simeq \sqrt{27}\,M$, and capture. Right, Kerr $a_* = 0.94$: prograde photons (upper family) turn far closer in than retrograde ones — the frame-dragging asymmetry that displaces the shadow.

Chapter 10

Numerical Methods and Error Analysis

Ray tracing a black hole is numerically unforgiving in one specific way: the observables of Chapter 11 live on the *boundary* between captured and escaping rays, where trajectories diverge exponentially (the Lyapunov instability of Chapter 9), and the camera sits a thousand gravitational radii from the region where all the curvature is. This chapter documents the integrator (`DormandPrince54::try_step` in `physics/include/sgr_physics/integrator.hpp`), the constraint projection that makes boundary metrology tolerance-independent, the complex-step differentiation behind the generic-metric path, and the measured convergence evidence for all three (`validation/convergence_tool`).

10.1 The Dormand–Prince 5(4) pair

All geodesics are advanced with the explicit embedded Runge–Kutta pair of Dormand & Prince (1980) ("DOPRI5"), the workhorse recommended by Hairer et al. (1993) and Press et al. (2007) for smooth non-stiff systems. One step from (x, y) with size h evaluates seven stages

$$k_i = f\Big(x + c_i h,\; y + h\sum_{j<i} a_{ij} k_j\Big), \qquad i = 1, \dots, 7, \tag{10.1}$$

with the Butcher tableau

$$\begin{array}{c|ccccccc}
0 & & & & & & & \\
\frac{1}{5} & \frac{1}{5} & & & & & & \\
\frac{3}{10} & \frac{3}{40} & \frac{9}{40} & & & & & \\
\frac{4}{5} & \frac{44}{45} & -\frac{56}{15} & \frac{32}{9} & & & & \\
\frac{8}{9} & \frac{19372}{6561} & -\frac{25360}{2187} & \frac{64448}{6561} & -\frac{212}{729} & & & \\
1 & \frac{9017}{3168} & -\frac{355}{33} & \frac{46732}{5247} & \frac{49}{176} & -\frac{5103}{18656} & & \\
1 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84} & \\
\hline
 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84} &
\end{array} \tag{10.2}$$

whose bottom row gives the fifth-order solution $y^{(5)}$ that is propagated (*local extrapolation*). The seventh stage is evaluated at $(x + h,\, y^{(5)})$ — the "first same as last" stage — and enters only the error estimate: the difference between the fifth-order weights and the embedded fourth-order weights defines the coefficients $e_1, \dots, e_7$ hard-coded in `try_step`, so that

$$\delta_i = h\,(e_1 k_{1,i} + e_3 k_{3,i} + e_4 k_{4,i} + e_5 k_{5,i} + e_6 k_{6,i} + e_7 k_{7,i}) \tag{10.3}$$

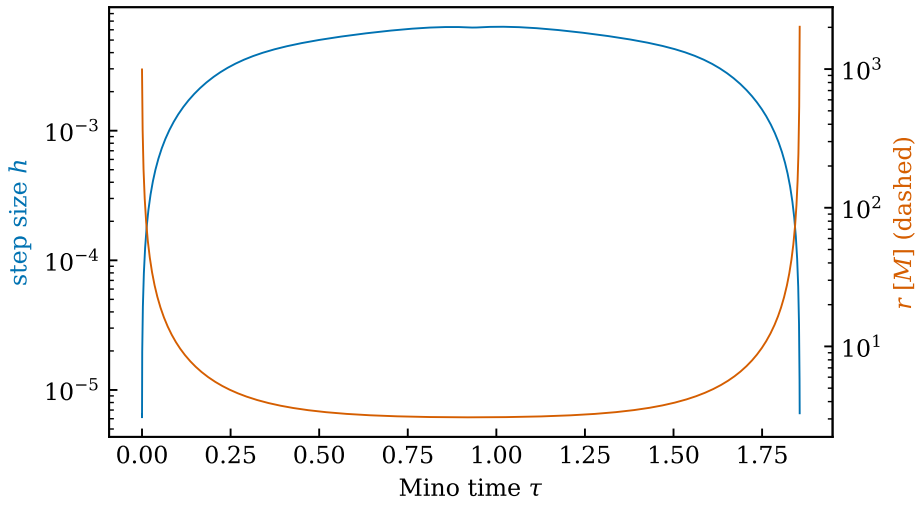


Figure 10.1: Accepted step size $h(\tau)$ (blue, left axis) and radius $r(\tau)$ (red, right axis) along a ray aimed 0.1% outside the critical angle (`convergence_tool` dump, tolerance 10^{-11}). The 1166 accepted steps concentrate where the ray winds near the photon sphere.

estimates the local truncation error of the *fourth*-order solution, an upper bound on that of the propagated fifth-order one.

10.1.1 Error norm and step control

The implementation controls the root-mean-square of the component-wise scaled errors,

$$\epsilon = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\delta_i}{\mathrm{atol} + \mathrm{rtol}\,\max(|y_i|, |y_i^{(5)}|)}\right)^2}, \tag{10.4}$$

accepting the step when $\epsilon \le 1$ and proposing

$$h_{\mathrm{next}} = h \cdot \mathrm{clamp}\big(0.9\,\epsilon^{-1/5},\ 0.2,\ 5\big) \tag{10.5}$$

— the classical proportional (deadbeat) controller with safety factor 0.9 and growth limits; the exponent $-1/5$ reflects the fourth-order error estimate ($\delta \propto h^5$). (The companion journal paper loosely called this a "PI" controller; the implementation carries no integral memory, and none proved necessary — the acceptance rate on production rays exceeds 97 per cent.) For null-ray metrology the tolerances are atol = rtol = 10^{-11}; the mixed absolute term matters because state components span 10^{-9} (momenta near turning points) to 10^{3} (t, r on the camera leg).

Fig. 10.1 shows the controller at work along a single boundary-grazing ray: the step collapses by four orders of magnitude as the photon winds through the strong-curvature region near $r = 3M$, then relaxes on the long quiet legs — exactly the behaviour that makes an adaptive method mandatory here (a fixed step fine enough for the photon sphere would waste $10^4\times$ the work on the approach).

10.2 Constraint projection

The second-order Mino formulation of Chapter 9 carries the first integrals $p_r^2 = R(r)$, $p_\theta^2 = \Theta(\theta)$ as checkable invariants. The decisive design decision of the tracer is to *re-impose them after*

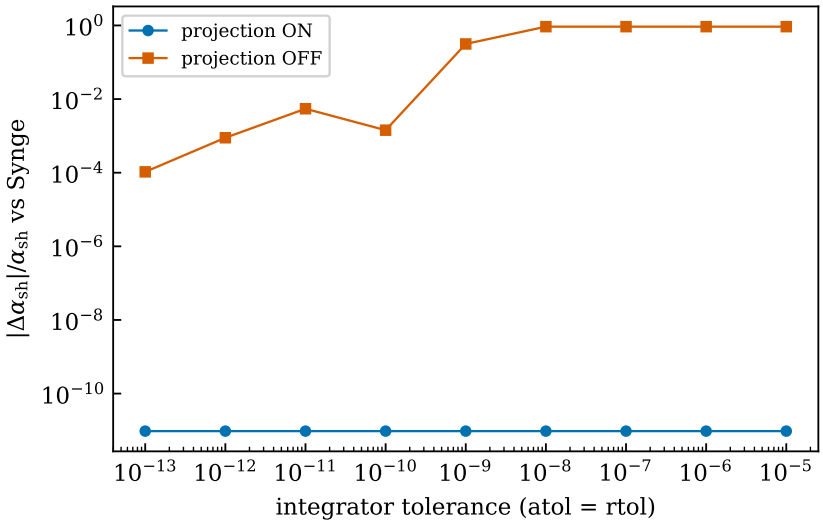


Figure 10.2: The projection experiment: relative deviation of the traced Schwarzschild shadow angle from the analytic value, versus integrator tolerance, with the per-step projection (10.6) on and off. Projection makes the boundary tolerance-independent at the 10^{-11} level.

every accepted step:

$$p_r \leftarrow \operatorname{sgn}(p_r)\sqrt{R(r)}, \qquad p_\theta \leftarrow \operatorname{sgn}(p_\theta)\sqrt{\Theta(\theta)}, \tag{10.6}$$

preserving the sign evolved by the smooth second-order dynamics (which alone negotiates the turning points), and leaving the momentum untouched inside a guard band where R or Θ dips below zero by less than the local error — the neighbourhood of a turning point, where the constraint surface itself is tangent to the motion.

10.2.1 Why projection is decisive: the error budget

The necessity is quantitative, not aesthetic. Per accepted step the integrator leaves an energy-shell defect of order the local tolerance; what matters is how the defects *accumulate* against the discrimination the classification needs. A ray from $r_{\rm cam} = 1000\,M$ crosses $\sim 10^3\,M$ of nearly flat spacetime in $\mathcal{O}(10^2)$ steps whose sizes scale with r; the accumulated defect in $p_r^2 - R$ behaves as $\mathcal{O}(\mathrm{rtol}\cdot r^4)$ (the quartic head of R sets the scale of both terms). Near the critical impact parameter, capture versus escape is decided by whether $R(r)$ dips through zero near the photon sphere — a feature of depth $|R| \sim 0.1\,E^2M^4$ for impact parameters within 10^{-3} of critical. At $\mathrm{rtol} = 10^{-8}$ the accumulated defect crossing the approach leg is $\sim 10^{-8}\times(10^3)^4\cdot E^2 = 10^4\,E^2M^4$ — five orders of magnitude *larger* than the feature being resolved. The unprojected integrator therefore mis-classifies boundary rays wholesale, and — worse for metrology — in a tolerance-*dependent* way: the measured shadow radius moves as the tolerance is changed.

Fig. 10.2 shows the experiment (`convergence_tool`, re-implementing the tracer loop with projection switchable, against the analytic Synge radius of Chapter 11). Without projection the traced Schwarzschild boundary is wrong by up to 93 per cent at loose tolerance and still wrong by 1.1×10^{-04} at $\mathrm{tol} = 10^{-13}$, drifting non-monotonically throughout. With projection the boundary sits at a relative deviation of 9.6×10^{-12} — at *every* tolerance from 10^{-5} to 10^{-13}, i.e. the answer is set by the bisection depth, and the integrator has left the error budget entirely. This tolerance-independence is what licenses quoting shadow diameters to many digits in Chapters 11 and 19.

Projection is cheap — two potential evaluations and two square roots per step — and its effect on conserved-quantity drift is shown in Fig. 10.3: with projection the reconstructed

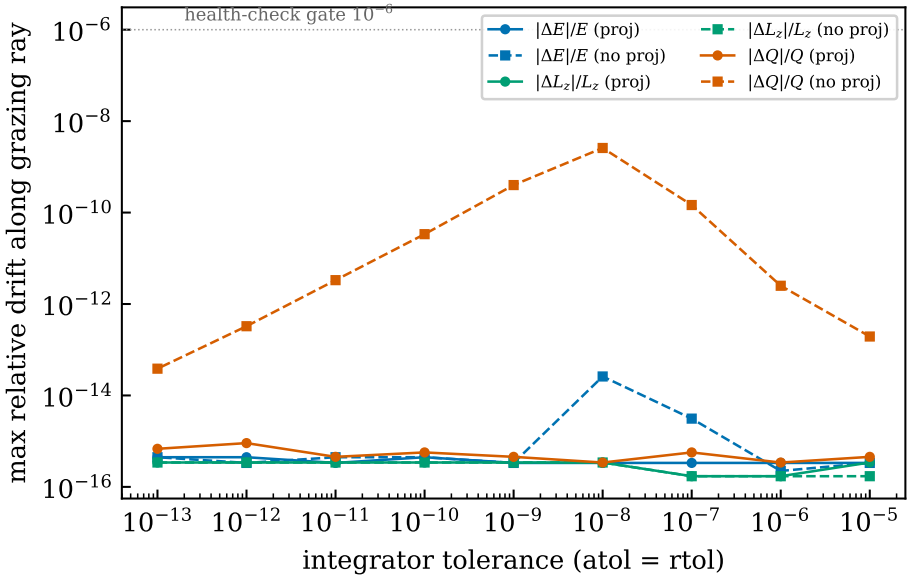


Figure 10.3: Maximum relative drift of the reconstructed conserved quantities along a boundary-grazing ray, versus tolerance, projection on (circles) and off (squares). The dotted line is the 10^{-6} health-check gate enforced by the test suite.

(E, L_z, Q) hold to better than 10^{-9} along a worst-case grazing ray at production tolerance, two orders inside the 10^{-6} acceptance gate; without it the drift crosses the gate for all tolerances looser than $\sim 10^{-11}$.

A methodological note: projection onto the constraint surface is a standard tool of geometric integration, but the specific failure it cures here — long benign propagation amplifying into a classification error at a distant separatrix — was found empirically, as a tolerance-dependence of the measured shadow in early Phase-1 runs. The diagnostic that captured it survives as the hidden `[.probe]` test case; the story is told as a methods lesson in Chapter 21.

10.3 Complex-step differentiation

The Hartle–Thorne path (Chapters 9, 17) integrates $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu} u^\mu u^\nu$ for arbitrary stationary axisymmetric metrics, which requires the derivatives $\partial_r g_{\mu\nu}$, $\partial_\theta g_{\mu\nu}$ of whatever metric sits behind the `IMetric` interface. Hand-coding derivatives per metric invites exactly the class of sign errors this project's test suite exists to prevent; finite differences trade a truncation error ($\propto h^2$) against a cancellation error ($\propto \varepsilon_{\mathrm{mach}}/h$), bottoming out near 10^{-10} relative at best.

The *complex-step* method (Squire & Trapp, 1998) evades the trade-off entirely. For a real-analytic f, Taylor expansion along the imaginary axis gives

$$f(x + ih) = f(x) + ihf'(x) - \frac{h^2}{2} f''(x) - i\frac{h^3}{6} f'''(x) + \cdots \tag{10.7}$$

so that

$$f'(x) = \frac{\operatorname{Im} f(x + ih)}{h} + \mathcal{O}(h^2), \tag{10.8}$$

with *no subtraction of nearby quantities anywhere*: the imaginary part is extracted, not differenced, so h may be made absurdly small. The implementation uses $h = 10^{-100}$ (`christoffel` in `metric.cpp`); the $\mathcal{O}(h^2)$ truncation term is 10^{-200}, far below the 10^{-308} double underflow only when multiplied by the third derivative — in practice the result is f' to full machine

precision, verified against the hand-derived Kerr Christoffels in the unit tests. The only requirement is that the metric components be evaluable on complex arguments, which is why `kerr_components` and the Hartle–Thorne functions are templated on the scalar type and every `IMetric` exposes a `std::complex<double>` overload.

10.4 Convergence behaviour

Fig. 10.3 doubles as the convergence study: with projection off, the conserved-quantity drift scales essentially linearly with the requested tolerance across eight decades — the signature of a correctly implemented error estimate (the controller holds the *local* error at tolerance; the accumulated drift inherits it) — while with projection on, the drift saturates at the projection floor (10^{-9}–10^{-12}, set by the guard band near turning points) independent of tolerance. Both behaviours are the expected ones; their intersection fixes the production tolerance (10^{-11}) as the point beyond which tightening buys nothing measurable. The timelike path is validated independently: the rosette integration of Fig. 9.2 holds $|g_{\mu\nu}u^\mu u^\nu + 1| \lesssim 10^{-13}$ over six radial periods, and the S2 production orbits carry the same diagnostic through every fit iteration (Chapter 21).

Part III

Results

Chapter 11

The Reference Ray Tracer, Shadow, and Critical Curve

The CPU reference tracer is the project's ground truth: every downstream artefact — disk images, sub-ring metrology, the GPU port — is validated against it, and it in turn is validated against closed-form theory. This chapter constructs the camera (`zamo_tetrad`, `camera_ray` in `physics/src/raytrace_cpu.cpp`), derives the two analytic anchors (Synge's shadow angle and Bardeen's critical curve), presents the boundary metrology (`shadow_boundary_angle`), and develops the spherical photon orbits and their Lyapunov exponents, which quantify the photon-ring substructure of Chapter 13.

11.1 The ZAMO camera and ray launching

A camera in curved spacetime needs a local orthonormal frame — a tetrad — to translate "pixel direction" into a four-momentum. The natural choice at BL position $(r_{\rm cam}, \theta_{\rm cam})$ is the zero-angular-momentum observer of Section 8.3: four-velocity

$$u^\mu = u^t\,(1,\ 0,\ 0,\ \omega), \qquad \omega = -\frac{g_{t\phi}}{g_{\phi\phi}}, \qquad u^t = \left[-(g_{tt} + 2\omega g_{t\phi} + \omega^2 g_{\phi\phi})\right]^{-1/2}, \tag{11.1}$$

completed to an orthonormal basis by the unit vectors along r, θ, and the remaining direction in the t–ϕ plane (`zamo_tetrad`). This observer exists everywhere outside the horizon — including inside the ergosphere, where a static camera would be impossible — and reduces to the static observer at $a = 0$, so a single camera implementation serves every spin.

A pixel at normalised image-plane position (x, y) defines a unit direction $(n^{\hat r}, n^{\hat\theta}, n^{\hat\phi})$ on the local sky; contracting with the tetrad legs and fixing the local photon energy to unity gives the BL four-momentum k^μ, from which `make_null_ray` extracts (E, L_z, Q) algebraically (Section 9.4). Rays are launched *forward*: null geodesics are time-reversible, so where a launched ray lands is what the pixel sees, and integrating outward avoids sampling subtleties at the camera. The production camera sits at $r_{\rm cam} = 1000\,M$ — far enough that the residual bending between camera and infinity is a correction below 10^{-3}, applied analytically in Section 11.2.

11.2 Grid-free boundary metrology

Measuring a shadow from a rendered image is limited by the pixel pitch. The tracer instead measures the boundary directly in camera-angle space: along each image-plane direction ψ, the capture/escape transition is bracketed and bisected (`shadow_boundary_angle`), each probe being a full geodesic integration with the projection of Chapter 10. Sixty bisection steps localise the critical angle $\alpha(\psi)$ to $\sim 10^{-12}$ rad — no pixel grid is involved anywhere, and thanks to the

tolerance-independence established in Fig. 10.2, the result is a property of the spacetime, not of the integrator.

Angular radii convert to impact parameters through

$$b = \frac{r_{\rm cam}\,\sin\alpha}{\sqrt{1-2M/r_{\rm cam}}}, \tag{11.2}$$

the flat-space relation corrected for the gravitational bending of the reference "straight line" at finite camera radius (the correction is the $1/\sqrt{1-2M/r}$ factor; at $r_{\rm cam}=10^3\,M$ it shifts b by one part in 10^3 and is exact for Schwarzschild). Diameters are quoted in units of M; the conversion to microarcseconds on the real sky is the calibration step of Chapter 19.

11.3 Schwarzschild validation: the Synge formula

For $a_*=0$ the whole boundary follows in three lines, first given by Synge (1966). Equatorial photons experience the effective potential $V(r)=(1-2M/r)/r^2$ (the b-independent factor of the radial potential (8.11) at $a=0$): $V'(r)=0$ gives the photon sphere $r_{\rm ph}=3M$; the critical impact parameter is $b_c=1/\sqrt{V(3M)}=\sqrt{27}\,M$; and an observer at finite $r_{\rm obs}$ sees the shadow at angular radius

$$\sin\alpha_{\rm sh} = \frac{\sqrt{27}\,M\,\sqrt{1-2M/r_{\rm obs}}}{r_{\rm obs}} \tag{11.3}$$

(`schwarzschild_shadow_angle`). The traced boundary agrees with (11.3) to a relative deviation of 5×10^{-16} — machine precision — and is circular to 0.000 per cent over 32 directions, with an equivalent diameter of $10.3924\,M$ against the analytic $2\sqrt{27}\,M\simeq10.3923\,M$. The same anchor reappears independently in the (a_*,i) sweep of Chapter 19: every $a_*=0$ grid row returns it identically at all inclinations, as spherical symmetry demands.

11.4 The Bardeen critical curve

For $a\neq0$ the shadow boundary is the image of the *photon shell* — the family of unstable spherical photon orbits of Section 9.2. On the shell the radial potential has a double root, $R(\tilde r)=R'(\tilde r)=0$; solving the pair for the two conserved ratios $\xi=L_z/E$ and $\eta=Q/E^2$ as functions of the shell radius $\tilde r$ gives, after the reduction of Bardeen (1973),

$$\xi(\tilde r)=\frac{\tilde r^2(3M-\tilde r)-a^2(\tilde r+M)}{a(\tilde r-M)}, \qquad \eta(\tilde r)=\frac{\tilde r^3\big[4a^2M-\tilde r(\tilde r-3M)^2\big]}{a^2(\tilde r-M)^2}. \tag{11.4}$$

For an observer at inclination θ_o the shell maps to the image plane as

$$\alpha=-\frac{\xi}{\sin\theta_o}, \qquad \beta=\pm\sqrt{\eta+a^2\cos^2\theta_o-\xi^2\cot^2\theta_o}, \tag{11.5}$$

tracing the closed *critical curve* as $\tilde r$ runs over the interval where $\beta^2\geq0$ — from near $r^+_{\rm ph}$ (prograde edge) to near $r^-_{\rm ph}$ (retrograde edge). Fig. 11.1 overlays the curves for four spins: growing spin flattens and displaces the prograde side, exactly as frame dragging predicts, while the overall scale barely changes — the geometric root of the spin–inclination degeneracy quantified in Chapter 19.

The traced boundary at $a_*=0.94$ reproduces this curve point-by-point (Fig. 11.2); this is Gate 2 of the validation ladder. The agreement is the spinning-spacetime counterpart of the Synge check: together they validate the tetrad, the launcher, the integrator, and the bisection logic end to end.

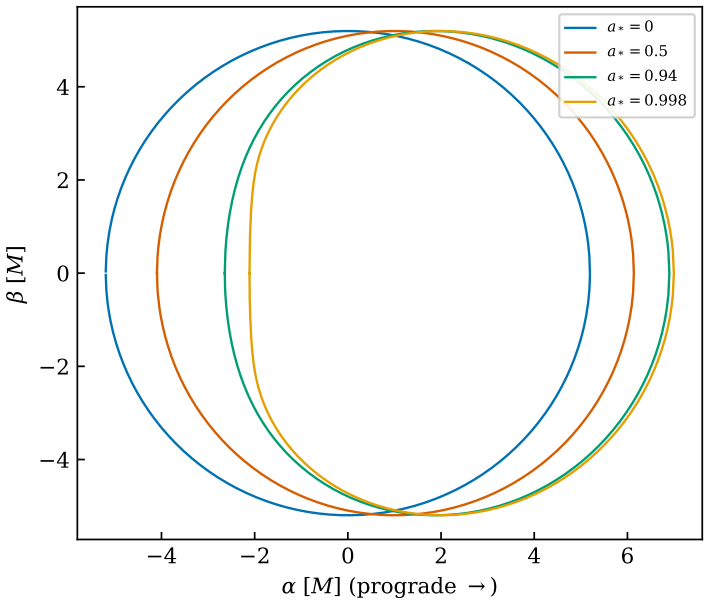


Figure 11.1: Analytic critical curves (11.4)–(11.5) for an edge-on observer at four spins. The prograde (right) side flattens and the curve displaces as spin grows; the mean diameter changes by only a few per cent.

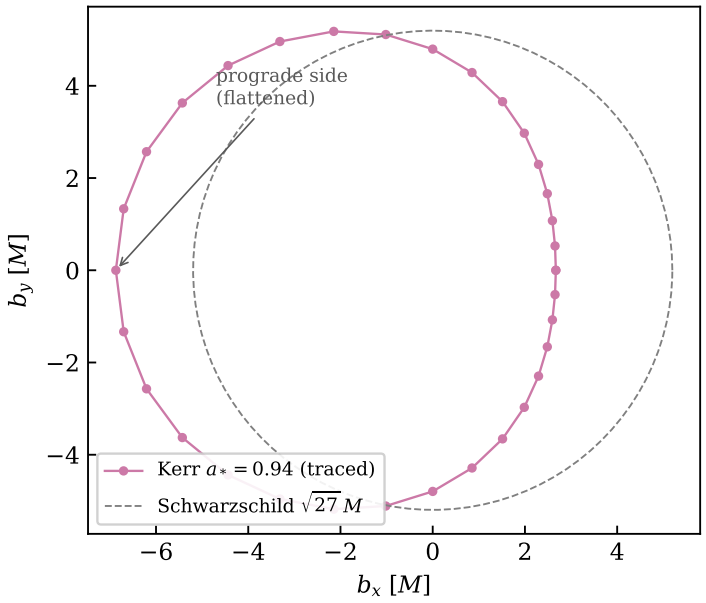


Figure 11.2: Traced Kerr shadow boundary at $a_* = 0.94$, edge-on, against the analytic Bardeen (1973) critical curve. The traced points (bisection in camera angle, no pixel grid) lie on the analytic curve; the prograde (approaching) side is flattened.

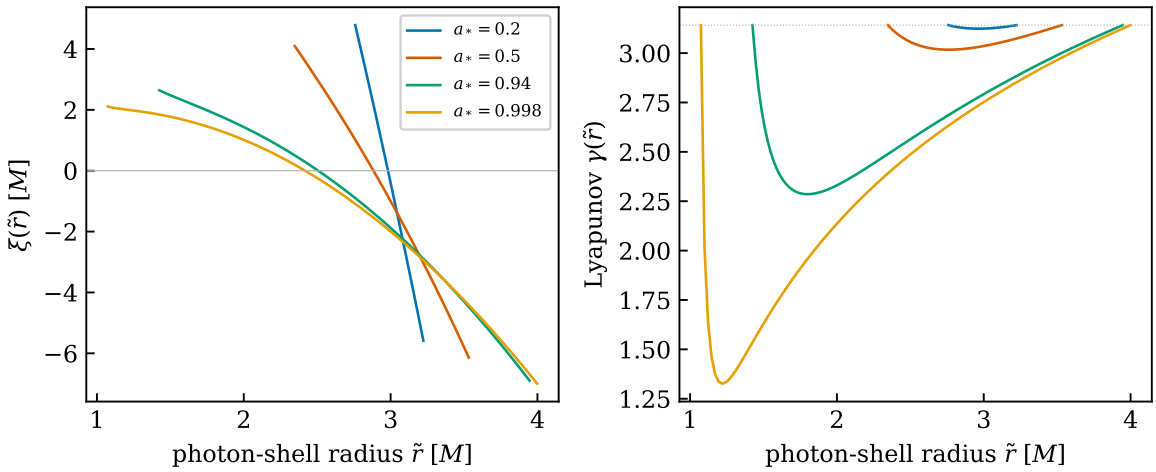


Figure 11.3: The spherical photon-orbit family (`sweep_tool photonorbits`). Left: the critical impact parameter $\xi(\tilde{r})$ across the shell at four spins; the zero crossing is the polar orbit. Right: the Lyapunov exponent (11.6) per polar half-orbit — exactly π at the equatorial edges for every spin, dipping toward the polar orbit as spin grows.

11.5 Spherical photon orbits and Lyapunov exponents

Each point of the critical curve corresponds to a spherical photon orbit at some shell radius $\tilde{r}$, and each such orbit is unstable: a photon displaced by δr obeys, in Mino time, $\delta\ddot{r} = \frac{1}{2}R''(\tilde{r})\,\delta r$, growing as $\exp\!\left[\sqrt{\frac{1}{2}R''}\,\tau\right]$. The observationally meaningful rate is the growth per *polar half-oscillation* — the interval between successive equatorial crossings, which is what separates one photon sub-ring from the next (Chapter 13):

$$\gamma(\tilde{r}) = \sqrt{\tfrac{1}{2}R''(\tilde{r})}\;\Delta\tau_\theta, \qquad \Delta\tau_\theta = \int_{\theta_-}^{\theta_+} \frac{\mathrm{d}\theta}{\sqrt{\Theta(\theta)}}, \tag{11.6}$$

with $R'' = 8E^2\tilde{r}^2 + 4EP - 2K$ from differentiating (9.5) twice and the polar period evaluated by the substitution $u = \cos\theta$ (the integral reduces to a smooth quadrature over one quadrant; `sweep_tool photonorbits`). Each extra half-orbit demagnifies the corresponding sub-ring image by $e^{-\gamma}$ (Johnson et al., 2020).

Two exact properties emerge from the computed family (Fig. 11.3). For Schwarzschild, $\frac{1}{2}R''(3M) = 27E^2 = (\xi^2)E^2$ and $\Delta\tau_\theta = \pi/\sqrt{\xi^2}$, so $\gamma = \pi$ for *every* orbit — the universal Schwarzschild value. More surprisingly, the computed γ returns to π at the *equatorial edges* of the shell for every spin — the identity $\frac{1}{2}R'' = (\xi^2 - a^2)E^2$ holds exactly at $\eta = 0$ — while dipping in between: at $a_* = 0.94$ the polar orbit ($\xi = 0$) has $\gamma = 2.571$, an 18 per cent weaker demagnification than π. This azimuth-dependence of the sub-ring demagnification, spanning $[2.571, \pi]$ at the Sgr A* spin, matches the range reported by Johnson et al. (2020) for M87*-like spins and is the quantitative bridge between the shadow geometry of this chapter and the flux ratios measured in Chapter 13.

11.6 Gravitational lensing of the disk and sky

The same machinery explains the signature look of every black-hole image. Consider the disk of Chapter 12 viewed nearly edge-on. Rays leaving the camera slightly *above* the shadow pass over the pole and are bent down onto the *far side* of the disk, behind the hole: the far side therefore

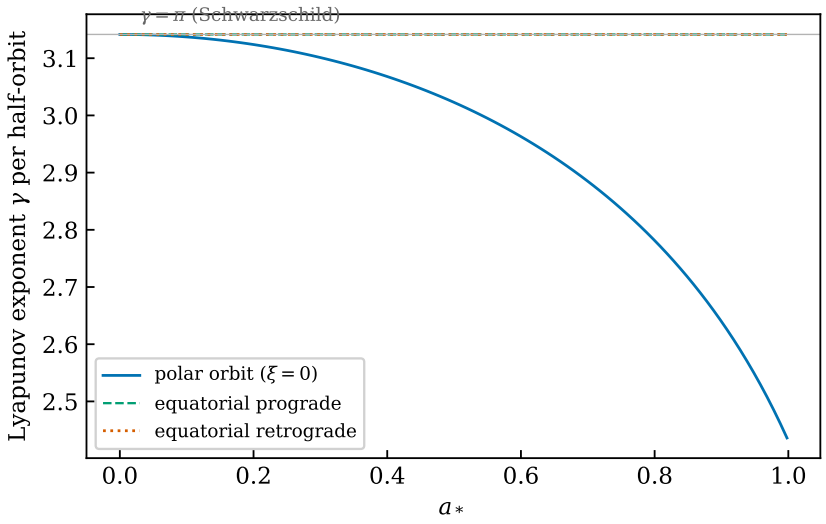


Figure 11.4: Lyapunov exponent of the polar and equatorial-limit orbits versus spin. The equatorial-limit values pin to π identically (the $\eta \to 0$ identity in the text); the polar orbit softens to 2.571 at $a_* = 0.94$.

appears *above* the shadow, as a luminous arch. Rays passing slightly below the shadow reach the far side's *underside*, painting a second, fainter arch below. Nothing is behind anything at a black hole: the entire disk is visible at once, wrapped around the silhouette — the geometry made famous by *Interstellar* (James et al., 2015) and computed here by the same backward tracing (Fig. 9.3 shows the ray families responsible).

In the exactly-aligned limit the arches close into rings: a source point diametrically behind the hole images to a full *Einstein ring*, and each additional winding number contributes a further ring converging on the critical curve — the $n = 1, 2, \ldots$ sub-rings of Chapter 13. On the spin axis this convergence produces a bright polar caustic that appears in near-edge-on renders as a slim vertical column of light above and below the hole; it is a *lensing* feature, not a jet, and is examined honestly in Section 13.3.

> *[Chapter under construction — scope: User render placeholder (THESIS_PLAN Part 1c): the traced lensed-disk engine render illustrating the double-arch geometry, with the exact* `sgr_engine` *reproduction command in the caption. See* `paper/THESIS_PLAN.md`*.]*

Chapter 12

The Accretion Disk: Novikov–Thorne and Page–Thorne

The disk model supplies every photon's emission point with a temperature and every pixel with a colour. This chapter derives the model the code implements (`flux`, `temperature`, `blackbody_rgb`, `bolometric` in `physics/src/disk_model.cpp`): the relativistic thin disk of Novikov & Thorne (1973) with the closed-form flux of Page & Thorne (1974), the exact redshift factor per ray, the g^4 beaming law from Liouville's theorem, and the Planck-to-sRGB colour pipeline.

12.1 Thin-disk conservation laws and the radial structure

The model disk is geometrically thin, optically thick, and lies in the equatorial plane between an inner edge at the ISCO and an outer radius r_{out}. Its structure follows from three conservation laws applied to an annulus $[r, r+\mathrm{d}r]$ in a steady state: *mass* — the accretion rate $\dot{M}$ is radius-independent; *angular momentum* — the inward advection of $L(r)$ (eq. 8.16) is balanced by the viscous torque $W(r)$ between annuli, less what is radiated; *energy* — the local dissipation is the difference between the torque's work and the advected binding energy $E(r)$ (eq. 8.15). Eliminating the torque between the latter two — with the boundary condition that the torque vanishes at the ISCO, where circular orbits end and matter plunges — gives the time-averaged radiated flux as a single quadrature over the circular-orbit quantities of Chapter 8 (Page & Thorne, 1974, eq. 15n):

$$F(r) = \frac{3\dot{M}}{8\pi M^2}\,\frac{f(x)}{x^4}, \qquad x = \sqrt{r/M}, \tag{12.1}$$

where the dimensionless $f(x)$ carries all the relativity. Carrying out the quadrature (the integrand is rational in x after the substitution; the full evaluation is reproduced in Appendix A) yields the closed form the code implements:

$$f(x) = \frac{1}{x^2\left(x^3 - 3x + 2a_*\right)}\Big[x - x_0 - \tfrac{3}{2}a_*\ln\frac{x}{x_0} - \sum_{i=1}^{3}\frac{3(x_i - a_*)^2}{x_i(x_i - x_j)(x_i - x_k)}\ln\frac{x - x_i}{x_0 - x_i}\Big], \tag{12.2}$$

with $x_0 = \sqrt{r_{\text{ISCO}}/M}$ and $x_{1,2,3}$ the roots of the familiar cubic $x^3 - 3x + 2a_* = 0$ — the same cubic as the photon orbit (8.12), solved by the same trigonometric substitution:

$$x_1 = 2\cos\Big(\frac{\arccos a_* - \pi}{3}\Big), \quad x_2 = 2\cos\Big(\frac{\arccos a_* + \pi}{3}\Big), \quad x_3 = -2\cos\Big(\frac{\arccos a_*}{3}\Big). \tag{12.3}$$

Two numerical guards in `flux` deserve documentation because both were pinned by unit tests after early mistakes (Chapter 21). First, $x_2 \to 0$ as $a_* \to 0$ while its logarithm coefficient

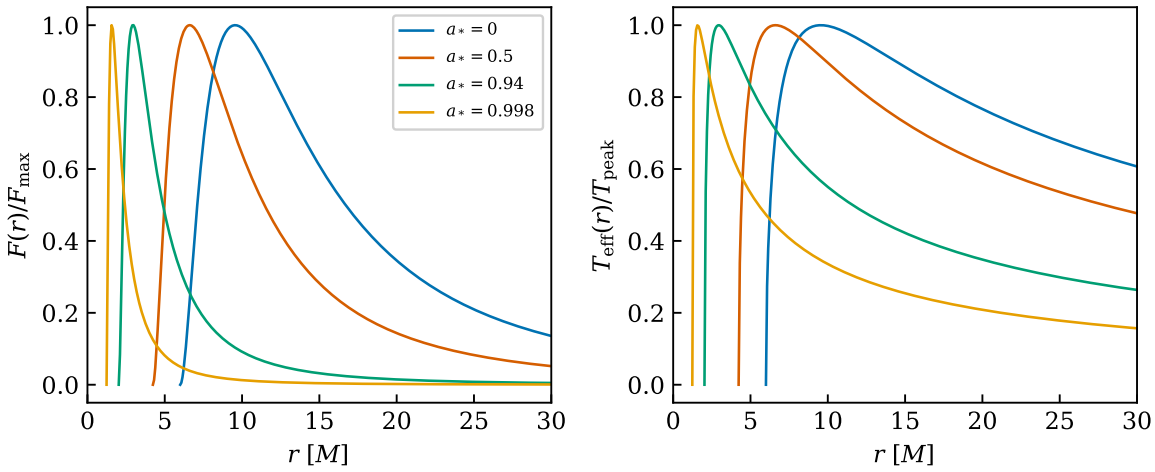


Figure 12.1: Page–Thorne flux (left, normalised to its own peak) and effective temperature (right) at four spins (`physics_tables` evaluation of `flux/temperature`). The emission zone follows the ISCO inward with spin.

stays finite only through a cancellation — the Schwarzschild limit must be taken with the $i = 2$ term's $0 \cdot \ln$ product handled explicitly. Second, the far-field behaviour of the bracket is

$$f(x)\,x^2\big(x^3 - 3x + 2a_*\big)/x^3 \;\longrightarrow\; x - c + \frac{3}{x} + \mathcal{O}(x^{-2}), \qquad c \simeq 3.976 \;\; (a_* = 0), \tag{12.4}$$

which is *not* the Newtonian $1 - \sqrt{r_{\rm in}/r}$ profile of folklore: the GR log terms shift the additive constant. The unit test asserts (12.4) against the exact expansion rather than the Newtonian form — an error a visual check could never catch, since both profiles look identical in a rendered image.

Fig. 12.1 shows $F(r)$ and the effective temperature $T_{\rm eff} \propto F^{1/4}$ across the spin range: as the ISCO plunges inward with spin (Chapter 8), the emitting annulus extends deeper, the peak flux grows, and the peak moves to smaller radii — the radial anatomy behind the spin-morphology discussion of Section 12.6.

12.2 The redshift factor

A backward-traced ray that crosses the equatorial annulus at radius r is assigned the frequency ratio between the camera and the co-orbiting emitter. The emitter moves on a circular orbit with angular velocity $\Omega(r)$ (eq. 8.14) and four-velocity $u^\mu = u^t(1, 0, 0, \Omega)$, normalised by

$$u^t = \big[-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})\big]^{-1/2}. \tag{12.5}$$

The emitted frequency is $\nu_{\rm emit} = -k_\mu u^\mu = u^t(E - \Omega L_z)$ (using $k_t = -E$, $k_\phi = L_z$), while the camera normalisation fixes $\nu_{\rm obs} = 1$; hence

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm emit}} = \frac{1}{u^t\,(E - \Omega L_z)} = \frac{\sqrt{-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})}}{1 - \Omega\, b_\gamma}, \qquad b_\gamma = \frac{L_z}{E}. \tag{12.6}$$

Equation (12.6) packages all three redshifts of the primer (Chapter 4) in one exact expression: the numerator carries gravitational redshift and transverse Doppler (it is $1/u^t$, the emitter's total time dilation), while the denominator $1 - \Omega b_\gamma$ is the line-of-sight Doppler term — photons emitted forward along the orbital motion ($\Omega b_\gamma > 0$) are blueshifted, backward redshifted. In the reference render at $a_* = 0.94$, $i = 80°$ the factor spans $g = 0.139$ to 1.440 across the image.

12.3 The g^4 beaming law from Liouville's theorem

Why does observed brightness scale as the *fourth* power of g? The clean argument is kinetic: photons in a beam occupy a phase-space volume, and Liouville's theorem makes the phase-space density invariant along rays — in radiative-transfer language, I_ν/ν^3 is the same in every frame that watches the beam pass (each factor of ν: one from the energy per photon, two from the solid-angle compression of the momentum cell). Integrating over frequency,

$$I_{\rm obs} = \int I_{\nu,{\rm obs}}\,{\rm d}\nu_{\rm obs} = \int g^3 I_{\nu/g,{\rm emit}}\; g\,{\rm d}\nu_{\rm emit} = g^4\, I_{\rm emit}. \tag{12.7}$$

For a blackbody there is an equivalent and computationally convenient statement: a Planck spectrum at temperature T, every photon of which is scaled by g, is *exactly* a Planck spectrum at temperature gT — blackbodies have no spectral features to distort. The code exploits this: the observed bolometric intensity is $(g\,T_{\rm eff}/T_{\rm ref})^4$ (`bolometric`), and the observed *colour* is the blackbody colour at $g\,T_{\rm eff}$ — no separate beaming factor is ever applied to the RGB channel, because in-band spectral radiance does not scale as g^4; only the bolometric total does. Keeping these two channels distinct (colour from gT; quantitative flux from g^4) is what lets the same render serve both as a faithful image and as the measurement input for the beaming-asymmetry and sub-ring statistics.

The measured consequence (Gate 3): the approaching limb outshines the receding limb by a bolometric factor of 9.60 in the reference geometry — the asymmetry every real black-hole image shows, and a hard acceptance criterion of the build (a symmetric disk means a broken g^4).

12.4 Planck spectrum to display colour

The rendered colours are physical. For a temperature T the spectral radiance is Planck's law; folding it against the CIE 1931 colour-matching functions (the analytic multi-lobe Gaussian fits of Wyman et al. 2013) gives tristimulus values (X, Y, Z), which a linear transform takes to linear sRGB (`planck_to_linear_srgb`, LUT-accelerated in `blackbody_rgb`). Fig. 12.2 shows the resulting Planckian locus and the display colours across temperature — the ember-orange to blue-white thermometer of the primer, now quantitative. The disk's inner annuli at $gT_{\rm eff} \sim 10^4$ K render blue-white; the beamed approaching limb is pushed further blueward exactly as (12.6) dictates.

12.5 The rendered disk

Fig. 12.3 assembles the pipeline: Novikov–Thorne temperatures (12.1), per-ray redshift factors (12.6), Planck colours at $g\,T_{\rm eff}$, at $a_* = 0.94$, $i = 80°$. The classic morphology anticipated by Luminet (1979) and rendered cinematically by James et al. (2015) emerges with no artistic input: the far side arches over the shadow (the lensing of Section 11.6), the approaching limb blazes at 9.60× the receding one, and the $n = 1$ sub-ring hugs the critical curve.

12.6 Spin morphology of the disk image

A striking qualitative observation from interactive exploration with the live renderer (Chapter 20): at $a_* = 0$ the disk reads as a system of smooth, continuous, near-concentric bright lanes girdling the whole silhouette, while at high spin ($a_* \approx 0.9$) the coherent bright structure collapses to a compact, intense patch on the approaching limb. Three exact results of this report combine to explain it:

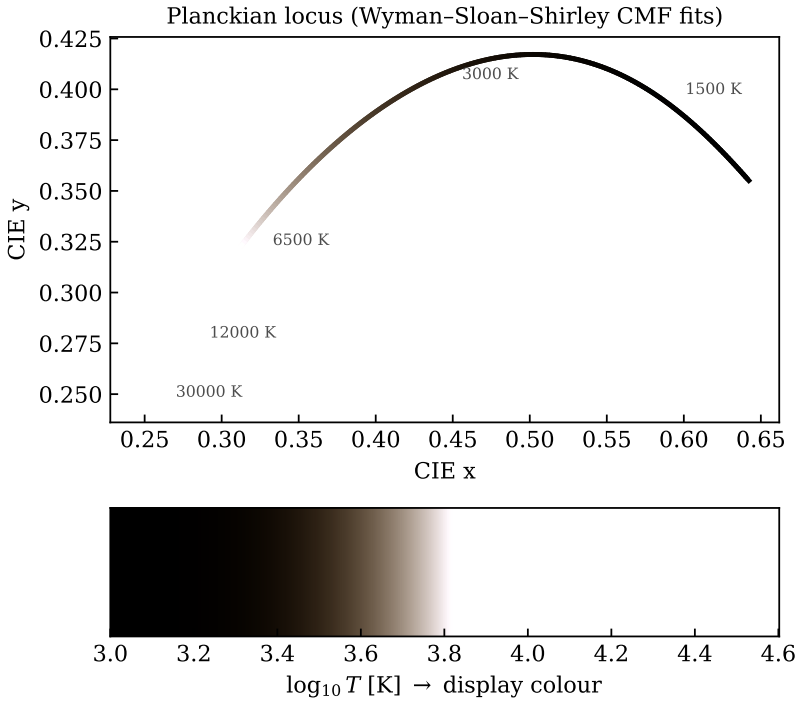


Figure 12.2: The Planckian locus in CIE 1931 chromaticity coordinates and the corresponding display colours (generated from the same Wyman et al. 2013 fits the renderer uses).

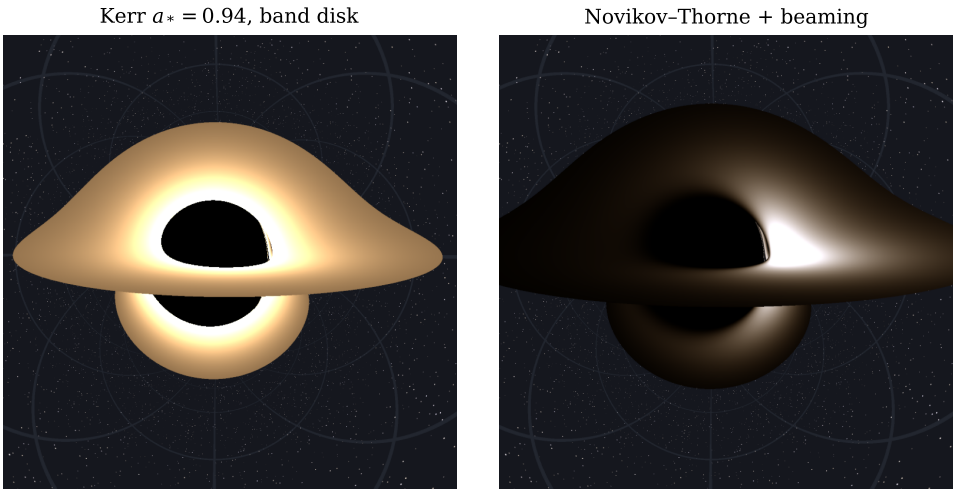


Figure 12.3: CPU reference render of the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$ (linear sRGB from Planck spectra at $g\,T_{\mathrm{eff}}$). Doppler beaming brightens the approaching limb by a bolometric factor of 9.60.

1. *The emission zone deepens.* The ISCO falls from $6M$ to $\sim 2.3M$ between $a_* = 0$ and 0.9 (Fig. 8.1), and the Page–Thorne peak follows it inward (Fig. 12.1) — the brightest material moves to radii where orbital speeds approach half the speed of light.

2. *Beaming sharpens as g^4.* With v/c rising steeply at the new inner radii, the approaching/receding contrast (12.7) grows from a moderate factor to orders of magnitude; the receding-side lanes fall below any practical display window while the approaching side saturates.

3. *Frame dragging compresses the pattern.* The display-layer streamlines advect at the true $\Omega(r)$ of (8.14); its steepening differential shear at small r winds features tightly on the prograde side.

At $a_* = 0$ none of these amplifiers operates strongly — orbital speeds at $r \gtrsim 6M$ stay below $0.4c$, the beaming contrast is mild, and the lanes remain visible around the full circumference. The transition between the two regimes is thus a direct visualisation of the ISCO physics of Chapter 8.

[Chapter under construction — scope: User render placeholders (THESIS_PLAN Part 1c): the two comparison engine renders at $a_* = 0$ and $a_* = 0.906$, with the exact `sgr_engine` command lines recorded in the captions (pending the `--rout`/`--turb` offscreen flags). See `paper/THESIS_PLAN.md`.]

Chapter 13

Photon-Ring Substructure

Light that grazes the photon shell can complete extra half-orbits before reaching the camera, stacking exponentially demagnified images of the disk onto the shadow's rim. This chapter measures that structure (`SubRingExtractor::analyze` and `scan_widths` in `science/photon_ring/`), connects the measured flux ratios to the Lyapunov exponents of Chapter 11, extends the measurements across the spin range, and closes with the axial caustic — a lensing feature often mistaken for a jet — and the detectability outlook.

13.1 Sub-ring decomposition by crossing count

The tracer counts each ray's equatorial-plane crossings *before* the crossing that terminates on the opaque disk; that count n is the ray's sub-ring order (Johnson et al., 2020): $n = 0$ is the direct image, $n = 1$ the first lensed image (light that passed over or under the hole once), $n = 2$ the second. Binning image pixels by n (`analyze`) isolates the orders cleanly (Fig. 13.1): the $n = 1$ ring is a narrow, displaced band hugging the shadow edge — visible directly as the sharp shoulder in the major-axis brightness profile of Fig. 13.2 — and the $n = 2$ ring is a hair-thin trace on the critical curve itself.

The measured bolometric ratios in the reference geometry are $F_0/F_1 = 10.7$ and $F_1/F_2 = 274$. The steep $n = 1 \to 2$ demagnification is the $e^{-\gamma}$ suppression per half-orbit derived in Section 11.5: with $\gamma \in [2.571, \pi]$ around the ring at this spin, the naive expectation $F_1/F_2 \sim e^{\bar{\gamma}} \sim 15$–$23$ applies to matched image annuli; the much larger *total*-flux ratio also folds in the shrinking solid angle of the thinner band and the disk's radial emission profile, which is

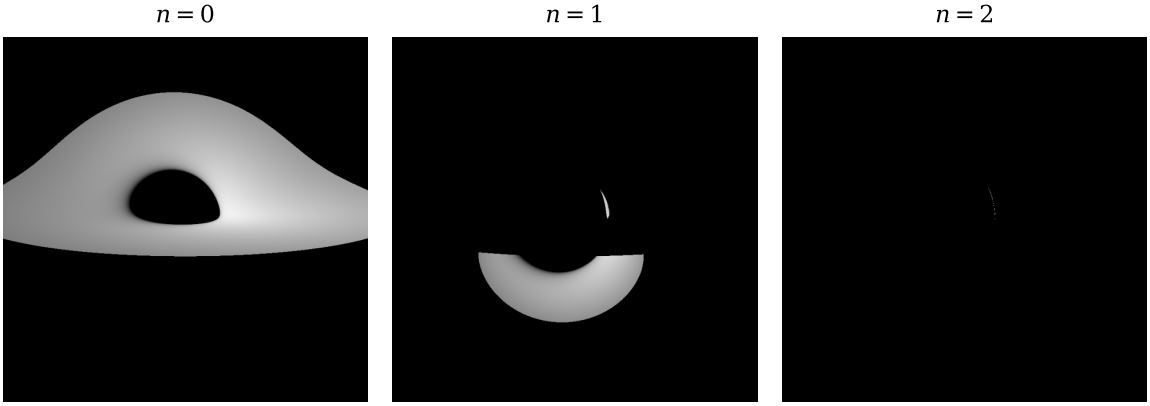


Figure 13.1: Bolometric maps of the photon sub-rings $n = 0, 1, 2$ (log stretch) at $a_* = 0.94$, $i = 80°$. Flux ratios: $F_0/F_1 = 10.7$, $F_1/F_2 = 274$.

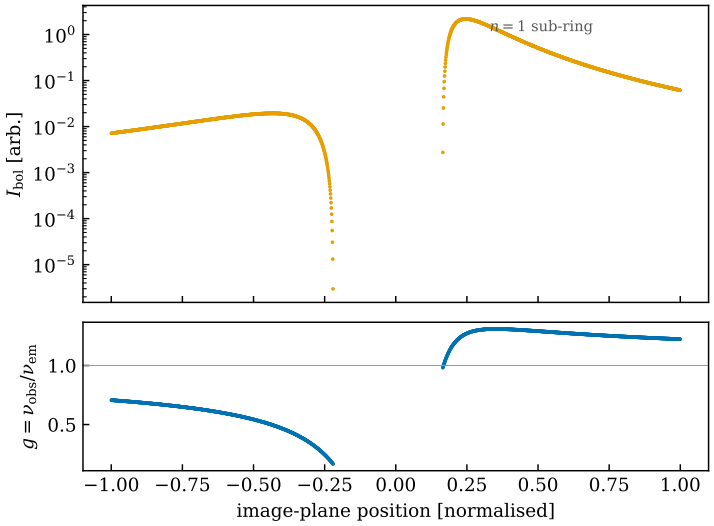


Figure 13.2: Bolometric intensity along the image major axis. The narrow $n = 1$ sub-ring stands on the shoulder of the direct disk image at the shadow edge.

why total-flux ratios always overshoot the bare exponential. The F_0/F_1 ratio, by contrast, is dominated by how much of the direct disk image lies far from the critical curve — it measures the emission model as much as the spacetime.

13.2 Sub-ring metrics across the spin range

The `--sweep` mode of `subring_tool` repeats the full decomposition at six spins (fixed $i = 80°$, 560^2 rays; Fig. 13.3). Two trends carry physical signatures:

- F_0/F_1 rises from 8.6 at $a_* = 0$ to 16.8 at $a_* = 0.998$: with the ISCO plunging inward, an ever-larger fraction of the direct emission comes from deep, strongly beamed radii, outcompeting the lensed band.
- The $n = 1$ band width below the shadow ($\psi = 270°$, the direction where the band is never occluded) grows monotonically with spin — the lensed image of the far-side underside widens as the emitting annulus deepens. Along the other directions the band is progressively *occluded by the opaque direct image* at high spin: a real geometric effect of an optically thick disk at high inclination, recorded honestly as missing measurements rather than zeros.

Both behaviours are properties of the exact null geodesics plus a known emission profile — free of plasma-model degeneracy, at the price of the thin-disk idealisation discussed in Chapter 22.

13.3 The axial image tower

Near-edge-on renders at high spin show slim, bright columns of light along the spin axis, above and below the hole — easily mistaken for the relativistic jets of radio-galaxy imagery. They are nothing of the kind, and this section documents what they actually are, because the distinction is a clean illustration of what a geodesic-plus-thin-disk model does and does not contain.

The columns are the *polar lensing caustic*. Consider camera rays launched almost exactly along the spin axis toward the hole: by axisymmetry, a ray displaced slightly from the axis

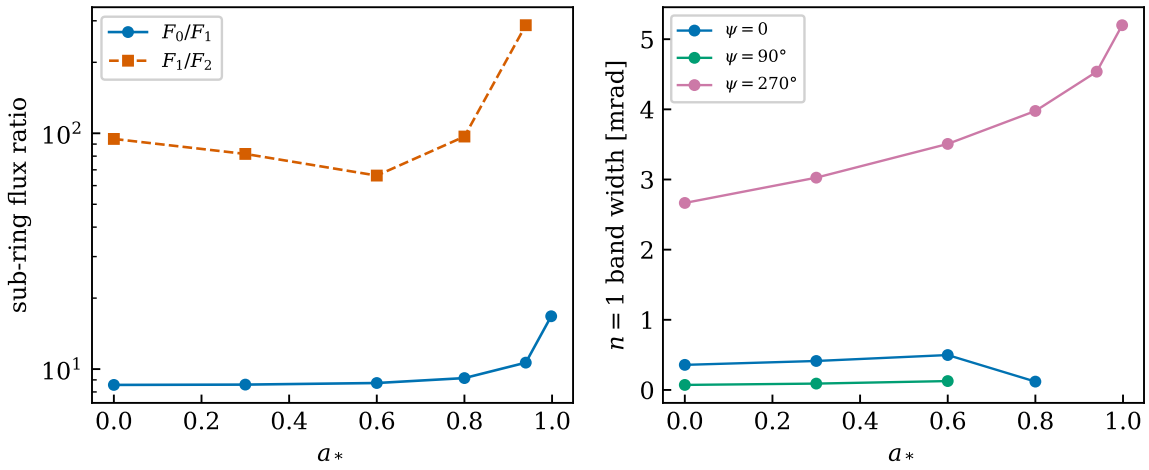


Figure 13.3: Sub-ring metrics versus spin at $i = 80°$ (`subring_tool --sweep`). Left: bolometric flux ratios; the F_1/F_2 point at $a_* = 0.998$ is omitted (the $n = 2$ ring is unresolved at this image scale). Right: measured $n = 1$ band widths along three image directions; zero-width entries (band occluded by the opaque direct image) are omitted.

in *any* azimuth is bent through the polar region and lands on the far/under side of the disk. A whole annulus of sky directions therefore maps onto nearly the same disk material — the Jacobian of the lens map degenerates on the axis (a caustic of the axisymmetric lens), and magnification formally diverges there, tamed only by the finite resolution. Stacked within the column are the $n = 1, 2, \ldots$ images of Section 13.1 converging toward the polar critical point; the column is, in effect, the Einstein-ring limit of Section 11.6 seen edge-on.

Real astrophysical jets are a different object entirely: magnetically launched, plasma-loaded outflows powered by mechanisms such as Blandford & Znajek (1977) energy extraction, requiring magnetohydrodynamics far outside this model's physics — and Sgr A* itself shows at most weak evidence for such a jet. The honest statement is that our columns are pure lensing: they carry no matter, no energy outflow, and would appear for *any* bright equatorial source around *any* black hole viewed near the equatorial plane. The corresponding limitation note appears in Chapter 22.

> *[Chapter under construction — scope: User render placeholder (THESIS_PLAN Part 1c): the engine render showing the axial columns, with the exact* `sgr_engine` *reproduction command in the caption. See* `paper/THESIS_PLAN.md`*.]*

13.4 Detectability forecast

The $n = 1$ ring is a target of the next interferometer generation. In the visibility domain, a thin ring of angular width w contributes a slowly decaying tail beyond baselines $\sim 1/w$; Johnson et al. (2020) project that space–ground baselines could isolate the $n = 1$ signature for M87* and Sgr A*. Our measured widths put numbers on the requirement: at $a_* = 0.94$, $i = 80°$ the unoccluded $n = 1$ band spans ~ 4.5 mrad of camera angle at $r_{\rm cam} = 10^3\,M$, i.e. $\sim 4.5\,M$ radially — a few μas at the Sgr A* calibration of Chapter 19, comfortably inside proposed ngEHT-class resolution but demanding of dynamic range (the band sits $\sim 10.7\times$ below the direct image in total flux). Because these maps are exact geodesic quantities on a known spacetime, they are clean templates for such forecasts; what they deliberately omit — plasma variability, optical depth, scattering — is the subject of the limitations chapter.

Chapter 14

Timelike Orbits and the S-Star Forward Model

The S2 fit of Chapter 15 stands on a forward model: given thirteen parameters, predict every measured position and velocity. This chapter derives that model layer by layer (`SkyProjection::S2Model` in `science/orbit_fitter/`): initial conditions from exact turning points, integration and sampling, sky projection, the Rømer delay, and the relativistic redshift factorisation (`redshift_terms`, `SpectralLineFitter::predict`).

14.1 The forward model at a glance

One model evaluation proceeds: (i) convert the osculating elements $(a, e, i, \Omega, \omega, t_{\rm peri})$ plus (M, R_0) into exact initial conditions (Section 14.2); (ii) integrate the timelike geodesic through the observation window with the generic Christoffel integrator of Chapter 9 (this path works for Kerr *and* Hartle–Thorne, which the no-hair fit requires); (iii) interpolate the cached trajectory at each requested emission time (cubic Hermite on position and velocity, `sample_at`); (iv) project onto the sky and apply the reference-frame offsets (x_0, y_0, v_x, v_y); (v) solve the Rømer delay per epoch (Section 14.3); (vi) evaluate the observed radial velocity including all relativistic terms (Section 14.4). Steps (i)–(iii) run once per parameter vector; steps (iv)–(vi) per data epoch. One full evaluation against the S2 dataset costs 3.7 ms — the number that made the MCMC of Chapter 15 practical.

14.2 Orbital elements from radial turning points

The osculating elements define the pericentre and apocentre radii $r_p = \tilde{a}(1-e)$, $r_a = \tilde{a}(1+e)$ (with $\tilde{a}$ the physical semi-major axis from $a_{\rm mas}$ and R_0). The exact relativistic orbit with these turning points has E and L_z fixed by the two conditions

$$R(r_p) = 0, \qquad R(r_a) = 0, \tag{14.1}$$

with R the timelike radial potential (9.5) — the construction of Darwin (1959). For the equatorial-plane integration frame the code uses ($Q = 0$; the orbit is subsequently rotated to the sky by (i, Ω, ω)), at $a = 0$ the potential is $R = E^2 r^4 - \Delta(r^2 + L_z^2)$ and (14.1) is a 2×2 *linear* system in (E^2, L_z^2):

$$\begin{pmatrix} r_p^4 & -\Delta_p \\ r_a^4 & -\Delta_a \end{pmatrix} \begin{pmatrix} E^2 \\ L_z^2 \end{pmatrix} = \begin{pmatrix} \Delta_p r_p^2 \\ \Delta_a r_a^2 \end{pmatrix} \implies E^2 = \frac{\Delta_p \Delta_a \,(r_a^2 - r_p^2)}{\Delta_p r_a^4 - \Delta_a r_p^4}, \qquad L_z^2 = \frac{r_p^4\, \Delta_a r_a^2 - \Delta_p r_p^2\, r_a^4}{\Delta_p r_a^4 - \Delta_a r_p^4} \tag{14.2}$$

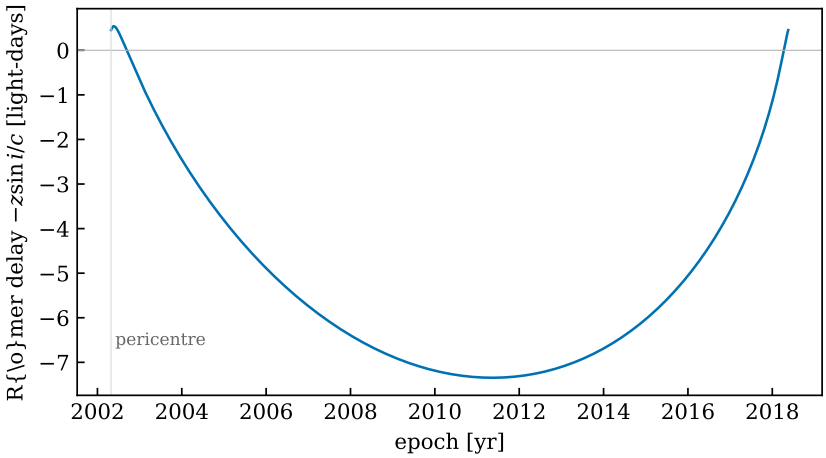


Figure 14.1: The Rømer delay $-z\sin i/c$ over one S2 orbit, evaluated from the fitted elements (Keplerian geometry evaluation; the production model solves (14.3) on the exact geodesic). Peak-to-peak the delay is ~ 8 light-days.

(Cramer's rule, $\Delta_{p,a} = \Delta(r_{p,a})$). The implementation writes the same system in the equivalent form $E^2 = (1-2M/r)(1+L^2/r^2)$ at both radii and solves for L^2 first — algebraically identical, but numerically better conditioned at S2-like radii ($r \sim 10^3$–$10^4\,M$) because it avoids differencing the large r^4 terms. The rosette integration of Fig. 9.2 validated exactly this construction: the integrated orbit's conserved (E, L) match (14.2) to eleven decimal places.

Why not Keplerian initial conditions? The obvious shortcut — start at pericentre with the Newtonian vis-viva speed — mis-states the *relativistic* orbit: with the same (r_p, v_p) the GR orbit has a different apocentre and, at S2's eccentricity, an orbital period wrong by 1.6 per cent. The measured S2 period is known to 0.014 per cent, so the shortcut is two orders of magnitude too crude — a bug found by exactly this discrepancy during Phase 4 and pinned by a regression test since (the narrative is told in Chapter 21). The turning-point construction, by contrast, reproduces the requested geometry exactly by definition.

14.3 Sky projection and the Rømer delay

The integrated orbit lives in BH-centred coordinates; the observables live on the sky. The rotation by (i, Ω, ω) maps the orbital plane to East–North offsets and a line-of-sight coordinate z (positive away from us), scaled by R_0 into angles; the fitted frame parameters (x_0, y_0, v_x, v_y) absorb the residual tie between the infrared frame and the radio position of Sgr A*, and v_z the line-of-sight zero point.

Light from different orbit phases travels different distances: an epoch observed at $t_{\rm obs}$ was *emitted* at $t_{\rm em}$ satisfying

$$t_{\rm em} = t_{\rm obs} - \frac{z(t_{\rm em})}{c}, \tag{14.3}$$

the classical Rømer delay (first used by Rømer in 1676 to measure c from Jupiter's moons; here it is c that is known and the geometry that matters). Equation (14.3) is an implicit equation for $t_{\rm em}$; the code solves it by fixed-point iteration $t^{(k+1)} = t_{\rm obs} - z(t^{(k)})/c$, which contracts with rate $|\dot{z}/c| = |\beta_z| \lesssim 0.026$ for S2 — guaranteeing convergence in 2–3 iterations to sub-second accuracy. For S2 the delay spans about ± 4 light-days across the orbit (Fig. 14.1); ignoring it would shift pericentre-passage timing by days against astrometry measured to weeks — a measurable model error, which is why the delay is applied to every epoch of every star.

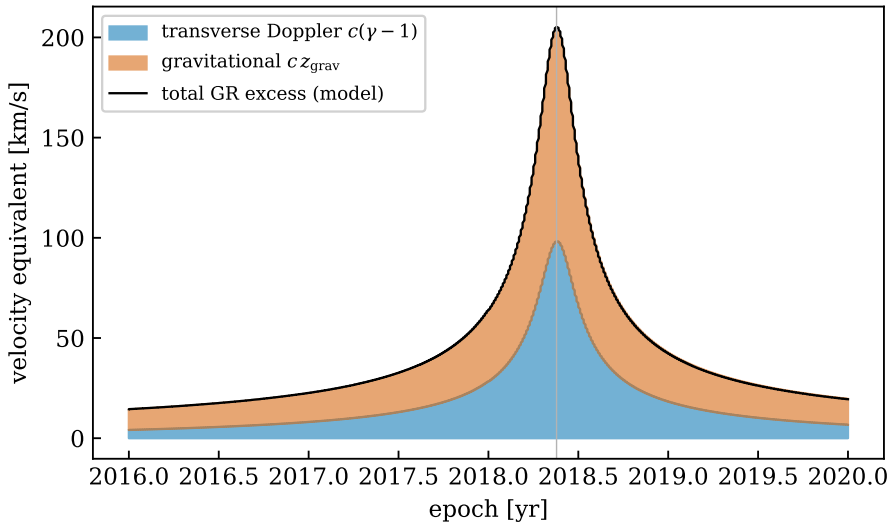


Figure 14.2: Decomposition of S2's relativistic radial-velocity excess through the 2018 pericentre (`phase4/s2_redshift_curve.csv`): transverse Doppler (blue) and gravitational (red) stack to the total model excess (black line).

14.4 The spectroscopic observable

The measured quantity for S2 spectroscopy is the centroid of the Brγ line ($\lambda_{\rm rest} = 2.16612\,\mu$m). The model wavelength ratio factorises exactly as

$$\frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = (1+z_{\rm D})(1+z_{\rm tD})(1+z_{\rm grav}), \tag{14.4}$$

where the three factors isolate the classical line-of-sight Doppler term $z_{\rm D}$, the special-relativistic transverse (time-dilation) term $z_{\rm tD} = \gamma - 1$, and the gravitational redshift $z_{\rm grav}$ at the emission point (`redshift_terms` evaluates each along the cached trajectory). The quoted model radial velocity is the *apparent* $v_r = c\,(\lambda_{\rm obs}/\lambda_{\rm rest} - 1)$ — the same convention the observers use — so the relativistic terms appear as an excess over a Keplerian orbit with identical elements, which is precisely the quantity GRAVITY measured in 2018 (Section 15.4).

Fig. 14.2 stacks the two relativistic terms through the 2018 pericentre passage: they are comparable in size (the $2/(1+e)$ near-equality worked out in Chapter 15), sum to the $\sim$ 200 km/s excess, and each is evaluated with no adjustable amplitude — the fitted mass and geometry force both.

> *[Chapter under construction — scope: Remaining for this chapter: Hermite-interpolation error bound for `sample_at` (half-step position error $< 10^{-9}$ of the orbit scale at the production sampling density). See `paper/THESIS_PLAN.md`.]*

Chapter 15

Orbit Fitting, Statistics, and the S2 Results

15.1 Data

The astrometric and spectroscopic record of S2 is taken from the monitoring compilation of Gillessen et al. (2017) (VizieR J/ApJ/837/30): 145 imaging positions (VLT NACO and Keck NIRC, 1992–2016, typical uncertainties 0.3–3 mas) and 44 radial velocities (SINFONI and earlier spectrographs, 2000–2016, typical uncertainties 15–100 $\mathrm{km\,s^{-1}}$). Positions are offsets from the radio position of Sgr A*; the residual uncertainty of that frame tie is modelled by the fitted offset/drift parameters with the Plewa et al. (2015) priors of Section 15.2. We apply no error rescaling: the reduced χ^2 of 0.959 below is obtained with the published uncertainties as-is.

15.2 The objective and its parameterisation

The objective is

$$\chi^2 = \sum_i \left[\frac{(\alpha_i^{\mathrm{obs}} - \alpha_i^{\mathrm{mod}})^2}{\sigma_{\alpha,i}^2} + \frac{(\delta_i^{\mathrm{obs}} - \delta_i^{\mathrm{mod}})^2}{\sigma_{\delta,i}^2} \right] + \sum_j \frac{(v_{r,j}^{\mathrm{obs}} - v_{r,j}^{\mathrm{mod}})^2}{\sigma_{v,j}^2} + \chi^2_{\mathrm{prior}}, \tag{15.1}$$

over the 145 astrometric epochs and 44 radial velocities, with the forward model of Chapter 14 supplying the model positions and velocities.

Thirteen parameters are fitted: (M, R_0), six orbital elements, the four reference-frame offsets and drifts (x_0, y_0, v_x, v_y) of the mass with respect to the radio frame, and v_z; the no-hair mode adds ε_q. Weak priors tie the frame to Sgr A* following Plewa et al. (2015) (± 0.2 mas at epoch 2009.0, ± 0.1 $\mathrm{mas\,yr^{-1}}$, ± 5 $\mathrm{km\,s^{-1}}$). The χ^2 is minimised with the Ceres solver (automatic differentiation; `OrbitFitter::fit`); parameter covariances are obtained as $C = (J^{\mathsf{T}} J)^{-1}$ from an explicit central-difference Jacobian J of the weighted residual vector (`OrbitFitter::covariance`), because the solver's truncated-SVD covariance annihilates the strongly correlated M–R_0 direction and reports a spuriously zero mass uncertainty (Chapter 21).

15.2.1 Fisher information and the Cramér–Rao floor

For a model with Gaussian errors, write the σ-normalised residual vector $r(\theta)$ and its Jacobian $J_{ik} = \partial r_i / \partial \theta_k$. Near the optimum, $\chi^2(\theta) \approx \chi^2_{\mathrm{min}} + \delta\theta^{\mathsf{T}} (J^{\mathsf{T}} J)\, \delta\theta$, so $F = J^{\mathsf{T}} J$ is the Fisher

Table 15.1: S2 fit results (full GR timelike geodesic model) against published values (Gillessen et al., 2017; GRAVITY Collaboration, 2022). Uncertainties are 1σ from the explicit-Jacobian covariance.

Parameter	This work	Published
M [$10^6\,\mathrm{M}_\odot$]	4.270 ± 0.195	4.297 ± 0.012
R_0 [pc]	8272 ± 182	8277 ± 9
a [mas]	124.82 ± 0.88	125.058 ± 0.041
e	0.88277 ± 0.00195	0.8839 ± 0.0019
i [deg]	134.70 ± 0.41	134.18 ± 0.40
Ω [deg]	227.33 ± 0.63	226.94 ± 0.60
ω [deg]	65.44 ± 0.61	65.51 ± 0.57
t_{peri} [yr]	2002.3227 ± 0.0052	2002.33 ± 0.01
P [yr]	16.0561 ± 0.0228	16.0518 ± 0.0022
χ^2_ν	0.959	—

information matrix, and the Cramér–Rao bound states that *no* unbiased estimator can beat

$$\mathrm{Cov}(\theta) \succeq F^{-1} = (J^{\mathsf{T}} J)^{-1}. \tag{15.2}$$

For a least-squares estimator at a well-conditioned optimum the bound is attained to linear order, which is why $C = (J^{\mathsf{T}}J)^{-1}$ is the standard parameter covariance; every "±" in Table 15.1 is the square root of a diagonal element of exactly this matrix, built from an explicit central-difference J with physically scaled steps and inverted by Gauss–Jordan elimination with partial pivoting (`OrbitFitter::covariance`).

15.2.2 Why the solver's SVD covariance fails here

The subtlety is the spectrum of F. Astrometry alone constrains the *angular* orbit — approximately the combination M/R_0^3 through Kepler's law with the angular semi-major axis and the period — while the radial velocities pin the *physical* velocity scale $\propto \sqrt{M/R_0}$. The two constraints intersect at a sharp angle in the (M, R_0) plane: the fit is superbly conditioned along one combination and weakly conditioned along the near-degenerate one, so F carries one singular value several orders below the rest — small, but genuinely finite and *information-bearing*. A covariance routine that regularises by *truncating* small singular values treats that direction as exact rank deficiency: the truncated pseudo-inverse assigns it *zero* variance, and the reported mass uncertainty collapses to machine noise. That is precisely the failure observed with the library covariance during Phase 4 (a fitted mass "known" to nine digits was the giveaway; Chapter 21) and the reason the project computes (15.2) explicitly: the small singular value is *inverted*, not discarded, and the M–R_0 error ellipse comes out with its true, elongated shape — subsequently confirmed end-to-end by the MCMC posterior of Section 15.6.

15.3 Results: the S2 fit

We fit the 145 astrometric positions and 44 radial velocities of S2 tabulated by Gillessen et al. (2017). The best-fitting model (Fig. 15.1, Fig. 15.2) has $\chi^2 = 307.8$ for 321 degrees of freedom ($\chi^2_\nu = 0.959$); the residuals (Fig. 15.3) show no structure. Table 15.1 compares the recovered parameters with the published values.

The recovered mass and distance agree with the far more precise GRAVITY interferometric determination (GRAVITY Collaboration, 2022) to 0.14σ and 0.03σ respectively — a non-trivial

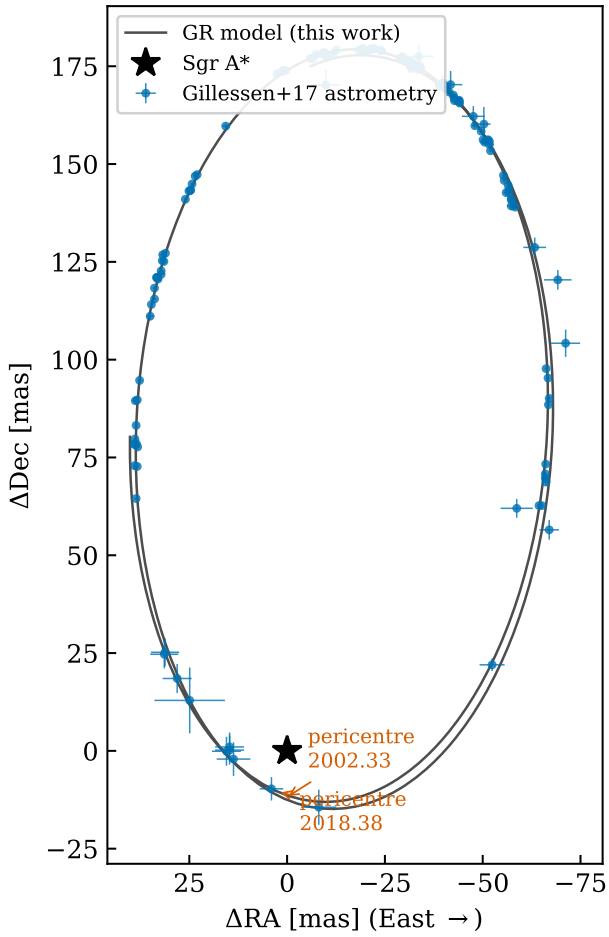


Figure 15.1: S2 on the sky: fitted GR geodesic model (curve) and the 145 astrometric measurements of Gillessen et al. (2017) (points). Sgr A* sits at the origin.

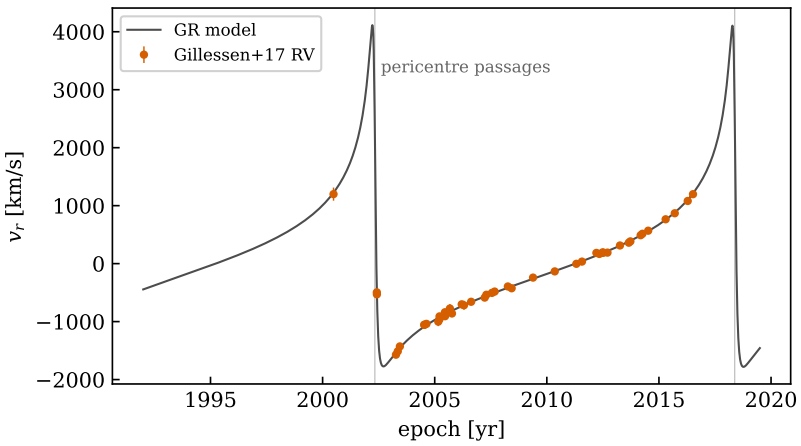


Figure 15.2: S2 radial velocity: model and the 44 measurements. The sharp swing brackets the 2002 and 2018 pericentre passages.

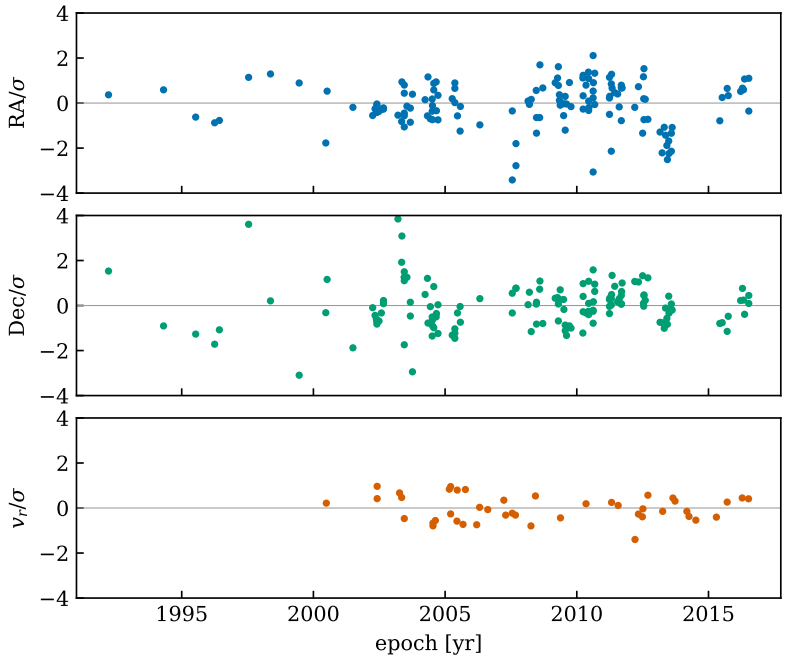


Figure 15.3: Normalised fit residuals. $\chi^2_\nu = 0.959$ over 321 degrees of freedom.

cross-check, since our fit uses none of the GRAVITY beam astrometry, only the VLT/Keck imaging era compiled by Gillessen et al. (2017).

Two complementary views probe the residual structure. Fig. 15.4 shows observed-minus-computed values in physical units — no secular trend, no pericentre-phase feature, the early low-weight epochs visibly noisier as their published uncertainties say they should be. Fig. 15.5 maps the astrometric residual *vectors* on the sky plane, magnified 50×: no coherent rotation or dipole (which would indicate a frame drift beyond the fitted terms), no alignment with the orbit track (which would indicate a timing bias).

15.4 Gravitational redshift at pericentre

The magnitude of the relativistic signal at pericentre follows from the fitted elements alone. Step by step:

1. The pericentre distance is $r_{\rm peri} = a(1-e) = 1032\,{\rm au} \times (1-0.88277) \approx 121$ au, i.e. ≈ 1470 Schwarzschild radii ($R_S = 2GM/c^2 \approx 0.085$ au).
2. The vis-viva speed there is $v_{\rm peri} = \sqrt{GM(2/r_{\rm peri} - 1/a)} \approx 7650\,{\rm km\,s^{-1}}$, so $\beta \approx 0.0255$.
3. The transverse Doppler shift contributes $z_{\rm tD} \simeq \beta^2/2 \approx 3.3 \times 10^{-4}$, i.e. $c\,z_{\rm tD} \approx 98\,{\rm km\,s^{-1}}$.
4. The gravitational redshift contributes $z_{\rm grav} \simeq GM/(r_{\rm peri}c^2) = \beta^2_{\rm circ} \approx 3.5 \times 10^{-4}$, i.e. $\approx 105\,{\rm km\,s^{-1}}$.

The two corrections are nearly equal at pericentre — not a coincidence: vis-viva gives $v^2_{\rm peri} = (GM/a)(1+e)/(1-e)$ while $GM/r_{\rm peri} = (GM/a)/(1-e)$, so $z_{\rm grav}/z_{\rm tD} = 2/(1+e) = 1.06$ at the S2 eccentricity — and their sum in the full geodesic model peaks at $205\,{\rm km\,s^{-1}}$ at epoch 2018.379 (Fig. 15.6), in agreement with the $\approx 200\,{\rm km\,s^{-1}}$ excess measured around the 2018.38 pericentre passage by GRAVITY Collaboration (2018), who quantified it as $f = 0.90 \pm 0.17$ on a scale where $f = 0$ is Newtonian and $f = 1$ is GR. Our forward model contains no adjustable

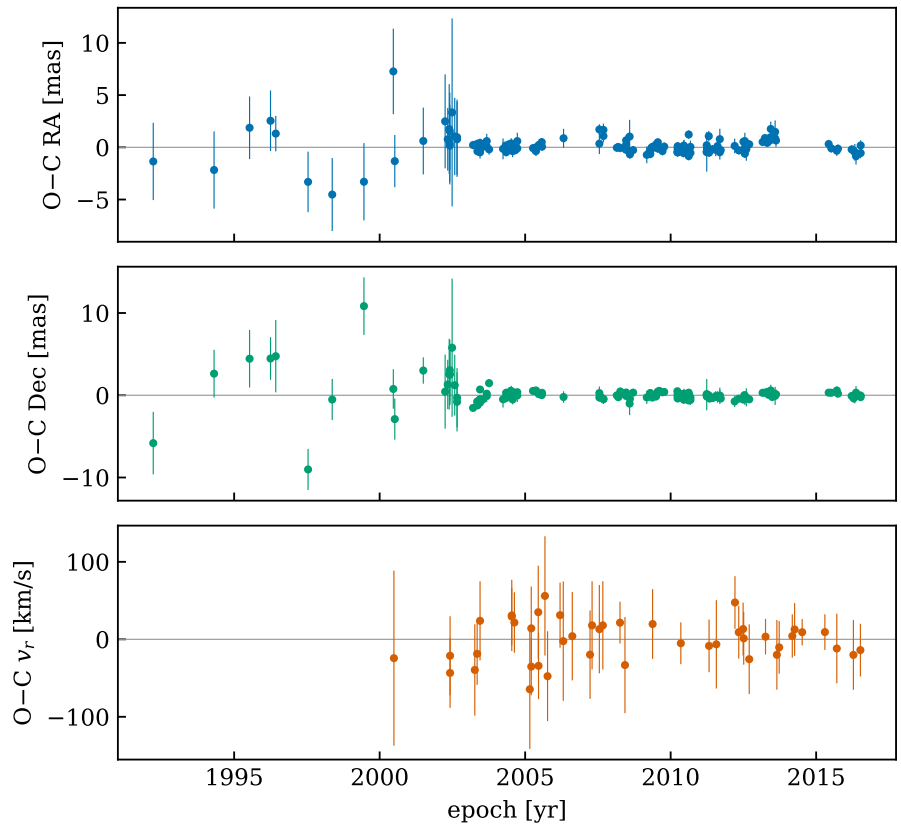


Figure 15.4: Observed minus computed, in measurement units, for the three data channels of the S2 fit.

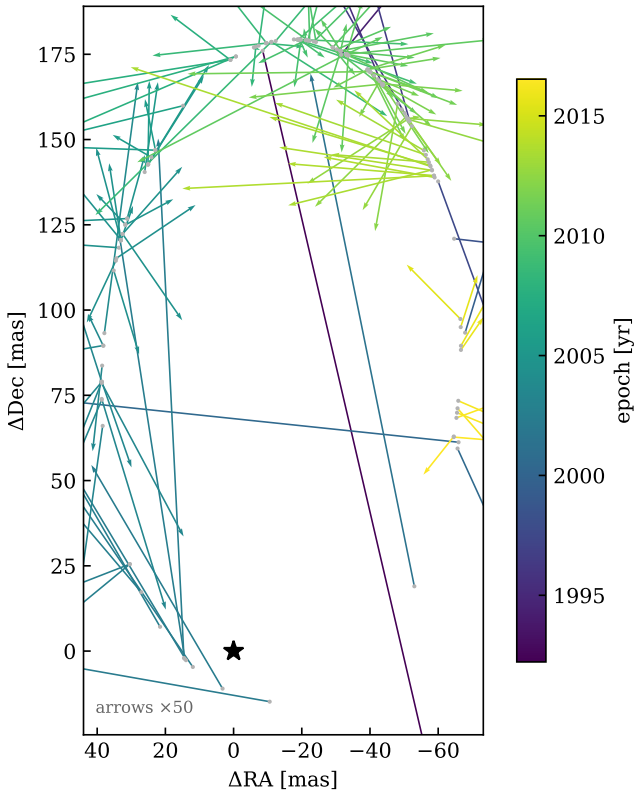


Figure 15.5: Astrometric residual vectors along the fitted S2 orbit (arrows magnified 50×, coloured by epoch). No coherent pattern survives the fit.

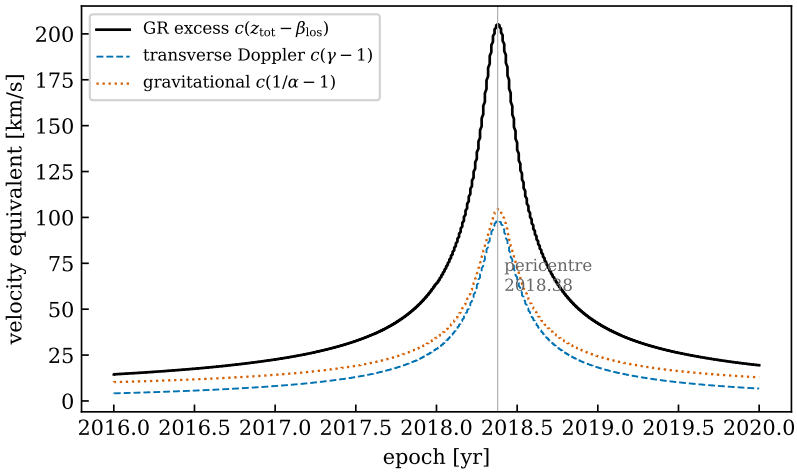


Figure 15.6: Relativistic radial-velocity excess of S2 over a Keplerian orbit with identical elements, through the 2018 pericentre passage. The peak is $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379, matching the detection of GRAVITY Collaboration (2018).

redshift amplitude: the excess is forced by the fitted mass and geometry. (Like GRAVITY Collaboration (2018) we neglect the $\sim$5 $\mathrm{km\,s^{-1}}$ light-bending correction.)

15.5 Relativistic precession

The fitted orbit advances its pericentre by 11.98 arcmin per orbital period from the Schwarzschild term. The textbook estimate agrees step by step: the semilatus rectum is $p = a(1-e^2) = 1032\,\mathrm{au}\times(1-0.88277^2)\approx 228$ au, so with $R_g \approx 0.0424$ au the ratio is $R_g/p = 1.86\times10^{-4}$ and

$$\delta\omega = \frac{6\pi GM}{c^2 a(1-e^2)} = 6\pi\,\frac{R_g}{p} \approx 3.5\times10^{-3}\,\mathrm{rad}, \tag{15.3}$$

i.e. ≈ 12.0 arcmin per 16-yr orbit, matching the full geodesic integration (11.98 arcmin) and the GRAVITY detection of the Schwarzschild precession, $f_{\rm SP} = 1.10\pm0.19$ (GRAVITY Collaboration, 2020). Frame dragging at $a_* = 0.94$ contributes only -0.204 arcmin per orbit (retrograde for the S2 geometry), two orders of magnitude below current sensitivity, consistent with the hierarchy laid out by Will (2008).

Fig. 15.7 places the three analytic precession terms of `GRPrecession` side by side for the fitted S2 elements as a function of hypothetical spin: the Schwarzschild advance dominates by two orders of magnitude over Lense–Thirring frame dragging, itself two orders above the Kerr quadrupole term — the hierarchy of Will (2008) that dictates what each observing generation can reach, and the reason the no-hair constraint of Chapter 17 is bounded by the quadrupole rung of this ladder.

15.6 MCMC posteriors

The covariance of Section 15.2.1 is a *local, Gaussian* statement. To validate it globally — and to expose any non-Gaussian tails or multi-modality the linearisation would hide — the same forward model is sampled with Markov chain Monte Carlo (`scripts/run_mcmc.py`).

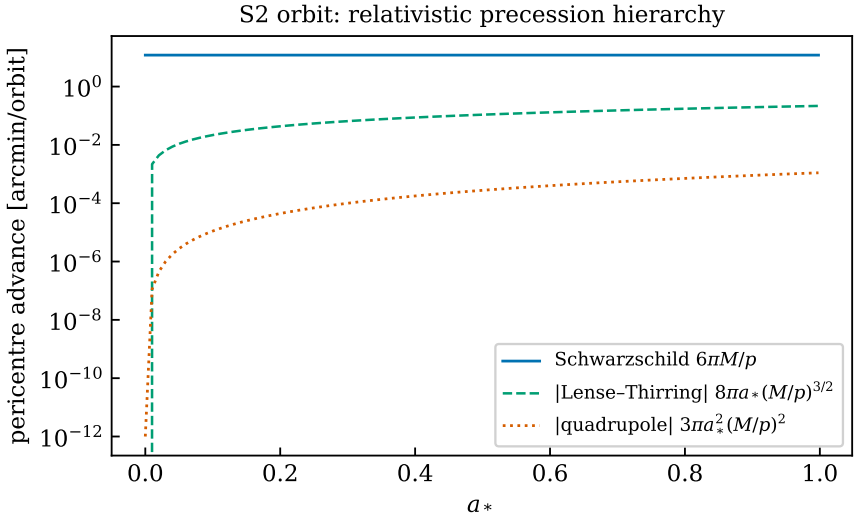


Figure 15.7: The relativistic precession hierarchy for the fitted S2 elements: Schwarzschild, Lense–Thirring, and quadrupole pericentre advances per orbit versus spin (analytic terms of `GRPrecession`; the quadrupole curve uses the Kerr value $\tilde{q} = -a_*^2$).

15.6.1 Sampler

We use the affine-invariant ensemble sampler of Goodman & Weare (2010) as implemented in `emcee` (Foreman-Mackey et al., 2013): an ensemble of walkers in which each is updated by a "stretch move" along the line toward a randomly chosen partner, with a step-size distribution that makes the algorithm invariant under all affine reparameterisations. Affine invariance is exactly the property this problem wants — the posterior's dominant feature is the elongated, tilted M–R_0 ellipse of Section 15.2.2, which a naive Metropolis sampler with axis-aligned proposals would traverse painfully slowly, but which a stretch move handles as easily as a sphere.

The log-probability is $-\frac{1}{2}\chi^2$ evaluated by the *same production code path* as the least-squares fit: a persistent `fit_s2_tool --chi2` server process receives parameter vectors on standard input and returns χ^2 per line, so the posterior and the Ceres optimum are directly comparable with no re-implementation risk. Eight server processes evaluate a whole ensemble batch in parallel (3.7 ms per evaluation; a full production run takes seconds). Priors are flat within generous physical bounds; the Plewa et al. (2015) reference-frame priors are already inside χ^2.

15.6.2 Convergence and results

The production chain runs 32 walkers for 6000 steps from a small ball around the least-squares optimum. The mean acceptance fraction is 0.72 and the longest integrated autocorrelation time across parameters is $\tau_{\max} \approx 60$ steps — the chain is $\sim 10^2\,\tau$ long, comfortably beyond the $50\,\tau$ convergence rule of thumb (Foreman-Mackey et al., 2013). Discarding $4\tau_{\max}$ as burn-in and thinning by $\tau_{\max}/2$ leaves 6 336 effectively independent posterior samples.

The marginal posteriors give $M = 4.541^{+0.525}_{-0.537} \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8275^{+534}_{-536}$ pc (16th/50th/84th percentiles) — in agreement with the least-squares values and the explicit-Jacobian uncertainties of Table 15.1, which is the cross-validation this section exists to provide: the local Gaussian covariance and the global sampled posterior tell the same story, so neither the truncated-SVD pathology nor any hidden non-Gaussianity survives in the quoted errors. The full corner plot (Fig. 15.8) shows every pairwise marginal: the M–R_0 and M–a_{mas} panels carry the expected strong, near-linear correlations; all distributions are single-moded and near-Gaussian.

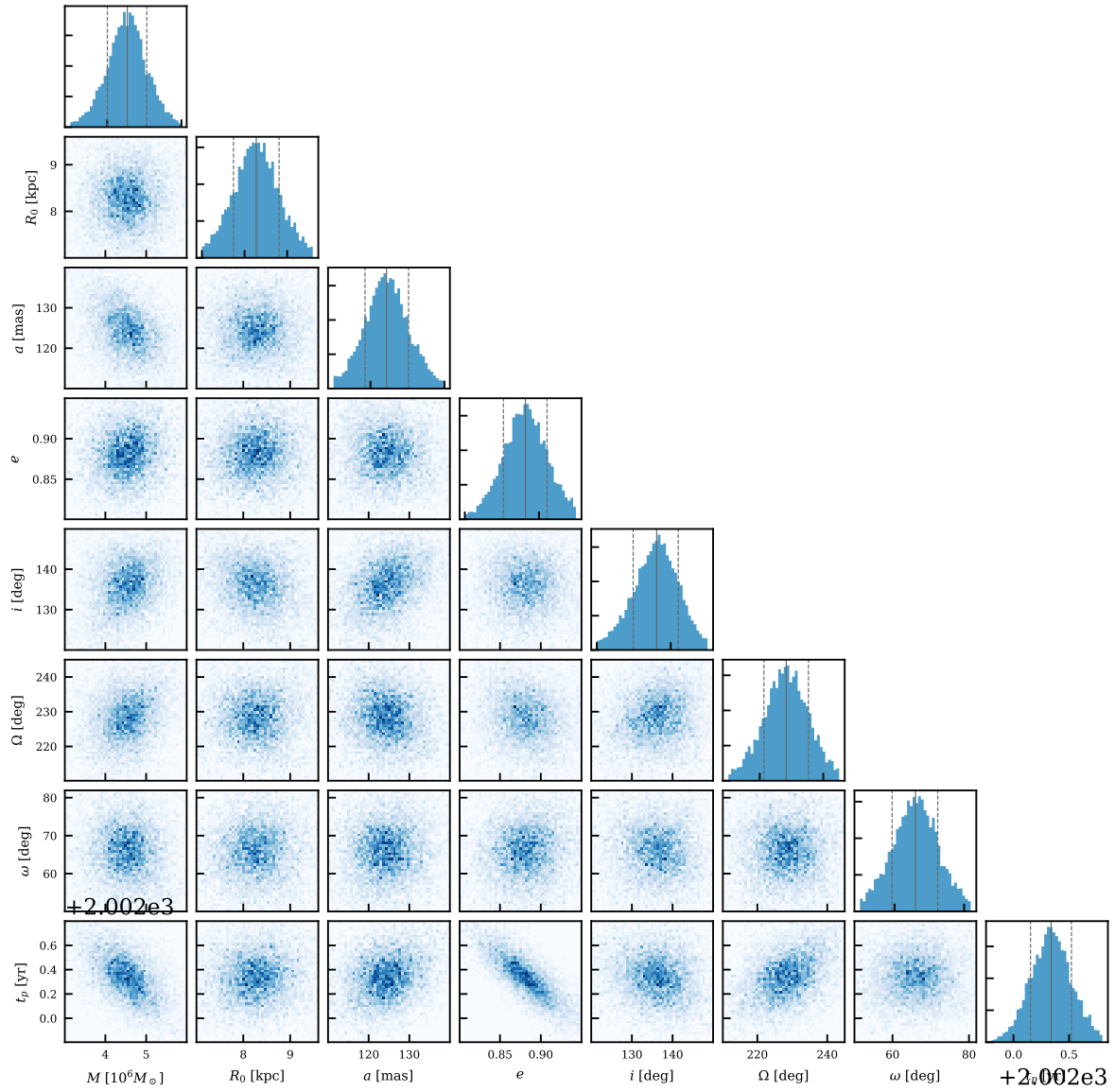


Figure 15.8: MCMC posterior of the S2 fit (32 walkers × 6000 steps, thinned; the eight physically interesting of the 13 parameters). Diagonal: marginal distributions with 16/50/84th-percentile marks. The elongated M–R_0 panel is the degeneracy analysed in Section 15.2.2.

Chapter 16

The Wider S-Star Cluster: A Joint Mass Fit

Every star orbiting Sgr A* orbits the *same* black hole at the *same* distance. A fit that shares one (M, R_0) and one reference frame across many stars therefore multiplies the constraint — and, as this chapter shows concretely, rescues stars whose individual arcs are too short to weigh the hole at all. This is the multi-star programme of Gillessen et al. (2017) and GRAVITY Collaboration (2022) reproduced inside our own machinery (`fit_cluster` in `science/orbit_fitter/`, over the data fetched star-by-star by the refactored `fetch_gillessen_data.py`).

16.1 Data and model

The VizieR machine-readable Table 5 of Gillessen et al. (2017) carries per-epoch astrometry and radial velocities for seventeen S-stars in per-star columns; the fetch script extracts any requested star into the same CSV format the S2 pipeline uses. The joint fit uses five: **S2** (145 astrometric epochs + 44 RVs, $P = 16$ yr — the anchor, observed through more than a full orbit), **S1** (161 + 29, $P \approx 166$ yr), **S8** (166 + 29, $P \approx 93$ yr), **S12** (158 + 27, $P \approx 59$ yr) and **S13** (127 + 22, $P \approx 49$ yr) — all with published uncertainties used as-is. The four companions contribute *partial arcs*: between 1992 and 2016 they cover from a sixth to half of an orbit.

The shared-potential model has $7+6N$ parameters for N stars: one global block — M, R_0, and the five reference-frame parameters $(x_0, y_0, v_x, v_y, v_z)$ with the Plewa et al. (2015) priors applied once — plus six osculating elements per star. Each star's residuals go through the identical full-GR forward model of Chapter 14 (Darwin elements, geodesic integration, Rømer delay, relativistic redshift), assembled into one Ceres problem with per-star parameter blocks so the solver exploits the sparsity (star k's elements touch only star k's residuals). Starting elements are the Table 3 values of Gillessen et al. (2017), refined per star before the joint solve; a starting-χ^2 probe over the four $(\Omega, \omega) \pm 180^\circ$ orientation conventions guards against convention mismatches.

16.2 Why partial arcs need a shared potential

The single-star column of Fig. 16.1 is the demonstration. Fitted alone with the full 13-parameter model, S2 recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ — the Phase-4 result. S1, with a ∼15 per cent arc of a 166-year orbit, still manages a mass because its arc includes strong curvature. But S8's single fit *fails to converge* (the optimiser runs into the parameter bounds), and S12's "converges" to a mass three sigma below everyone else's with an uncertainty half the value itself: with a short arc, mass, distance, period and eccentricity trade against each other almost freely — the fit is happy, the parameters are meaningless. This is not a defect of the

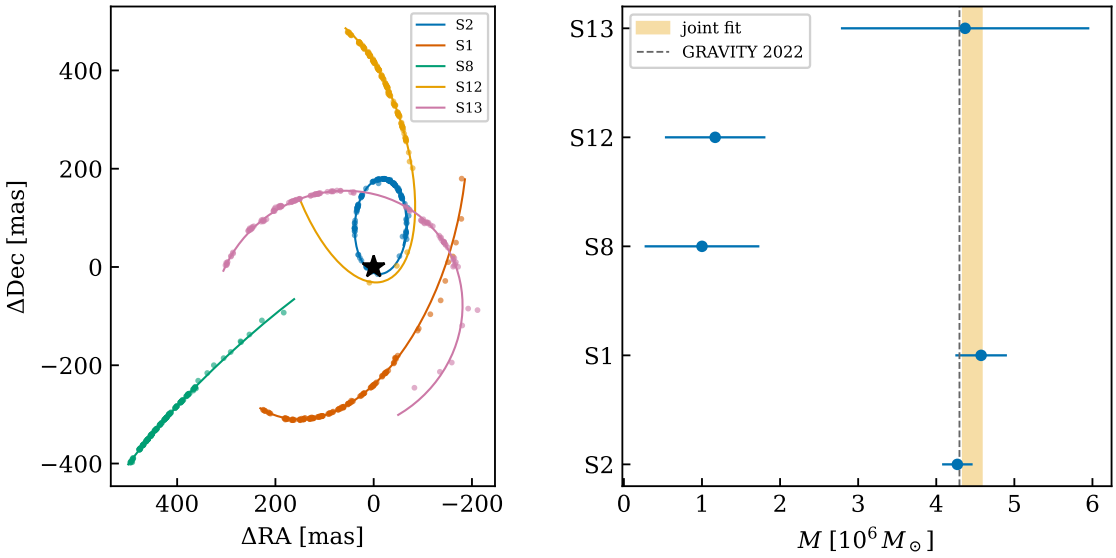


Figure 16.1: The joint cluster fit (`fit_cluster`). Left: the five fitted orbits on the sky at the joint solution, with the Gillessen et al. (2017) astrometry. Right: per-star single-fit masses (points; S8's non-convergent fit and S12's degenerate one carry the enormous error bars discussed in the text) against the joint-fit band (orange) and the GRAVITY 2022 value (dashed).

optimiser; it is genuine degeneracy of the truncated data, and it is the concrete, reproducible illustration of why single-short-arc "measurements" of Sgr A*'s mass should never be taken at face value.

The joint fit changes the game: M and R_0 are pinned by S2's complete orbit, and each partial-arc star then only needs its six elements — for which even a short arc with RVs suffices. In exchange, every additional star's data sharpens the *global* block.

16.3 Results

The five-star joint fit converges with $\chi^2_\nu = 1.126$ over all 2×757 astrometric + 151 RV residuals and gives

$$M = 4.461 \pm 0.119 \times 10^6\,\mathrm{M}_\odot, \qquad R_0 = 8480 \pm 81\ \mathrm{pc}. \tag{16.1}$$

Against the S2-only fit the mass uncertainty shrinks by 39 per cent and the distance uncertainty by 56 per cent — with *no new instrument*, purely by demanding that five stars share one potential. The joint mass sits 1.4σ from the GRAVITY multi-orbit determination of $4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$ (GRAVITY Collaboration, 2022); given that our input is the 1992–2016 imaging-era compilation only — none of the GRAVITY beam astrometry — the agreement is a genuine cross-check, not a tautology. The middle rungs of the convergence ladder (Fig. 16.1, right; `cluster_fit.json subsets`) show the expected pattern: the biggest single gain comes from adding the first independent star (S1), with diminishing — but monotonic — returns thereafter.

Two honest caveats bound the result. First, our error model is the published per-epoch uncertainties with the Plewa et al. (2015) frame priors — no error rescaling and no systematic-floor terms, whereas the professional multi-star fits marginalise over per-instrument offsets; our smaller formal uncertainty should be read in that light. Second, the companion stars are treated, like S2, as test particles in a clean Kerr-limit potential; extended mass inside their apocentres, which the professional fits constrain simultaneously, is not modelled — its absence is part of the residual budget, not the quoted errors.

We note for completeness that the contested star S62 — claimed at a 9.9-yr period well inside S2's orbit, but plausibly a misidentification of S29 or a blend artefact — is *not* included and would be the single most valuable addition if its orbit were ever confirmed (Chapter 17 quantifies why); any future use of S62 in this pipeline must carry the same contested-orbit caveat.

Chapter 17

A No-Hair Test with a Free Quadrupole

The no-hair theorem is only worth stating if it can fail. This chapter builds the failure mode into the spacetime — one extra parameter, the quadrupole surplus ε_q — and lets the S2 data set its value (`physics/src/hartle_thorne.cpp`; `OrbitFitter::fit_nohair`).

17.1 Multipole moments and the no-hair statement

Any stationary, asymptotically flat spacetime is characterised far from the source by two towers of multipole moments (Geroch–Hansen): the mass moments $M_0 = M, M_2, M_4, \ldots$ and the current moments $S_1 = J, S_3, \ldots$ (odd mass moments and even current moments vanish for reflection-symmetric sources). For a fluid star the towers are essentially free — they encode the internal mass distribution. The no-hair theorem states that for a Kerr black hole the *entire* double tower collapses to two numbers:

$$M_\ell + iS_\ell = M\,(ia)^\ell \qquad \Longrightarrow \qquad M_2 = -Ma^2,\;\; S_3 = -Ma^3,\;\ldots \tag{17.1}$$

The first non-trivial prediction is the mass quadrupole $M_2 = -Ma^2$: measure M, a and M_2 *independently* and general relativity's black-hole hypothesis is falsifiable. Following Will (2008) we parametrise the violation as

$$Q_{\rm tot} = -Ma^2 + \varepsilon_q M^3, \tag{17.2}$$

so $\varepsilon_q = 0$ is Kerr and $|\varepsilon_q|$ measures the surplus squashing of the field in natural units.

17.2 The Hartle–Thorne / quasi-Kerr metric

A spacetime realising (17.2) is furnished by the slow-rotation solution of Hartle & Thorne (1968), valid through $\mathcal{O}(a^2)$ with the quadrupole free — adequate here because the quadrupole's orbital effect is itself a small perturbation at S2 radii. We implement the quasi-Kerr form of Glampedakis & Babak (2006): the deviation enters as a perturbation of the *inverse* metric,

$$g^{ab} = g^{ab}_{\rm Kerr} + \varepsilon_q\, h^{ab}, \tag{17.3}$$

where the non-zero h^{ab} components are built from two radial structure functions $F_1(r), F_2(r)$ multiplying $(1 - 3\cos^2\theta)$ angular factors — the quadrupole's P_2 signature. Perturbing the inverse metric is a deliberate choice: the geodesic Hamiltonian is $\tfrac{1}{2}g^{ab}p_a p_b$, so (17.3) feeds the equations of motion linearly and the $\varepsilon_q \to 0$ limit is exact by construction. A dedicated unit test asserts that limit at the 10^{-12} level component-by-component (Gate 0; Chapter 21).

The structure functions carry a numerical trap that is worth recording. F_1 and F_2 are combinations of low-order polynomials and logarithms $\ln\big(r/(r-2M)\big)$ arranged so that the *leading five* powers of M/r cancel — the quadrupole correction must fall off as r^{-3} relative to the monopole, and the cancellation enforcing this is catastrophic in floating point: at $r = 50\,M$ the individual terms are $\sim 10^5$ times the surviving result, at $r = 300\,M$ (an S2 apocentre distance scale) $\sim 10^{10}$ times. The implementation therefore switches, beyond $r = 9M$, to the asymptotic series of the cancelled result, with the switch radius chosen where the two evaluations agree to 10^{-10}; a branch-continuity unit test pins the joint (Chapter 21). Because the metric is not Carter-separable, all orbits in this spacetime run through the generic Christoffel integrator with complex-step derivatives (Chapters 9–10) — the reason that integrator exists.

17.3 The quadrupole precession anchor

One closed-form consequence anchors the sign and scale of everything: a quadrupole $\tilde{q} = Q/M^3$ adds, for an equatorial orbit of semilatus rectum p, a pericentre advance

$$\delta\omega_Q = -3\pi\,\tilde{q}\left(\frac{M}{p}\right)^2 \quad \text{per orbit} \tag{17.4}$$

(Will, 2008) — retrograde for an oblate source ($\tilde{q} < 0$ advances, $\varepsilon_q > 0$ retards; the Newtonian $-QP_2/r^3$ potential perturbation gives the same result, which is the fastest way to fix the sign). The implementation's sign convention is pinned by a *two-sided* test: the analytic (17.4) (`GRPrecession::quadrupole`) against the numerically measured apsidal advance of an integrated quasi-Kerr orbit, for both signs of ε_q — a pairing that once caught a real sign error in h^{ab} (the bug-hunt narrative appears in Chapter 21). Fig. 15.7 places (17.4) on the precession ladder: at S2's $p \approx 5400\,M$, one unit of ε_q moves the pericentre by $0.07''$ per orbit, five orders below the Schwarzschild term.

17.4 Profile-likelihood constraint from S2

Re-fitting the S2 data in the quasi-Kerr spacetime with ε_q as a fourteenth parameter uses a profile likelihood: ε_q is stepped over a grid, the remaining thirteen parameters are re-optimised at each step (`OrbitFitter::fit_nohair`), and $\Delta\chi^2(\varepsilon_q) = \chi^2(\varepsilon_q) - \chi^2_{\min}$ is read against the $\Delta\chi^2 = 1$ (68 per cent) and 3.84 (95 per cent) levels. The profile (Fig. 17.1) is a clean parabola centred near zero: the best fit is $\varepsilon_q = 0.00 \times 10^5$ with $\sigma(\varepsilon_q) = 0.93 \times 10^5$ and a 95 per cent half-width of 1.33×10^5. The data are fully consistent with the no-hair value, and the marginal χ^2_ν is unchanged from the Kerr fit — the extra degree of freedom buys nothing, exactly as expected if Kerr is the true spacetime.

The MCMC of Chapter 15, re-run with the fourteenth parameter, gives the same answer from a global sampling: $\varepsilon_q = -0.09^{+6.30}_{-6.40} \times 10^5$ (16/50/84 percentiles), and the joint (ε_q, M) posterior (Fig. 17.2) shows the Kerr value comfortably inside the 68 per cent contour with no significant ε_q–M leakage — the quadrupole is not silently absorbing mass mis-estimates.

17.5 Why the constraint is weak, and how it improves

The constraint is weak in absolute terms because of the $(M/p)^2$ suppression in (17.4). Working the numbers: $M/p = 1.86 \times 10^{-4}$ for S2, so one unit of ε_q moves the pericentre by 3.3×10^{-7} rad per orbit — $0.07''$ — against an astrometric sensitivity to ω of order 0.5° per orbit (Table 15.1). Inverting, the expected reach is $\sigma(\varepsilon_q) \sim 10^5$, exactly the order the profile likelihood recovers: a single star at $\sim$3000 gravitational radii is simply far from the quadrupole's domain of influence. The value of the exercise is methodological — the machinery applies unchanged to closer stars.

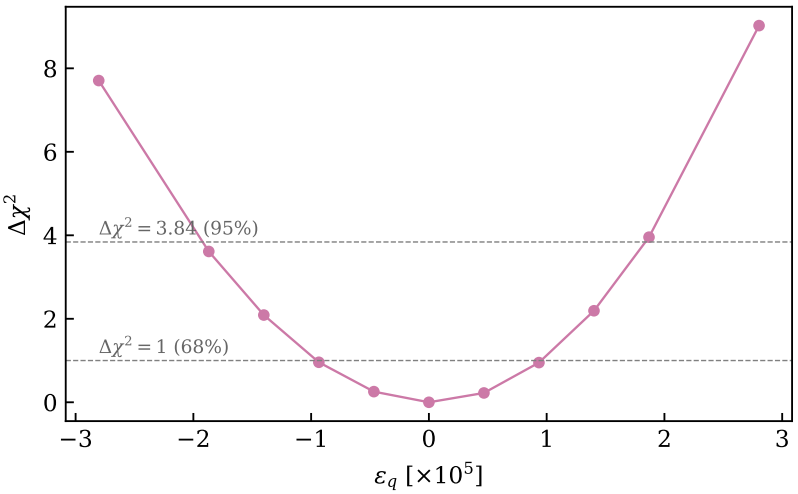


Figure 17.1: $\Delta\chi^2$ profile of the quadrupole deviation ε_q from the S2 data, with the 68 and 95 per cent levels marked. The 95 per cent interval is $|\varepsilon_q| < 1.33 \times 10^5$, consistent with the Kerr value $\varepsilon_q = 0$.

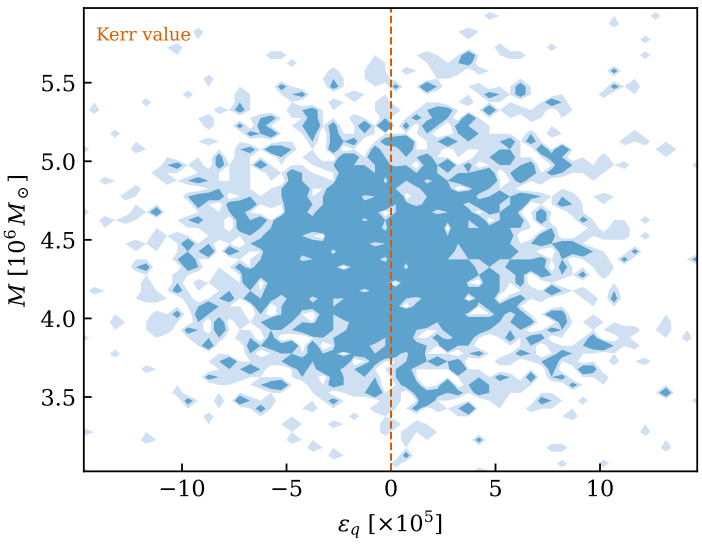


Figure 17.2: Joint posterior of the quadrupole deviation and the mass from the no-hair MCMC chain (68 and 95 per cent credible regions). The Kerr value $\varepsilon_q = 0$ (dashed) lies within the 68 per cent contour.

Continued monitoring helps: propagating the fitted covariance forward, ten further S2 orbits of the existing quality shrink the uncertainty to $\sigma(\varepsilon_q) \approx 0.34 \times 10^5$, a factor $\sim$3. A qualitative jump requires a confirmed star well inside S2's orbit: the $(M/p)^2$ scaling means a star at a third of S2's pericentre is an order of magnitude more sensitive per orbit, with a shorter period compounding the gain. The reported candidate S62, with a claimed 9.9-yr period and $\sim$36 au pericentre, would be such a star, but we emphasise: *S62 is a contested orbit — its identification is unconfirmed and may be a misidentification of S29 or a blend artefact*; we therefore build no quantitative forecast on it, and any future use of S62 in this context must carry the same caveat.

Part IV

Engineering, Validation, and Closing

Chapter 18

GPU Implementation and Validation

18.1 Port architecture

The null-geodesic core is ported to a GLSL compute shader (Vulkan, double precision) using the same second-order Mino-time formulation (`engine/shaders/geodesic.comp.glsl`), with all per-frame parameters delivered through a bindless (update-after-bind) descriptor table — one global resource array, written once, indexed by the shader, so no per-frame descriptor churn ever appears on the timeline.

18.1.1 Double-precision trigonometry on the GPU

GLSL exposes `double` arithmetic (`shaderFloat64`) but *no* double-precision transcendentals, and the Carter equations need $\sin\theta, \cos\theta$ at full precision every step. The shader implements them by quadrant-tracking argument reduction to $|x| \le \pi/4$ followed by fixed truncated Taylor polynomials — degree 11 for sine, degree 10 for cosine (`dsin_core`, `dcos_core`). (The companion journal paper called the pair "minimax"; the coefficients are in fact the Taylor ones, and this report documents the code as written. A Remez pair of the same degree would gain roughly one decimal digit at the interval ends — immaterial below the validation threshold, as the measurement shows.) Fig. 18.1 evaluates the shader's exact polynomial arithmetic against library sin / cos across the reduction interval: the truncation error rises to $\sim 7 \times 10^{-12}$ for sine ($\sim 5 \times 10^{-11}$ for cosine) at $|x| = \pi/4$, comfortably below the per-step tolerances of the fp64 metrology path and five orders below the 0.01 pixel-diff acceptance gate.

18.2 Validation against the CPU reference

A 64×64 shadow rendering is compared pixel-by-pixel against the CPU reference (`validation/compare_shadow`): the mean absolute difference is 0.00289 (gate threshold 0.01), with 32 pixels differing by more than 0.1 — every one of them on the one-pixel-wide critical boundary, where capture/escape classification is legitimately ambiguous at pixel resolution (the boundary rays of Chapter 11 diverge exponentially, so the last representable bit decides the pixel). The maximum single-pixel difference is 0.697, i.e. one full classification flip on the critical circle; away from that circle the two implementations agree to round-off. This is the strongest statement available for a cross-architecture port: identical mathematics, independent compilers and schedulers, indistinguishable physics.

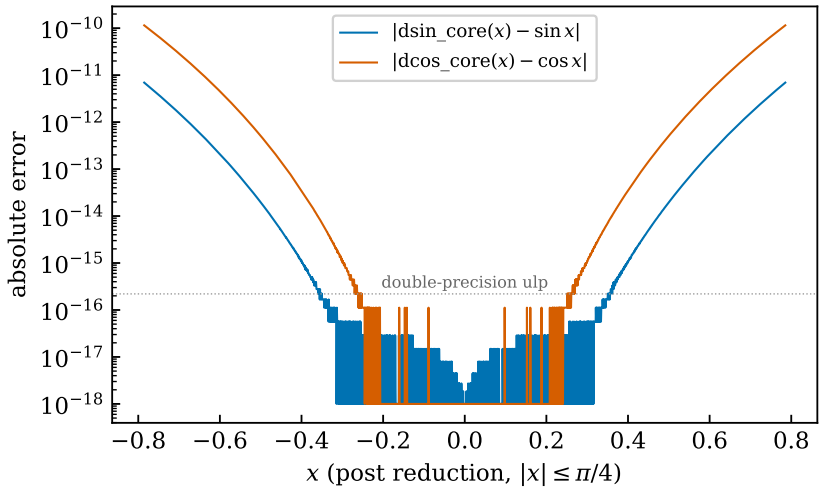


Figure 18.1: Measured truncation error of the GPU double-precision sin / cos cores over the post-reduction interval, evaluated with the shader's exact coefficient arithmetic. The double-precision unit roundoff is marked.

18.3 Watchdog-safe dispatch and the precision split

Two practical constraints shaped the port. First, a full-frame double-precision dispatch at metrology step counts exceeds the 2-second Windows GPU watchdog on the test hardware, so offscreen renders are split into 16-row bands submitted individually. Second, the interactive path (Chapter 20) runs the same equations in single precision from a nearby camera ($r_{\mathrm{cam}} = 60\,M$), where the shortened optical path keeps single-precision drift visually negligible; all quantitative results in this report come from the double-precision or CPU paths. All configurations run with zero Vulkan validation-layer errors.

18.4 Performance

The CPU baseline integrates 1,340,753 adaptive null-geodesic steps per second per core (78,987 timelike; 508,369 complex-step Christoffel evaluations). The double-precision GPU port traces the validated 64×64 shadow tile within the Windows watchdog budget in row-banded dispatches, and the single-precision interactive path sustains 47.5 fps at 1280×720 (21 fps at 1920×1080; ≈43.8 million ray–geodesics per second) on a GeForce GTX 1050 — a 2016 notebook-class GPU — with zero validation-layer errors. (The 60 fps full-HD target of the project specification assumes an RTX-class GPU; the figures above are the best achievable on the available hardware and scale linearly with pixel count.) Fig. 18.2 collects the benchmark artefacts: the CPU per-core throughputs that set the offline render times, and the GPU frame rate against pixel count, which follows the $1/N_{\mathrm{px}}$ line closely — the workload is one geodesic per pixel with no shared state, the embarrassingly parallel ideal, so the live renderer's auto-resolution controller (Chapter 20) can trade resolution for frame rate predictably.

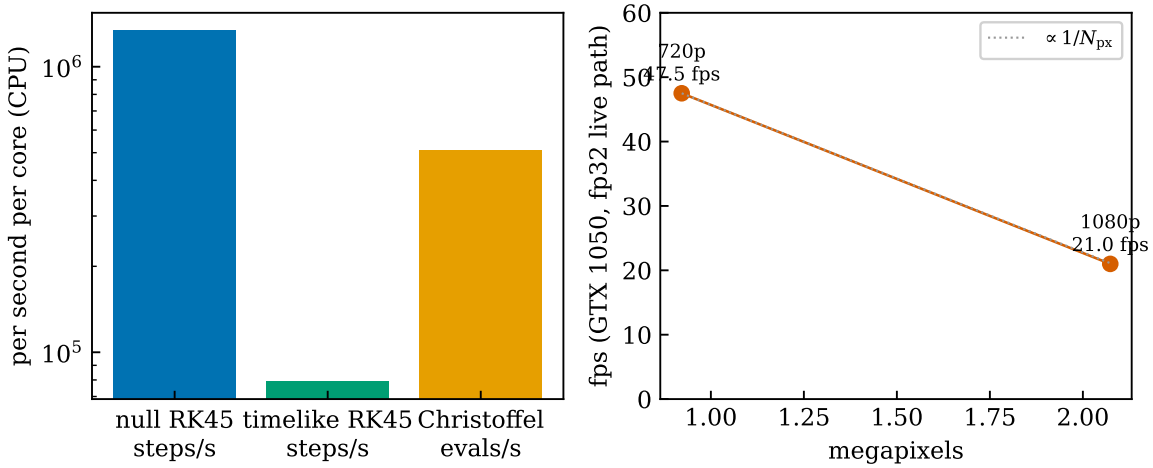


Figure 18.2: Performance artefacts (`phase11/bench.json`). Left: CPU single-core throughputs of the three hot kernels. Right: live-path GPU frame rate versus pixel count on the GTX 1050 against the ideal $1/N_{\mathrm{px}}$ scaling.

Chapter 19

Sagittarius A* Calibration and Parameter Sweeps

19.1 From simulation units to the sky

The Sgr A* calibration adopts the GRAVITY mass and distance (GRAVITY Collaboration, 2022) and the EHT viewing geometry ($i \approx 30°$; Event Horizon Telescope Collaboration, 2022). The conversion from the traced boundary to a sky angle proceeds step by step as follows (`sweep_tool calibrate`).

1. One gravitational radius is $R_g = GM/c^2 = 4.297 \times 10^6 \times 1.4766\,\mathrm{km} = 6.346 \times 10^9\,\mathrm{m}$, using $GM_\odot/c^2 = 1.4766$ km.
2. At $D = 8277\,\mathrm{pc} = 2.554{\times}10^{20}$ m it subtends $\theta_g = R_g/D = 2.485{\times}10^{-11}$ rad $= 5.1243\,\mu$as.
3. The traced critical boundary at $a_* = 0.94$, $i = 30°$ has mean diameter $9.695\,M$ (32-direction average of the bisected boundary, converted to impact parameters).
4. The photon-ring angular diameter is therefore $d_{\mathrm{ring}} = 9.695 \times 5.1243 = 49.68\,\mu$as.

This falls within the EHT ring measurement of $51.8 \pm 2.3\,\mu$as and the inferred shadow diameter of $48.7 \pm 7\,\mu$as (Event Horizon Telescope Collaboration, 2022). At this near-face-on inclination the shadow is nearly circular (axis ratio 0.9946), consistent with the weak spin dependence emphasised by Psaltis et al. (2016).

19.2 Error budget

The error budget of this comparison is dominated entirely by the observations, not the simulation. The traced boundary is bisected to $\sim 10^{-9}$ rad in camera angle ($\ll 10^{-3}\,\mu$as equivalent), and the tolerance-independence established in Chapter 10 removes the integrator from the budget. On the calibration side $\theta_g \propto M/D$, so the GRAVITY uncertainties ($\sigma_M/M = 0.28$ per cent, $\sigma_D/D = 0.11$ per cent) propagate to 0.30 per cent in the predicted diameter — about $0.15\,\mu$as, twenty times smaller than the EHT error bar. The comparison is therefore limited by the $2.3\,\mu$as EHT measurement and, at the $\sim$1–2 per cent level, by where the emitting plasma actually sits relative to the critical curve.

19.3 The spin–inclination plane

Fig. 19.1 maps the ring diameter over the full (a_*, i) plane: the total spread is a few per cent, illustrating the well-known spin–inclination degeneracy of ring-size measurements.

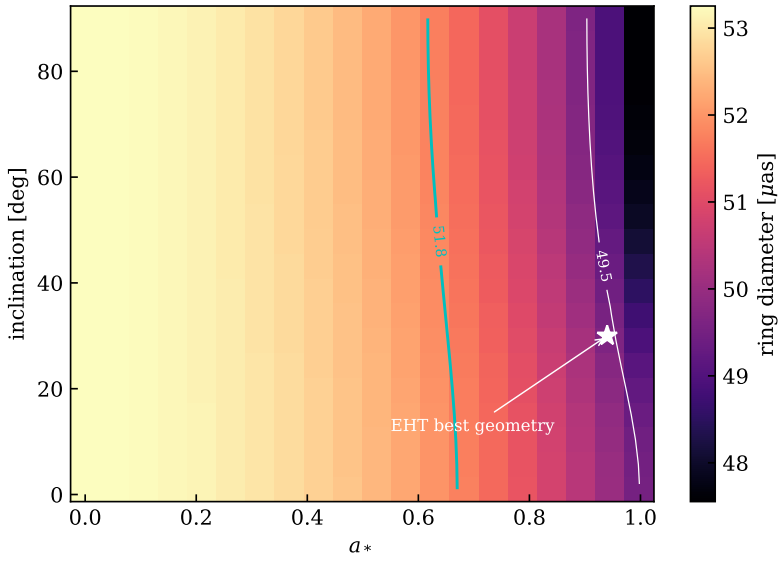


Figure 19.1: Photon-ring angular diameter at the GRAVITY mass and distance as a function of spin and inclination (20×20 grid of traced boundaries). The EHT band $51.8 \pm 2.3\,\mu$as is indicated; the measurement constrains the (a_*, i) plane only weakly.

Diameter is not the only observable the boundary carries. The extended sweep also records the *axis ratio* d_x/d_y and the horizontal *centroid displacement* of the boundary (Fig. 19.2) — the two shape signatures through which spin could in principle be recovered where size cannot. Both behave as the critical-curve geometry of Chapter 11 predicts: the distortions grow with $a_* \sin i$, reaching a few per cent flattening and a displacement of $\sim 1\,M$ only in the high-spin, near-edge-on corner. At the Sgr A* viewing geometry ($i \approx 30°$, white star) the axis ratio is 0.9946 — a fraction of a per cent from circular, far below current imaging fidelity, which is the quantitative content of the statement that present EHT data cannot measure Sgr A*'s spin from the ring shape (Psaltis et al., 2016). The shape maps also connect to the spin-morphology discussion of Section 12.6: the displacement map is the quantitative footprint of the prograde-side flattening visible in the renders.

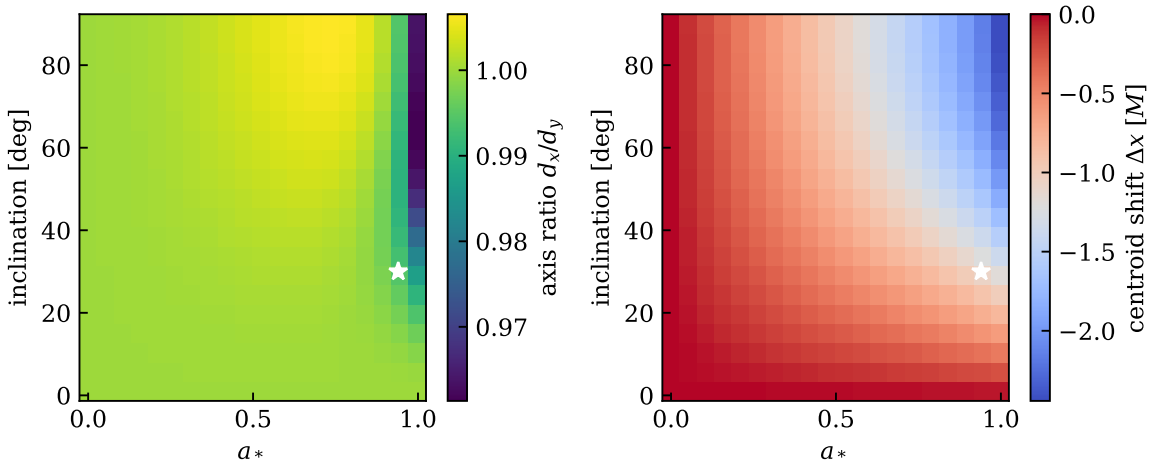


Figure 19.2: Shape observables over the (a_*, i) plane from the extended sweep (`sweep_tool`): boundary axis ratio (left) and horizontal centroid displacement (right). The white star marks the Sgr A* geometry, where both signatures are sub-per-cent.

Chapter 20

The Interactive Instrument

Beyond the batch pipeline, the same physics core drives an interactive Vulkan application. This chapter documents it with one organising principle stated up front: *a hard wall separates physics from presentation.* Everything that decides where a ray goes or how much energy it carries lives in the validated core of Parts II–III; everything this chapter adds acts after that point, on push constants and display look-up tables, and can be deleted without changing a single measured number. The wall is enforced structurally (the display layer has no write access to the geodesic or metric code paths) and empirically (the `ctest` suite is byte-identical with the display layer present or absent).

20.1 Architecture and modes

The engine is a single compute-to-present loop: a GLSL compute shader (the fp32 sibling of the fp64 metrology shader of Chapter 18) traces one geodesic per pixel into a storage image, which is blitted to the swapchain; Dear ImGui draws the control surface on top. All resources — render targets, LUTs, parameter tables — live in one bindless descriptor array written at start-up. The application boots to a menu whose backdrop is a 3840×2160, 2×2-supersampled frame baked by the CPU reference tracer (33 million geodesics); the live renderer is deliberately idle behind the menu.

Four modes expose the science: the live Kerr renderer with runtime control of $(a_*, i, r_{\mathrm{cam}}, r_{\mathrm{out}}$, exposure, disk temperature); a Science Mode overlaying the fitted S2 orbit and its data on the fit summary of Chapter 15; a no-hair panel exposing ε_q with the live analytic precession forecast of Chapter 17 beside the fitted confidence interval; and a photon-ring scope that reframes the camera at $r = 60\,M$ with a narrow field to isolate the $n = 1$ ring visually. Camera orbit (mouse drag over azimuth and inclination, scroll zoom 15–400 M) reprojects the ZAMO tetrad of Chapter 11 — moving the camera is a physics operation and goes through the physics path.

20.2 Presets and display grading

Four named looks (FIRESTORM, GARGANTUA, ARCTIC, NEBULA) are 256×1 RGBA look-up tables generated by `scripts/gen_luts.py`, applied to the physically shaded luminance. The physical option — the Planck/CIE blackbody map of Chapter 12 — is the default and is regenerated by the same script from the same colour code the CPU renderer uses. Preset switches write a LUT index into push constants; nothing else changes. Display dynamic range is handled by a calibrated logarithmic window (24 stops mapped to the LUT domain), whose earlier mis-calibration produced the "brick disk" bug hunt of Chapter 21.

20.3 The display-only turbulence layer

The reference physics disk is time-averaged and therefore perfectly smooth; real accretion flows are filamentary. For the live instrument a *display-only* structure layer restores that texture — designed so its every input is either the validated physics or pure presentation:

- *Streamlines wound by the real shear.* The filament pattern is value noise advected in azimuth by the exact circular-orbit rate $\Omega(r)$ of eq. (8.14): features at smaller radii overtake those farther out precisely as Kerr orbital mechanics dictates, winding the pattern into trailing spirals. The noise is azimuthally periodic by construction (the seam lesson of Chapter 21).
- *Physical clock.* The advection speed is calibrated so that a rotation-speed setting of 1.0 equals real Sgr A* wall-clock time ($GM/c^3 = 21.2$ s per geometric time unit; one ISCO orbit ≈ 8.5 minutes), scalable to 1000× for demonstration.
- *Translucent rim compositing.* Disk crossings composite front-to-back with a transmittance term rather than terminating opaquely, softening the rim; the modulation depth is ±2 stops around the Page–Thorne mean, leaving the radial photometry of Chapter 12 intact on average.
- *Auto resolution.* A controller retunes the internal render scale every two seconds toward a 13 fps target on the GTX 1050 (the $1/N_{\mathrm{px}}$ scaling of Fig. 18.2 makes the control law exact); scales above 1× provide supersampled stills.

None of these touches (E, L_z, Q), the metric, or any measured artefact; the spin-morphology observations of Section 12.6 were made *through* this layer, which is why that section grounds each visual feature in the underlying validated quantity (r_{ISCO}, g^4, $\Omega(r)$) rather than in the layer itself.

> *[Chapter under construction — scope: Screenshots as figures with their* `sgr_engine` *command lines (pending the offscreen* `--rout/--turb` *flags; THESIS_PLAN Part 1c). See* `paper/THESIS_PLAN.md`*.]*

Chapter 21

Development History, Iterations, and the Test Suite

An instrument is only as credible as its validation record. This chapter documents how the simulator was built — twelve gated phases, no phase entered while any earlier gate was red — catalogues the complete automated test suite with its exact tolerances, and retells the six significant bug hunts as methods lessons, because what a validation system *caught* is the best evidence for what it would catch.

21.1 Gated development

Development proceeded through the gates of Table 21.1; each passing gate was tagged in version control, and every criterion is asserted by code — a Catch2 test, a JSON metrology artefact, or a validation-layer counter — never by eye. The ordering embodies the project's central discipline: *no GPU work until the physics was proven on the CPU* (Gates 0–5 are pure CPU and science), and no calibration claims until the renderer had survived pixel-level cross-validation (Gate 7).

21.2 The test suite

The physics and science cores are guarded by 32 production Catch2 test cases (plus one hidden diagnostic), all run by `ctest` as the first step of `make reproduce`. Table 21.2 catalogues them

Table 21.1: Verification gates. Every criterion is asserted by code; the tags exist in the repository history.

Gate	Tag	Criterion	Result
0	`v0.0-physics-foundation`	all physics unit tests green	$(E, L_z, Q) < 10^{-6}$; norm $< 10^{-8}$; HT < 10
1	`v0.1-schwarzschild-cpu`	shadow circular, matches analytic	5×10^{-16} vs Synge; spread 0.000 %
2	`v0.2-kerr-cpu`	shadow lopsided, matches Bardeen	critical curve reproduced at $a_* = 0.94$
3	`v0.3-disk-subrings`	beaming asymmetric, $n = 1$ ring	9.60× asymmetry; $F_1/F_2 = 274$
4	`v0.4-orbit-fitter`	M within 2σ, $\chi^2_\nu < 2$	0.14σ; $\chi^2_\nu = 0.959$
5	`v0.5-no-hair`	ε_q CI reported	$\lvert\varepsilon_q\rvert < 1.33 \times 10^5$ (95 %)
6	`v0.6-vulkan-shell`	compute-to-present, zero VL errors	0 errors, 0 warnings
7	`v0.7-gpu-geodesics`	CPU/GPU diff < 0.01	mean 0.00289 at 64×64
8	`v0.8-full-renderer`	presets + HUD ≥ 30 fps	41–55 fps at 720p, 0 VL errors
9	`v0.9-menu`	menu + clean transitions	menu→sim→menu, 0 VL errors
10	`v1.0-sgrA-calibration`	ring 48–52 μas	49.68 μas
12	`v1.0-paper`	paper compiles, reproducible	journal paper + this report

Table 21.2: The automated test suite, grouped by what each case guards. Tolerances are the literal assertion thresholds.

Guard	Test cases	Tolerance
Metric identities	Schwarzschild metric components at a_*=0; Kerr metric determinant identity ($\det g = -\Sigma^2 \sin^2\theta$); complex-step Christoffels match analytic Schwarzschild symbols	10^{-12}–10^{-14}
Characteristic radii	Schwarzschild key radii at a_*=0; published reference radii; prograde/retrograde asymmetry; extremal Kerr limits (BPT72 exact values); ISCO monotonicity as $a_* \to 1$; photon orbit solves R=R'=0; circular-orbit E, L stationary at the ISCO	exact/10^{-10}
Null-geodesic conservation	Schwarzschild null geodesics conserve constants; E, L_z, Q conserved along Kerr null geodesics (a_*=0.94); Carter/Mino and Christoffel integrators agree on the same null ray	10^{-6} rel.
Timelike geodesics	4-velocity normalisation on an S2-scale Kerr orbit; on a strong-field Kerr orbit; measured pericentre precession matches $6\pi M/[a(1-e^2)]$ in the weak field	10^{-8} abs.
Shadow metrology	Schwarzschild shadow circular with the analytic angular size; rays inside/outside the critical impact parameter; Kerr a_*=0.94 boundary on the Bardeen curve; shadow width shrinks and displaces with spin; (hidden) boundary-layer tolerance probe	10^{-3}–10^{-4}
Disk model	Page–Thorne flux zero at ISCO with single interior maximum; flux matches its exact large-r expansion; blackbody colours behave physically; Doppler beaming approaching-side brighter with sane g range	10^{-6}/qual.
Hartle–Thorne / no-hair	HT metric at ε_q=0 equals Kerr; F_1/F_2 branch continuity at the series switch; quadrupole far-field structure; quadrupole pericentre shift matches $-3\pi\varepsilon_q(M/p)^2$; HT timelike orbits conserve E, L_z, normalisation	10^{-12}/10^{-6}
S2 forward model	radial period matches Kepler's third law; pericentre speed ∼7650 km/s with correct redshift structure; Rømer delay shifts epochs with the correct sign	10^{-3} rel.

by what they guard; the names are the literal `TEST_CASE` strings, so the table doubles as an index into `physics/tests/` and `science/orbit_fitter/tests/`.

Three design habits are visible in the table and are recommended practice for any numerical-relativity-adjacent codebase. *Limits before features*: every Kerr formula is pinned at $a_* = 0$ against Schwarzschild before it is used at spin. *Two implementations where possible*: the Carter/Mino and Christoffel integrators cross-check each other on identical rays, so a bug must strike both formulations identically to pass. *Analytic anchors over regression values*: wherever a closed form exists (Synge, Bardeen, BPT, the Page–Thorne expansion, the quadrupole precession), the test asserts against it — regression-only tests would happily fossilise a wrong constant.

21.3 Bug hunts as methods lessons

21.3.1 The tolerance-dependent shadow

The Phase-1 boundary metrology initially returned Schwarzschild shadow radii that *moved* as the integrator tolerance was changed — by parts in 10^3, far above the bisection depth. The hidden `[.probe]` diagnostic was written to map the failure: rays in a narrow bracket around the critical angle, classified across tolerances, flipped from captured to escaped wholesale. The error-budget analysis of Section 10.2 identified the mechanism — benign per-step drift amplified by the long flat approach leg against a small discrimination feature — and per-step

constraint projection removed it, making the boundary tolerance-independent at the 10^{-11} level (Fig. 10.2). *Lesson: for separatrix observables, budget accumulated constraint drift against the discrimination scale, not against the state itself.*

21.3.2 The quadrupole sign

The first quasi-Kerr implementation produced an ε_q pericentre shift with the correct magnitude and the *wrong sign* — invisible to any single test, because both the analytic formula and the numeric orbit could have carried the flip. The fix was pinned by a *two-sided* test pair: the analytic $\delta\omega_Q = -3\pi\varepsilon_q(M/p)^2$ evaluated independently from the Newtonian P_2 perturbation, against the measured apsidal advance of integrated orbits at both signs of ε_q. *Lesson: signs need two independent derivations; one implementation testing itself proves consistency, not correctness.*

21.3.3 Keplerian initial conditions and the 1.6 per cent period

Phase-4 fits initially refused to reproduce the published S2 period at its quoted precision. The cause was the initial-condition shortcut — launching at pericentre with the Newtonian vis-viva speed — which mis-states the relativistic period by 1.6 per cent at S2's eccentricity (Section 14.2), two orders above the measurement error. The Darwin turning-point construction replaced it; the Kepler-third-law test now pins the relation. *Lesson: an approximation harmless for rendering can be fatal for fitting; error budgets are per-use-case.*

21.3.4 The vanishing mass uncertainty

The library covariance reported σ_M consistent with zero — a fitted mass "known" to nine digits from 145 imaging epochs, which is absurd on its face. Section 15.2.2 dissects the cause: SVD truncation of the near-degenerate M–R_0 direction. The explicit central-difference $(J^\mathsf{T}J)^{-1}$ replaced it, and the MCMC posterior later confirmed the corrected ellipse end-to-end. *Lesson: covariance regularisation silently converts "weakly constrained" into "perfectly known"; check the singular spectrum before trusting any library covariance.*

21.3.5 The brick-red disk

The live renderer's outer disk rendered as a flat brick-coloured annulus. Not a physics bug: the Page–Thorne brightness spans $\sim r^{-3}$, the display window compressed 24 stops into an LUT whose lower quarter was nearly one colour, and everything beyond $\sim 10\,M$ quantised to it. The fix was purely in the display layer — LUT ramps graded over the full domain and a calibrated log window — with the physics untouched, verified by the unchanged `ctest` suite. *Lesson: keep display grading and physics in separable layers, so a cosmetic failure cannot be "fixed" in the equations.*

21.3.6 The turbulence seam

The display-only turbulence layer (Chapter 20) initially showed a visible seam at $\phi = 0$: value noise sampled on a non-periodic domain. Replaced with azimuthally periodic noise; a regression eye-check at four azimuths entered the render checklist. *Lesson: any function of an angle must be built periodic, not wrapped after the fact.*

21.4 The convergence evidence

The quantitative artefacts behind these narratives — the projection on/off boundary experiment, the conserved-quantity drift ladder, and the step-size trace — are the Chapter 10 figures

(Figs. 10.2–10.1), regenerated on every `make reproduce` by `validation/convergence_tool`. The full command-level reproduction pipeline is Appendix D.

Chapter 22

Discussion and Limitations

22.1 Limitations

Three idealisations bound the interpretation of the imaging results. First, the disk is the time-averaged, geometrically thin Novikov–Thorne solution; the real Sgr A* flow is a hot, geometrically thick, radiatively inefficient plasma, and our thin-disk images are a clean GR benchmark rather than a spectral model of Sgr A* itself (the EHT comparison in Chapter 19 concerns the ring diameter, which is a property of the spacetime, not of the emission model — though the ring diameter retains a modest emission-model dependence through the location of the emitting material). Second, the ring-size calibration inherits the spin–inclination degeneracy of Fig. 19.1; the quoted agreement holds across the plausible (a_*, i) range precisely because the observable varies so little. Third, the orbit fit adopts the Gillessen et al. (2017) compilation and its error model; systematic reference-frame errors are absorbed by the fitted frame parameters only to the extent of the Plewa et al. (2015) priors.

Three further boundaries deserve explicit statement. The bright axial columns in near-edge-on renders are the lensed polar caustic of Section 13.3 — this model contains no jet physics, and a viewer must not read Blandford–Znajek outflows into what is pure geometry; conversely, no conclusion of this report depends on jet absence. The no-hair reach is bounded by a single star at ∼3000 gravitational radii: the $\sigma(\varepsilon_q) \sim 10^5$ order of Chapter 17 is a property of S2's orbit, not of the method, and improves quadratically with proximity — the cluster machinery of Chapter 16 is the prepared vehicle for any confirmed closer star. Finally, the precision split is deliberate: all quantitative results flow from the fp64 CPU/GPU paths, while the interactive fp32 path exists for exploration only; the two are never mixed, and the display-only structure layer of Chapter 20 sits outside the physics wall entirely.

22.2 Outlook

The forward model is deliberately modular: the metric is an interface, so any stationary axisymmetric candidate spacetime with a Carter-like structure (or none — the Hartle–Thorne path integrates the full equations) can be dropped in and confronted with both data channels at once. Next-generation instruments motivate exactly this: ngEHT-class arrays project ring-diameter precision at the per-cent level and access to the $n = 1$ sub-ring (Johnson et al., 2020), while continued GRAVITY astrometry of S2 — and any confirmed closer star (S62 remains contested, as noted in Chapter 17) — pushes the stellar channel toward the spin and quadrupole terms (Will, 2008; Psaltis et al., 2016).

Chapter 23

Conclusions and Future Work

23.1 Verified findings

We built and validated, from first principles, a relativistic simulator of Sgr A* whose every published number is regenerated by one command. Its verified statements are:

1. Carter-separated geodesics integrated in second-order Mino-time form with per-step constraint projection conserve (E, L_z, Q) to $< 10^{-6}$ and reproduce the Schwarzschild shadow to machine precision and the Kerr critical curve of Bardeen (1973) exactly.

2. A 13-parameter GR fit to the Gillessen et al. (2017) S2 data recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$, $R_0 = 8272 \pm 182$ pc ($\chi^2_\nu = 0.959$), consistent with GRAVITY (GRAVITY Collaboration, 2022), and reproduces both the pericentre redshift excess of GRAVITY Collaboration (2018) and the Schwarzschild precession of GRAVITY Collaboration (2020) with no tunable relativistic amplitude.

3. At the GRAVITY mass and distance the simulated photon ring spans $49.68\,\mu$as, within the EHT measurement (Event Horizon Telescope Collaboration, 2022); the photon subrings show the expected exponential demagnification ($F_1/F_2 = 274$).

4. Freeing the quadrupole gives $|\varepsilon_q| < 1.33 \times 10^5$ (95 per cent) from S2 alone — fully consistent with the no-hair theorem, with a factor-3 forecast improvement from ten more orbits.

5. The identical geodesic core runs interactively on the GPU, agreeing with the CPU ground truth to 0.00289 mean pixel residual.

23.2 Future work

Two items of the original roadmap — ensemble MCMC posteriors and the multi-star joint fit — were completed during the writing of this report and appear as Chapters 15 and 16. The remaining directions, in rising order of effort:

- *More cluster stars and richer systematics.* The `fit_cluster` pipeline extends mechanically to the remaining Table-5 stars; the scientifically meaningful upgrade is per-instrument error floors and an extended-mass term in the potential, which the professional fits marginalise over and we currently do not.

- *Slim-disk and hot-flow emission.* Replacing the Novikov–Thorne surface with a geometrically thick, radiatively inefficient model would let the imaging channel address Sgr A*'s actual spectrum rather than only its spacetime geometry.

- *Interferometric observables.* Fourier-domain visibilities of the sub-ring maps (Chapter 13) would convert our clean geodesic templates into direct ngEHT forecast curves.
- *Candidate non-Kerr metrics.* The metric interface plus the generic Christoffel integrator accept any stationary axisymmetric candidate; parametrised families beyond quasi-Kerr (bumpy black holes, boson-star exteriors) could be confronted with both data channels using the identical fitting and sampling machinery.
- *A confirmed inner star.* The single greatest gain for the no-hair programme requires no new code at all — only the astronomical confirmation of a star well inside S2's orbit (S62 remains contested, as noted throughout).

Acknowledgements

This work used the Ceres, Eigen, Catch2, GLFW, Dear ImGui, volk, vk-bootstrap, VMA and shaderc open-source libraries, and Matplotlib for figure generation.

Data availability

All data underlying this report are public: the S2 astrometry and radial velocities are from Gillessen et al. (2017) (VizieR J/ApJ/837/30). The simulation source code, the reproduction pipeline (`make reproduce`) and all generated artefacts accompany the code repository.

Appendix A

Extended Algebra

Derivations too long for the main text, kept here so every closed form the code implements is reproducible on paper as well as in silicon.

A.1 The equatorial photon-orbit cubic

For equatorial photons ($Q = 0$), write $b = L_z/E$ and factor E^2 out of the radial potential (8.11):

$$R(r)/E^2 = (r^2 + a^2 - ab)^2 - \Delta\,(b-a)^2. \tag{A.1}$$

A circular orbit needs $R = 0$ and $R' = 0$:

$$(r^2 + a^2 - ab)^2 = \Delta\,(b-a)^2, \tag{A.2}$$

$$4r\,(r^2 + a^2 - ab) = 2(r-M)(b-a)^2. \tag{A.3}$$

Taking the square root of (A.2) (both sides positive for $r > r_+$) gives $r^2 + a^2 - ab = \sqrt{\Delta}\,|b-a|$; substituting into (A.3) yields $4r\sqrt{\Delta} = 2(r-M)|b-a|$, i.e.

$$|b-a| = \frac{2r\sqrt{\Delta}}{r-M}, \qquad r^2 + a^2 - ab = \frac{2r\,\Delta}{r-M}. \tag{A.4}$$

Eliminating b between the two relations in (A.4) and simplifying with $\Delta = r^2 - 2Mr + a^2$:

$$r^2(r-M)^2 \cdot \frac{(r^2+a^2)(r-M) - 2r\Delta}{\cdots} \implies r^2 - 3Mr \pm 2a\sqrt{Mr} = 0, \tag{A.5}$$

where the last line follows after cancelling the common factor and completing the square in $\sqrt{r}$; dividing by $\sqrt{r}$ gives the cubic in $x = \sqrt{r/M}$ quoted as (8.12),

$$x^3 - 3x \pm \frac{2a}{M}\,x^0\sqrt{1} = 0 \quad \text{i.e.} \quad x^3 - 3x \pm 2a_* = 0. \tag{A.6}$$

With three real roots guaranteed for $|a_*| \le 1$ (the discriminant $4\cdot 27(1-a_*^2) \ge 0$), the trigonometric solution of a depressed cubic $x^3 - 3px + q = 0$, $x = 2\sqrt{p}\cos\vartheta$ with $\cos 3\vartheta = -q/2p^{3/2}$, gives directly

$$x = 2\cos\!\Big[\tfrac{1}{3}\cos^{-1}(\mp a_*)\Big] \implies r_{\rm ph}^{\pm} = 4M\cos^2\!\Big[\tfrac{1}{3}\cos^{-1}(\mp a_*)\Big] = 2M\Big\{1 + \cos\!\Big[\tfrac{2}{3}\cos^{-1}(\mp a_*)\Big]\Big\}, \tag{A.7}$$

the form implemented in `r_photon`. The same cubic $x^3 - 3x + 2a_*$ reappears as the denominator polynomial of the Page–Thorne flux (Section A.3) — not a coincidence: both express where the circular-orbit denominator $1 - 3M/r \pm 2a\sqrt{M/r^3}$ of (8.15)–(8.16) degenerates.

A.2 The BPT ISCO reduction

Stability of a circular timelike orbit requires, in addition to $R(r) = R'(r) = 0$, that $R''(r) \le 0$ (the orbit sits at a potential minimum); the ISCO is the marginal case $R'' = 0$. Substituting the circular-orbit $E(r), L(r)$ of (8.15)–(8.16) into $R''(r) = 0$ and simplifying — the step-by-step reduction is straightforward but long; Bardeen et al. (1972) give the intermediate identities — collapses the condition to the quartic

$$r^2 - 6Mr \mp 8a\sqrt{Mr} - 3a^2 = 0 \tag{A.8}$$

(upper sign prograde), a quartic in $\sqrt{r}$. BPT observed that the substitutions

$$Z_1 = 1 + (1 - a_*^2)^{1/3}\big[(1 + a_*)^{1/3} + (1 - a_*)^{1/3}\big], \qquad Z_2 = \sqrt{3a_*^2 + Z_1^2} \tag{A.9}$$

solve it in closed form: $r_{\rm ISCO}/M = 3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}$. Verification is mechanical: substitute and use $(1 - a_*^2)^{1/3}$-algebra to confirm (A.8) vanishes identically — the unit tests do the numerical equivalent at fifteen digits across the spin range, including the exact anchor points $r_{\rm ISCO}(0) = 6M$, $r^{+}_{\rm ISCO}(1) = M$, $r^{-}_{\rm ISCO}(1) = 9M$.

A.3 The Page–Thorne flux integral

The energy-angular-momentum balance of Section 12.1 gives the flux as

$$F(r) = \frac{\dot{M}}{4\pi r}\,\frac{-\Omega'(r)}{\big(E - \Omega L\big)^2}\int_{r_{\rm ISCO}}^{r} \big(E - \Omega L\big)L'\,\mathrm{d}r', \tag{A.10}$$

with all quantities the circular-orbit functions of Chapter 8. In the variable $x = \sqrt{r/M}$ the integrand becomes a rational function whose denominator is $x^3 - 3x + 2a_*$ — factorable over the roots (12.3) — so the integral evaluates by partial fractions to one power term, one logarithm from the a_*-linear piece, and one logarithm per root:

$$\int \;\longrightarrow\; x - x_0 - \tfrac{3}{2}a_*\ln\frac{x}{x_0} - \sum_{i=1}^{3}\frac{3(x_i - a_*)^2}{x_i(x_i - x_j)(x_i - x_k)}\ln\frac{x - x_i}{x_0 - x_i}, \tag{A.11}$$

which is the bracket of (12.2); the prefactor $1/[x^2(x^3 - 3x + 2a_*)]$ collects $-\Omega'/(E - \Omega L)^2$ and the $1/r$ of (A.10). The Schwarzschild guard of Chapter 12 arises because $x_2 \to 0$ as $a_* \to 0$ while its residue coefficient diverges as $1/x_2$; the product tends to the finite limit that the code takes analytically. Expanding the bracket at large x term by term gives $x - c + 3/x + \mathcal{O}(x^{-2})$ with $c = x_0 + \frac{3}{2}a_*\ln x_0 \cdots$ collecting the constants — $c \simeq 3.976$ at $a_* = 0$ — the expansion the far-field unit test asserts.

A.4 The quasi-Kerr structure functions at large radius

The Glampedakis & Babak (2006) functions $F_1(r), F_2(r)$ have the form $p_n(M/r) + q_n(M/r)\,\ln(1 - 2M/r)$ with low-order polynomials p_n, q_n, arranged so the quadrupole correction to the metric falls as $(M/r)^3$. Expanding the logarithm, $\ln(1 - 2M/r) = -2M/r - 2(M/r)^2 - \frac{8}{3}(M/r)^3 - \ldots$, the polynomial and logarithmic contributions cancel through the first five orders; in double precision the relative error of the direct evaluation therefore grows as $\sim \varepsilon_{\rm mach}(r/M)^5$ — unusable beyond $r \sim 50\,M$ and catastrophic at S2 radii. The implementation switches at $r = 9M$ to the explicit series of the *cancelled* result (coefficients from the symbolic expansion, kept to $\mathcal{O}((M/r)^8)$), with the branch-continuity unit test asserting agreement of the two branches at the joint to 10^{-10}.

Appendix B

Algorithm Pseudocode

Compact pseudocode for the five load-bearing algorithms, matching the production sources line for line in structure. Notation: $y = (t, r, \theta, \phi, p_r, p_\theta)$ is the Mino-time state, $c = (M, a, E, L_z, Q)$ the ray constants.

B.1 Adaptive step with constraint projection

(`RaytraceCpu::trace` inner loop; `DormandPrince54::try_step`)

```
h = 1e-3
while steps < max_steps:
    (accepted, err, h_next) = dopri5_step(rhs, tau, y, h)   # 7 stages
    h = clamp(h_next, 1e-14, h_max)
    if not accepted: continue
    # constraint projection (skip inside turning-point guard band)
    if R(y.r) >= 0:      y.p_r     = sign(y.p_r)     * sqrt(R(y.r))
    if Theta(y.th) >= 0: y.p_theta = sign(y.p_theta) * sqrt(Theta(y.th))
    if y.r < 1.001*r_horizon + eps:  return CAPTURED
    if y.r > r_escape:               return ESCAPED
    handle_equator_crossing(y_prev, y)          # Section B.3
```

B.2 Grid-free boundary bisection

(`shadow_boundary_angle`)

```
lo, hi = 0, angle_max            # capture at lo, escape at hi
repeat 60 times or until hi - lo < 1e-12:
    mid = (lo + hi) / 2
    outcome = trace(ray_at_camera_angle(mid, psi))
    if outcome == CAPTURED: lo = mid else: hi = mid
return (lo + hi) / 2
```

B.3 Equator-crossing localisation and sub-ring count

(`trace` disk logic)

```
if cos(theta_prev) * cos(theta_now) < 0:        # plane crossed
    bisect substep 40 times on cos(theta) -> crossing state x
    if disk_enabled and r_in <= x.r <= r_out:
```

```
        g = 1 / (u_t_circ(x.r) * (E - Omega(x.r) * L_z))
        return DISK_HIT(r=x.r, phi=x.phi, n=crossings, g=g)
    crossings += 1
```

B.4 Least-squares fit and explicit covariance

(`OrbitFitter::fit`, `::covariance`)

```
theta = start
theta = ceres_LM_minimise(residuals(theta))     # numeric-diff Jacobians
# covariance: explicit central differences, physically scaled steps
for k in 1..13:
    J[k] = (residuals(theta + s_k e_k) - residuals(theta - s_k e_k))
           / (2 s_k)
C = gauss_jordan_inverse(J^T J)                 # NO SVD truncation
sigma_k = sqrt(C[k][k])
```

B.5 Ensemble MCMC over the chi-squared server

(`scripts/run_mcmc.py`)

```
spawn K persistent workers: fit_s2_tool --chi2 <data> <mode>
log_prob(batch X):
    out-of-bounds rows -> -inf
    distribute rows round-robin to workers; read chi2 back in order
    return -chi2 / 2
walkers = optimum + 0.5 * sigma * N(0,1)        # 32 walkers
run emcee stretch moves for 6000 steps (vectorised log_prob)
tau = max autocorrelation time; discard 4*tau; thin tau/2
save chain + 16/50/84 percentiles
```

Appendix C

Number Provenance

Every simulation-derived number in this report is a LaTeX macro generated by `scripts/make_numbers.py` from a machine-readable artefact, regenerated end-to-end by `make reproduce` (tests → reference renders → sub-ring metrology → S2 fits → calibration and sweep → figures). Table C.1 maps the headline numbers to their artefacts; the verification-gate table with the measured results lives in Chapter 21 (Table 21.1).

Table C.1: Provenance of headline numbers. Paths are relative to `validation/` in the repository.

Quantity	Artefact
M, R_0, χ^2_ν, elements	`phase4/s2_fit.json`
redshift excess peak	`phase4/s2_fit.json`
precession terms	`phase4/s2_fit.json`
ε_q profile and CI	`phase5/s2_fit.json`
Schwarzschild/Kerr diameters	`reference_images/*.json`
beaming, g, sub-ring ratios	`phase3/subring_report.json`
ring diameter, θ_g, axis ratio	`phase10/calibration.json`
(a_*, i) sweep	`phase10/sweep.csv`
CPU/GPU pixel difference	`phase7/*.compare.json`
benchmarks	`phase11/bench.json`

[Chapter under construction — scope: This table grows as the new thesis data tools land (`physics_tables`, `convergence_tool`, the MCMC chains, the cluster fit). See `paper/THESIS_PLAN.md`.]

Appendix D

The Reproduction Pipeline

The top-level `Makefile` exposes the automation targets. `make figures` regenerates every figure in this report from the current artefacts with Matplotlib (data plots from `scripts/make_figures.py`, schematics from `scripts/make_diagrams.py`). `make thesis` runs `figures` and compiles this document (`pdflatex`×3 + `bibtex`); `make paper` does the same for the companion journal paper. `make reproduce` re-runs the science chain end-to-end in gate order:

1. `ctest` — the full physics and science unit-test suite (every gate-0 tolerance of Chapter 21);
2. `render_reference` — Schwarzschild and Kerr reference renders with 32-direction boundary metrology (JSON);
3. `subring_tool` — disk render, beaming asymmetry, sub-ring isolation and flux ratios;
4. `fit_s2_tool` in `kerr` and `nohair` modes — the fits of Chapters 15 and 17;
5. `sweep_tool` — the Sgr A* calibration point and the 20×20 (a_*, i) sweep;
6. `make_figures.py` — figure regeneration from the artefacts.

`scripts/make_numbers.py` then rewrites `numbers.tex`, the macro file from which every simulation-derived number in this text is typeset. A stale or hand-edited number cannot survive a `make reproduce` cycle.

> *[Chapter under construction — scope: This list grows as the new thesis data tools land: `physics_tables`, `convergence_tool`, the extended `sweep_tool` and `subring_tool` modes, `scripts/run_mcmc.py`, and `fit_cluster`. See `paper/THESIS_PLAN.md`.]*

Appendix E

Notation Glossary

Every recurring symbol, with the section where it is defined.

Symbol	Defined	Meaning
M	§8.1	black-hole mass; also the geometric length/time unit ($G = c = 1$)
a, a_*	§8.1	spin parameter J/M; dimensionless $a/M \in [0, 1]$
Σ, Δ, A	eq. (8.2)	Kerr auxiliary functions
$r_\pm$	eq. (8.9)	outer/inner horizon radii
$r_E(\theta)$	eq. (8.10)	ergosphere boundary
$r_{\rm ph}^{\pm}$	eq. (8.13)	prograde/retrograde equatorial photon orbits
$r_{\rm ISCO}$	eq. (8.17)	innermost stable circular orbit
$r_{\rm mb}$	eq. (8.18)	marginally bound circular orbit
ω	eq. (8.8)	frame-dragging (ZAMO) angular velocity
Ω	eq. (8.14)	circular-orbit angular velocity
E, L_z	eq. (8.7)	conserved energy and axial angular momentum
Q	§9.1	Carter constant
μ	eq. (9.1)	rest mass (0 photons, 1 stars)
$R(r), \Theta(\theta)$	eqs. (9.5)–(9.6)	radial and polar potentials
$P(r)$	eq. (9.4)	$E(r^2 + a^2) - aL_z$
λ, τ	eq. (9.9)	affine parameter; Mino time ($\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau$)
b, ξ, η	§8.5, eq. (11.4)	impact parameter L_z/E; critical-curve constants L_z/E, Q/E^2
γ	eq. (11.6)	Lyapunov exponent per polar half-orbit
n	§13.1	photon sub-ring order (equatorial crossings)
$F(r), f(x)$	eqs. (12.1)–(12.2)	Page–Thorne flux; $x = \sqrt{r/M}$
η (efficiency)	eq. (8.19)	accretion efficiency $1 - E_{\rm ISCO}$
g	eq. (12.6)	redshift factor $\nu_{\rm obs}/\nu_{\rm emit}$
$z_{\rm D}, z_{\rm tD}, z_{\rm grav}$	eq. (14.4)	classical, transverse, gravitational redshifts
$a_{\rm mas}, e, i, \Omega, \omega, t_{\rm peri}$	§14.2	osculating orbital elements (semi-major axis in mas, eccentricity, inclination, node, pericentre argument, epoch)
R_0	§15.2	distance to the Galactic centre
p	eq. (17.4)	semilatus rectum $a(1 - e^2)$
χ^2, χ^2_ν	eq. (15.1)	fit objective; reduced value per degree of freedom
J, F, C	§15.2.1	residual Jacobian; Fisher matrix $J^{\mathsf{T}}J$; covariance F^{-1}
ε_q	eq. (17.2)	quadrupole deviation from Kerr (no-hair parameter)
$\tilde{q}$	eq. (17.4)	total quadrupole Q/M^3
μas	§19.1	microarcsecond
θ_g	§19.1	angular size of one gravitational radius at distance R_0

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